

Diophantine Laws of Probabilistic, Quantum and Continuous Computation

Normalization budgets and unique quartic certificates,
four-dimensional quantum mortality,
and a continuous three-outcome trichotomy

Research report merged from three manuscripts prepared for Vladimir Reshetnikov
(batch 60, manuscripts 07, 08 and 09, all dated 30 September 2026)

AI-assisted mathematical development and reproducible exact-arithmetic implementations; unrefereed

Abstract

[write] Which quantitative assertions about computation that is *not* a plain discrete execution log admit finite natural-number witnesses of integer polynomials? Part I (manuscript 08, the base text) represents the output law of every effective probabilistic or quantum program as the dyadic mass of a prefix-free projection of one fixed quartic zero set; determines the exact mass budget of a prescribed history of normalized output laws, the product of the successive largest coordinate ratios; compiles its feasibility into an explicit quartic with $5Nm + m + 4N + 3$ natural witnesses and exactly one witness per valid feasible input; and proves that normalized two-outcome probabilities with any success floor below one are exactly the weakly computable reals, whereas strict conditional comparisons are Σ_2^0 -complete. Part II adds exact Clifford+ T arithmetic and an operator-valued budget (08), and (manuscript 07) a rational Gram-packing construction under which deciding whether a seven-outcome rational quantum instrument in dimension four has an impossible outcome word is undecidable, given Neary's published matrix-mortality theorem, with bijective quartics of $71n + 32$ witnesses for impossible words of length n . Part III (manuscript 09) computes the critical per-step noise radius of a continuous rational piecewise-affine map to within $(L + 1)/(2N)$ independently of time, gives complete quartic safety certificates for positive radius, and builds one map of $[0, 1]^3$ that realizes every disjoint pair of computably enumerable sets as exact acceptance and robust rejection, the rest being noise-fragile divergence.

Status. [write] All three manuscripts are AI-assisted and unrefereed. They give ordinary mathematical proofs, apart from the external inputs they name (MRDP; Neary's six-generator matrix-mortality theorem; Meyer's and Lagrange's theorems on quadratic forms; the existence of a universal Turing machine; the max–min representation of piecewise-affine functions), and finite exact-arithmetic checks. No result of this report is formalized in Lean or Rocq; placement beside ProveIt's Lean project on Hilbert's tenth problem confers no formal status (Section 1.3). No historical priority, minimal dimension or optimal constant is claimed, and no general single-fold or finite-fold MRDP question is settled. The status statements of the three sources are printed in full in Section 1.6.

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1 Research target and relation to ProveIt

1.1 What is being represented?

A computational substrate may manipulate tape symbols, lambda terms, fractions, rewrite graphs, amplitudes, or physical-looking states. If its finite executions are effective, existential arithmetic can describe their occurrence. This observation alone does not preserve the quantitative semantics of probabilistic computation. A polynomial may have many witnesses for one execution, and infinitely many distinct executions may contribute to an output probability. Quantum computational paths can interfere, so summing their squared amplitudes individually is generally wrong.

This article separates four questions. First, can one recover the *whole discrete output law* from integer polynomial solvability? Second, which probability comparisons are themselves existential Diophantine? Third, what is the exact probability mass required to realize a changing conditional output distribution? Fourth, can this last optimization problem have a small, explicit polynomial certificate with uniquely determined witnesses? The answers form a coherent extension of the finite-trace viewpoint.

Our strongest concrete construction is not merely an application of MRDP. It is a direct quartic compiler for optimal normalization feasibility, including precise witness and equation counts, invalid-input rejection, canonical tie resolution, and a uniqueness proof. The infinite and quantum extensions explain why that finite compiler is useful.

1.2 Repository boundary

The inspected ProveIt snapshot is

f608f1cb3c5be8a736df1328c01b293aabfccf4e.

Its Hilbert-tenth-problem development exposes the theorems `mrdp`, `mrdp_iff`, and `mrdp_dioph_iff` in the `Diophantine` namespace. In particular, the source of `Diophantine/MRDP.lean` states the equivalence between recursively enumerable subsets of $\mathbb{N}$ and projections of zero sets of integer polynomials with a finite natural witness dimension, chosen before the input. The statement includes input zero [1, 2]. We use exactly that mathematical interface, not the correctness of every other research claim in the repository.

The repository also supplies pairing, primitive-recursive graph, and trace modules adjacent to this interface. Those are natural integration points for the proposed compiler. This manuscript does not report a local rebuild of ProveIt or an axiom audit of newly supplied declarations: it supplies no new kernel-checked declarations. Its mathematical proofs are independent of a choice of proof-assistant syntax.

[write] **Manuscript 07's reading of the repository.** The inspected repository snapshot is

f608f1cb3c5be8a736df1328c01b293aabfccf4e.

The relevant material is in `Computability/HilbertTenthProblem`. Its MRDP guide states an interface that takes a recursively enumerable subset of $\mathbb{N}$ to the projection of the natural zero set of a fixed integer polynomial with finitely many witnesses. The guide explicitly distinguishes this existence theorem from a practical polynomial compiler with numerical resource bounds [14]. Its

shared trace module also exposes Diophantine closure under bounded universal quantification, exact iteration, and finite reachability [15].

This manuscript supplies an application at two levels. At the explicit level, matrix multiplication histories become quartic equations with counted, canonical witnesses. At the unbounded level, an enumerator of impossible words fits the existing MRDP interface. The rational Gram argument is a separate mathematical component and is not claimed to have already been formalized in the repository. No repository file has been changed, and no repository build has been run for this manuscript.

[write] Manuscript 09’s reading of the repository. The inspected ProveIt modules provide both directions of MRDP over natural inputs and natural witnesses. The declaration `Diophantine.mrdp` supplies an integer polynomial with finitely many witness coordinates, uniformly before the input is chosen; `Diophantine.mrdp_iff` supplies the converse [1, 14]. Thus, for a computably enumerable set $S \subseteq \mathbb{N}$,

$$n \in S \iff \exists w \in \mathbb{N}^k P(n, w) = 0, \quad P \in \mathbb{Z}[X, Y_1, \dots, Y_k]. \quad (1.1)$$

Input zero and zero-valued witnesses are allowed. Polynomial evaluation is in $\mathbb{Z}$, so subtraction never means truncated natural subtraction.

The additional work here lies before that interface: an exact rational abstraction, a complete finite safety checker, and a universal realization with quantitatively controlled noise. We do not use the repository’s optimized universal-equation operation counts or infer witness bounds from them. The guide explicitly distinguishes its existential MRDP endpoint from a practical polynomial-extraction algorithm. We preserve that distinction.

The repository was inspected on 30 September 2026. The head observed during inspection was `f608f1cb3c5be8a736df1328c01b293aabfccf4e`. The source declarations were read, but the repository was not rebuilt in this session. Accordingly, repository build and axiom-audit claims remain repository provenance, not a fresh verification receipt for this manuscript.

1.3 [write] What ProveIt has formalized, and what this report uses

This report continues the Lean project `Computability/HilbertTenthProblem` only through MRDP. Its fixed-arity statements (Theorems 4.3, 22.1 and 26.3, Corollaries 4.5 and 29.2) use `Diophantine.e.mrdp`, `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` (`Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`) as a mathematical interface. Manuscript 07 also names the trace interfaces `Diophantine.boundedForall_dioph`, `Diophantine.exactIter_dioph` and `Diophantine.existsExactIter_dioph` (`.../Lean/Diophantine/Common/DiophantineTrace.lean`) as a possible route; distinct declarations with the same short name `boundedForall_dioph` exist elsewhere in the project, so the full names are the ones meant. These interfaces assert that representations *exist*; they say nothing about how many witnesses a representation has. **None of the report’s own theorems is formalized**, and the Python programs are finite checks, not proofs.

Part II’s undecidability in dimension four (Theorem 18.2) rests on Neary’s published theorem that mortality of six integer 3×3 matrices is undecidable (STACS 2015, Corollary 12; our Theorem 18.1) and, for the $d + 3$ row bound, on Meyer’s theorem. ProveIt keeps Neary’s paper (`Computability/HilbertTenthProblem/Lean/LIPIcs.STACS.2015.649.pdf`) and formalizes, for its own purposes, the *binary-tag layer* of his construction: a binary tag system $0 \rightarrow 0, 1 \rightarrow u$ simulating

a tag system with deletion d , after his Lemma 9 (`TagBinary91.lean`), the universality of the resulting 91-operation tag certificate (`Jones1980.tag91_re`, `TagUniversal91.lean`), and the undecidability of the encoded tag family (`Jones1980.tag91_undecidable`, `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1980/TagDecide91.lean`); see the Neary rows of `Computability/HilbertTenthProblem/Lean/STATUS.md` (lines 416–418 at the placement commit). The Post-correspondence and matrix-mortality steps of Neary’s Corollary 12, Meyer’s theorem, the rational Gram-packing theorem and everything downstream of them are *not* formalized. Lagrange’s four-square theorem, the only number-theoretic input of the elementary $4d$ route, is available in Mathlib as `Nat.sum_four_squares`, which the project already uses.

All three manuscripts inspected ProveIt at `f608f1cb3c5be8a736df1328c01b293aabf4e`. Nothing under `Computability/` changed between that commit and the placement commit `7498484af`, so every repository statement quoted in this report is still accurate; the sentences of manuscripts 07 and 08 that describe the repository were checked against the current files.

1.4 Established ingredients and contribution ledger

The MRDP equivalence is an established theorem [9]. Formalized reductions through FRACSTRAN already exist [3]; hence a bare assertion that FRACSTRAN halting is Diophantine would not be a new result. Weakly computable reals and their bounded-variation characterization also predate this work [4, 5]. Max-relative entropy and its operator-order formula are established [8]. Finally, the arithmetic hardness of probabilistic termination and expectation comparison has substantial prior literature [6].

Component	Status in this manuscript
MRDP; prefix allocation; weak computability; max-relative entropy	Established ingredients, stated with attribution and used with explicit proofs of the needed consequences.
Exact normalization budget	Derived here as a finite/infinite optimization theorem, with a canonical minimal realization.
Unique quartic compiler	Explicit construction and proof here; implemented and checked on finite instances. Historical priority is not asserted.
Sharp binary and operator bounds	Proved here; the underlying entropy notions are not claimed as new.
High-success arithmetic boundary	Explicit realization and hardness reductions here, situated relative to known computability classifications.
Universal output-law polynomial	Existence and degree reduction proved; its full coefficients and witness count are not generated.

The questions answered here are precisely formulated mathematical questions about the proposed extension. They are not presented as previously published open problems whose priority history has been exhaustively established.

1.5 [write] The three sources and the plan of this report

This report merges three manuscripts of batch 60 of ProveIt’s incoming-report intake. They were written independently, from the same repository snapshot, and none cites another.

Source	Archive and title	Printed here
08 (base)	<code>ProveIt_Diophantine_Quantum_No</code> <code>rmalization: Diophantine Laws of</code> <i>Probabilistic Computation</i>	Sections 1, 3–9 (Part I), 11–12 (Part II), 33, and its parts of Sections 34–37 and the appendices
07	<code>quantum_diophantine_research:</code> <i>Rational Gram Packing, Four-Dimensional Quantum Mortality, and Diophantine Certificates</i>	Sections 13–22 (Part II), and its parts of Sections 34–37 and Appendix A.2
09	<code>ProveIt_Diophantine_Barriers:</code> <i>Diophantine Barriers for Continuous Computation</i>	Sections 23–32 (Part III), and its parts of Sections 34–37 and the appendices

The three share no theorem. They meet in three places, each discussed where it occurs: two different *quantum models* (08’s exact Clifford+ T amplitudes and operator budgets, 07’s rational real Kraus instruments; Section 10); three different *fixed MRDP polynomials* (Remark 2.1); and the absence of computable *witness or margin bounds* (Corollaries 18.4 and 30.3, Theorem 30.1). Section 2 fixes conventions and a dictionary of letters used differently by the sources. Appendix F records the provenance of the merge and every choice it made.

Abstract of manuscript 08. The statement that the halting set of an effective computational substrate is Diophantine follows from MRDP. For probabilistic and quantum computation, the more discriminating questions concern output *laws*, conditional probabilities, and the cost of realizing prescribed normalized histories. We develop a substrate-independent representation of an effective discrete output law as the dyadic mass of a prefix-free projection of a fixed quartic integer zero set. This construction counts projected events, not polynomial witnesses, and therefore requires no finite-fold hypothesis.

The central constructive result determines the exact mass budget for a sequence of strictly positive probability vectors. Its cost is the product of the successive largest coordinate ratios. For a finite rational history with N transitions and m outcomes, we construct one quartic polynomial with $5Nm + m + 4N + 3$ natural witnesses. A valid feasible input has exactly one witness tuple. A canonical earliest-maximum rule removes ties without increasing the degree. We also prove a sharp binary variation bound and an operator version governed by max-relative entropy.

Normalization changes the arithmetic boundary. Conditional two-outcome probabilities with any fixed success floor below one realize exactly the weakly computable reals in $[0, 1]$, whereas almost-sure return realizes exactly the computable reals. Strict conditional comparisons remain Σ_2^0 -complete even when success is guaranteed arbitrarily close to one. Explicit integer arithmetic for Clifford+ T circuits supplies a quantum front end. The accompanying code exports a concrete quartic and performs exhaustive, symbolic, and exact-arithmetic tests. We distinguish these proved constructions from established ingredients, unexpanded MRDP polynomials, and formalization work that has not been performed.

Abstract of manuscript 07. We develop a quantitative route from integer matrix mortality to exactly normalized rational quantum measurements and then to polynomial equations over the natural numbers. The central construction packs the rows of a rational Gram factor across existing Kraus operators. If an input consists of k rational $d \times d$ matrices, the resulting instrument has $k + 1$ outcomes and dimension

$$D = d + \left\lceil \frac{d + 3}{k} \right\rceil.$$

Mortality is preserved in both directions, arbitrary insertions of an invertible idle outcome do not create false witnesses, and the shortest nonempty mortal-word length increases by at most one. Combining this construction with the published six-generator, 3×3 matrix-mortality theorem gives undecidability for rational quantum instruments in dimension four with seven outcomes. An elementary four-square variant gives dimension five with seven outcomes without using Meyer’s theorem.

For fixed history length n , we give a fully specified quartic polynomial whose natural solutions are in bijection with the impossible outcome words. In dimension four with seven outcomes the construction uses $71n + 32$ natural witnesses and $72n + 48$ quadratic residuals. A separate MRDP argument gives a fixed-arity Diophantine representation of the unbounded property, without claiming that this compression preserves unique witnesses. We prove an exact length-dependent probability gap, explain its limited robustness meaning, and identify concrete interfaces to ProveIt’s Diophantine library. The accompanying standard-library Python implementation passes 32,418 finite exact-arithmetic checks. The mathematical proofs are not Lean-certified, and priority of the resource improvement is not asserted.

Abstract of manuscript 09. We study which infinite-time properties of an unconventional universal computer admit finite Diophantine witnesses. For a continuous rational piecewise-affine map of a compact cube, we define the critical per-step perturbation needed to reach a closed rational target. A finite weighted grid computes this real number with an error bounded by $(L + 1)/(2N)$, independently of trajectory length. A bottleneck/cut duality converts positive critical radius into a complete family of explicit quartic sum-of-squares certificates. Encoding the finite checker and applying ProveIt’s MRDP interface then gives one fixed finite-dimensional Diophantine representation of robust safety.

We construct a single continuous rational piecewise-affine map $F : [0, 1]^3 \rightarrow [0, 1]^3$ and a fixed rational target box that realize every disjoint pair (A, B) of computably enumerable sets: A is exact acceptance, B is positive-radius avoidance, and the complement is nonacceptance vulnerable to arbitrarily small perturbations. The construction combines a packed Cantor-tape simulation, separated terminal slabs, and an expanding emergency coordinate. Rejection after T steps satisfies explicit exponential upper and lower bounds on the critical radius. Consequently the radius is a computable real-valued function, while its positivity is Σ_1^0 -complete and fragile divergence is Π_1^0 -complete. The two enumerable outcomes are even recursively inseparable. No existential Diophantine formula represents the third outcome, and no computable positive gap, certificate-size bound, or robustness-resolution bound exists in general.

1.6 [write] Status statements of the sources

Each source’s own status statement is kept verbatim.

Manuscript 08. Status. This is a research manuscript with complete ordinary mathematical proofs and executable finite checks, not a claim of independently established historical priority or a Lean/Rocq-verified extension. The most concrete deliverable is the explicit single-fold finite-history compiler in Section 6. No general finite-fold MRDP conjecture is assumed or settled.

Manuscript 07. Scope. This is a research contribution with explicit dependencies, not a claim to have newly discovered quantum undecidability or to have solved general single-fold Diophantine

representability. The main contribution offered for further review is the packed normalization theorem and its quantitative consequences.

Manuscript 09. Status and scope. The manuscript supplies complete mathematical proofs for the core results, apart from classical universality and MRDP. The neural corollary also uses the standard max–min representation theorem. The proposed contribution is the joint realization–radius–certificate package, not the already known facts that continuous dynamics can be universal or that suitable robustness implies decidability. Global priority and minimal dimension are not established. The companion exact-rational code passes 5,660 checks; neither these results nor the compiler have been Lean-verified. No expanded universal machine table or fixed MRDP polynomial is claimed.

2 [write] Conventions and notation of the report

2.1 Domain

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Every polynomial witness ranges over $\mathbb{N}$; every polynomial has integer coefficients, and subtraction in a displayed polynomial is subtraction in $\mathbb{Z}$, never truncated subtraction. A *residual system* is a finite list of integer polynomials required to vanish; its single equation is the sum of their squares, of degree at most four when every residual is at most quadratic. Probabilities are exact rationals, exact elements of $\mathbb{Q}(\sqrt{2}, i)$, or computable reals, as stated in each Part; no floating-point quantity enters a theorem (the binary-sharpness diagnostics of manuscript 08’s software are illustrative only, Section 34).

A representation is *single-fold* if each parameter tuple has at most one witness, *finite-fold* if it has finitely many, and *parsimonious* (or *bijective*) for a set of semantic objects if its witnesses are in a specified bijection with them. The explicit certificates of Theorems 6.1, 21.1 and 26.2 are single-fold or bijective *for each fixed size parameter*; the fixed-arity MRDP polynomials of this report carry no multiplicity claim.

2.2 Words used in several senses

- *Certificate*. Here, a natural witness tuple of an explicit polynomial, or (Part III) the finite graph cut it encodes. In ProveIt’s Hilbert’s-tenth-problem project a “straight-line certificate” is a program whose arithmetic operations are counted; no result of this report concerns operation counts.
- *Quartic*. Total degree at most four. Part I’s quartic $F_{N,m}$, Part II’s $\mathcal{Q}_{D,K,n}$ and Part III’s Q_N are three unrelated explicit families; the fixed-arity polynomials obtained from MRDP are not explicit and have no stated degree unless Lemma 3.3 is applied.
- *Universal*. Theorem 4.3 (a fixed quartic for all effective output laws), Theorem 22.1 (a fixed polynomial for $\text{QMort}_{4,7}$) and Theorem 28.1 (a fixed three-dimensional map realizing all disjoint computably enumerable pairs) use “universal” for three different objects; none is a universal pair (ν, δ) in the sense of Jones’s articles.
- *Mortal / impossible / zero-probability*. In Part II a word is *mortal* for a matrix list when its ordered product is zero; for a positive definite input state this is the same as the outcome

word having probability zero (Lemma 14.2). In Part I “nonreturn” is missing probability mass, not a zero-probability word.

- *Robust.* Part III’s robust avoidance is positivity of the critical *per-step* noise radius (Section 24); Part II’s robustness discussion (Proposition 20.2) is a perturbation of Kraus operators at fixed word length. Neither is a probabilistic noise model.
- *Budget.* Part I’s normalization budget G_N is a product of coordinate ratios; Part II’s operator budget is its noncommutative analogue. Part III’s radius is a noise tolerance, not a mass.

2.3 Letters used differently by the sources

Each Part keeps its source’s letters, so that a symbol is always read in the Part where it occurs; the table records the letters whose meanings are most easily confused across the sources. Where two meanings met inside one Part, one was renamed (Section 2.4).

Letter	Part I (08)	Part II (08 in Sections 11–12; 07 from Section 13)	Part III (09)
N	horizon (number of transitions)	08: horizon in Theorem 12.1; 07: common denominator $H_j = NA_j$	mesh $N = D2^n$
m	number of output labels	07: number of active letters of a word; witness dimension in Theorem 22.1	number of interpreter states
K	—	07: number of outcome matrices in $\mathcal{Q}_{D,K,n}$ (so $K = k + 1$ for a packed instrument)	the cube $[0, 1]^d$; K_0 a halting set
k	witness dimension	07: number of active outcomes	witness dimension
D	D_s products of denominators; $D_\infty, D_{\max}$ divergences	07: quantum dimension; 08: $D_{\max}$	denominator-clearing integer
H	—	08: the Hadamard gate; 07: integer numerator matrices H_j, H_w	target set
A, B	A_s, B_s selected ratio witnesses	08: integer norm coefficients in $A + B\sqrt{2}$; 07: Kraus operators A_j , Gram factor B , blocks B_i	the disjoint computably enumerable pair (A, B) ; Lipschitz bounds in Theorem 31.1
T	—	08: the T gate; 07: the tensor matrices T_i	time; T_N grid targets
S	—	08: the S gate	a cut; $S_T(L)$ a geometric sum
μ	output law μ_e	07: shortest mortal length $\mu(\mathcal{M})$	envelope $\mu(n)$
δ	initial mass	08: initial mass; 07: gap δ_n , Kronecker δ_{ab}	a margin
α, β	input fraction α/β	07: $\alpha = 4/(5c)$, $\beta = 3/5$	α_j state offsets, $\beta = 1/(3m)$
L	—	08: number of gates	Lipschitz bound
Q	matrix of a Σ_2^0 normal form	07: positive definite form	quartic Q_N
R, G	R_s largest ratio, G_s product	08: operator ratio R_s and product G_s (renamed); 07: a residual block R	G_N the grid; r_N a bottleneck
ρ	—	density operator / accumulated ensemble	critical radius

The tempting false readings are these. Part II’s K is a count, never the accumulation factor (which is $R(\sigma, \tau)$ here). The symbols H , T , S without subscript in Section 11 are gates, not matrices of manuscript 07. Part III’s K is a set, and its N is a mesh, not a horizon. The single-fold quartic of Part I has $W(N, m)$ witnesses; Part II’s bijective quartic has $W(D, K, n)$; the two functions W and E are different, and Part III’s E_N is an edge set. Other letters (C , M , V , h , p , r , s , t , u , v , w , x , y , z among them) are local to the section that defines them; for example C_s are Part I’s numerator products while $C = L + 1$ in Part III, and h_s is a heralded mass in Part I while $h_N = 1/(2N)$ in Part III.

2.4 Renamed symbols and macros

- In Section 12 (manuscript 08’s operator budget), the domination factor written $K(\sigma, \tau)$, K_s in the source is written $R(\sigma, \tau)$, R_s here, and the product written H_s is written G_s here. The renaming removes clashes with 07’s outcome count K , 07’s numerator matrices H_j and 08’s own Hadamard gate H , and shows that the operator quantities specialize to Part I’s R_s and G_s for commuting (diagonal) states. No normalization changed. The same renaming is applied to research question 6.
- In Section 16, manuscript 07’s local $S = \sum_i M_i^T M_i$ and $T = \text{tr}(S)$ are written $S_{\mathcal{M}}$ and $t_{\mathcal{M}}$, to keep them apart from the S and T gates.
- Manuscript 09 defines `\ce` as “computably enumerable” while 08 defines it as “c.e.”; 09’s macro is `\cenum` here, and the printed words are unchanged. The three definitions of `\code` (08, 07, 09) are unified as a breakable typewriter path; 09’s escaped underscores inside it were unescaped. The pinned commit, typeset by three differently named macros in the sources, is printed by one.
- File names are the shipped names of this report (`code/NN-...`, `data/NN-...`); the delivered names are listed in the README, which also explains how to rerun each suite.

2.5 How the merge is marked

Text added in writing this report is marked [write]. Everything else is the text of the source named at the head of its Part or section, with labels, file names and the symbols of Section 2.4 changed. Cross-references between the sources were added only in [write] remarks.

Remark 2.1 ([write] Three fixed polynomials). Theorem 4.3 (08) represents the relation “interpreter e allocates word code c to label z ”; Theorem 22.1 (07) represents the set $\text{QMort}_{4,7}$ of mortal four-dimensional seven-outcome instruments; Theorem 26.3 (09) represents positive critical radius and exact acceptance. All three come from MRDP applied to a semidecision procedure; none is expanded, none has a stated witness count, and none is claimed single-fold or finite-fold. The explicit, counted polynomials of the three Parts are the size-indexed families $F_{N,m}$, $\mathcal{Q}_{D,K,n}$ and Q_N .

Part I

Probabilistic computation (manuscript 08)

3 Effective output laws and quartic arithmetic

3.1 Conventions

We take $\mathbb{N} = \{0, 1, 2, \dots\}$. A relation on natural tuples is *Diophantine* when it has the form

$$R(\mathbf{x}) \iff \exists \mathbf{w} \in \mathbb{N}^k P(\mathbf{x}, \mathbf{w}) = 0, \quad P \in \mathbb{Z}[\mathbf{X}, \mathbf{W}]. \quad (3.1)$$

All polynomial witnesses in this article are natural numbers. Negative coefficients are allowed. This is not a statement about rational witnesses, or about unrestricted integer witnesses unless an explicit translation is given.

A real a is *left-c.e.* if a computable nondecreasing rational sequence converges to a . A discrete *subprobability law* is a function $\mu : \mathbb{N} \rightarrow [0, 1]$ with $\sum_z \mu(z) \leq 1$. Missing mass is allowed to represent nonreturn. An effective family of output laws is uniformly left-c.e. in its program and output arguments.

Definition 3.1 (Effective first-return presentation). An effective first-return presentation consists of an effective enumeration of finite terminal histories, a computable output label for each history, and uniformly computable nonnegative real history weights. Histories represent pairwise disjoint terminal outcomes, and the sum of their weights is at most one. Each actual finite terminal history occurs once in the presentation, or is canonically deduplicated.

For a quantum machine, a history in this definition is a sequence of actual measurement outcomes together with the classical control information. Interfering basis paths between measurements must already have been combined when computing its weight.

Lemma 3.2 (Effective first returns give lower approximations). *Every effective first-return presentation has a uniformly left-c.e. output law.*

Proof. For each history j , compute rational lower bounds on its weight, replace negative bounds by zero, and take running maxima. At stage s , sum the bounds for the first s enumerated histories with the specified output label. These sums are computable, nondecreasing, and bounded by the true output mass. Every finite collection of history weights is approximated arbitrarily well, so the limit is their full sum. The construction is uniform. Rounding down on successively finer dyadic grids gives dyadic lower approximations. For infinitely many labels, expose only finitely many labels at each stage; all lower masses still have sum at most one. $\square$

The assumptions permit an infinite state space and an unbounded running time. They do not permit noncomputable real transition constants hidden in the source language. They also do not turn an infinite limiting configuration into a finite first return.

3.2 A degree-reduction interface

Lemma 3.3 (Quartic lowering). *Every Diophantine relation has a representation by a single integer polynomial of total degree at most four. A fixed input polynomial gives a fixed finite new witness dimension.*

Proof. Evaluate the input polynomial by a finite arithmetic circuit. Introduce variables for its intermediate values. Each addition has a linear residual, and each multiplication has a quadratic residual. Signed intermediate integers can be represented by differences of natural variables; a product of two such differences is still quadratic. Equate the output difference to zero and sum the squares of all residuals. An integer sum of squares is zero exactly when every residual is zero. Conversely, a genuine evaluation supplies intermediate values and hence natural positive/negative parts. The resulting sum has total degree at most four. No uniqueness of the introduced witnesses is required for this lemma. $\square$

The polynomial in Lemma 3.3 can be much larger than the original one. Degree reduction is not an efficiency theorem. The direct construction in Section 6, in contrast, controls its arithmetic circuit size and natural witness multiplicity.

4 Output laws as masses of Diophantine projections

4.1 Canonical allocation of fair-coin events

A finite binary word σ specifies the cylinder of infinite fair-coin streams beginning with σ . Its probability is $2^{-|\sigma|}$. A prefix-free set of words specifies disjoint cylinders.

Lemma 4.1 (Effective labeled allocation). *Suppose a computable enumeration requests labeled binary words of lengths $n_1, n_2, \dots$, and $\sum_j 2^{-n_j} \leq 1$. There is a deterministic effective allocation of prefix-free words with exactly these lengths and labels.*

Proof. Maintain a finite set of free words, initially containing the empty word. Maintain the invariant that there is at most one free word of each length. For a request of length n , choose the longest free word of length at most n . Remove it. Repeatedly split its cylinder into its left child, which is continued, and its right child, which is put in the free set, until length n is reached. Allocate the final left descendant to the request.

There was no other free word with length strictly between the chosen length and n , by maximality. Thus the one-free-word-per-length invariant is preserved. Free and allocated cylinders remain pairwise disjoint and together have total mass one. If there were no suitable free word, all free words would have distinct lengths greater than n . Since the free set is finite, its total mass would be strictly less than $\sum_{k>n} 2^{-k} = 2^{-n}$. The requested mass would then exceed the remaining mass, contrary to the hypothesis. Every allocation therefore succeeds. $\square$

Theorem 4.2 (Fair-coin realization). *A uniformly given discrete subprobability law is realizable as the terminal output law of a fair-coin program if and only if it is uniformly left-c.e.*

Proof. A fair-coin machine supplies an effective enumeration of disjoint finite halting cylinders, so the forward implication follows directly, or from Lemma 3.2.

For the reverse implication choose monotone dyadic lower approximations $d_s(z)$, with finite support at each stage and total mass at most one. Each increment $d_s(z) - d_{s-1}(z)$ is a nonnegative dyadic rational and can be decomposed into finitely many requests for masses 2^{-n} , all labeled by z . The total requested mass is $\sum_z \mu(z) \leq 1$. Lemma 4.1 allocates their cylinders. A program dovetails this allocation enumeration with reading its random stream and returns z when it discovers an allocated cylinder containing that stream. Its output events are exactly the allocated cylinders, and their mass for label z is $\lim_s d_s(z) = \mu(z)$. $\square$

The program may read additional random bits before an allocated cylinder is recognized. The allocated word is an event code, not necessarily its literal minimal runtime leaf. The theorem preserves the terminal output law, not runtime, space, or interactive behavior.

4.2 One quartic zero set for all effective laws

Encode a binary word by

$$c(\sigma) = 2^{|\sigma|} + \text{value}_2(\sigma), \quad \ell(c) = \lfloor \log_2 c \rfloor \quad (c \geq 1). \quad (4.1)$$

The empty word has code one. Fix a universal allocation interpreter. It accepts requests only while preserving the prefix-free allocation invariant; malformed or overbudget requests are discarded or leave the source stalled. Thus arbitrary program codes still specify a valid subprobability law. Valid source compilers from Theorem 4.2 never trigger this protection.

Theorem 4.3 (Universal projected-root representation). *There are a fixed k and a fixed integer polynomial $U(E, Z, C, W_1, \dots, W_k)$ of total degree at most four such that every effective discrete output law has, effectively from its presentation, an index e with*

$$\mu_e(z) = \sum_{c \geq 1} 2^{-\ell(c)} \mathbf{1}_{\{\exists \mathbf{w} \in \mathbb{N}^k \ U(e, z, c, \mathbf{w})=0\}}. \quad (4.2)$$

For fixed e , the word codes appearing in this sum across all labels are prefix-free, and each code has at most one label.

Proof. The relation “interpreter e allocates word code c to label z ” is c.e.: run the interpreter and enumerate its allocations. Apply MRDP to this relation, using an effective coding of its input tuple, and then apply Lemma 3.3. The resulting polynomial and its witness dimension depend only on the fixed interpreter. Prefix-freeness and label uniqueness are properties of that interpreter. Equation (4.2) is the disjoint-cylinder probability formula. Every law from Theorem 4.2 has a source for the interpreter. $\square$

Remark 4.4 (Projection is essential). Equation (4.2) assigns one weight to a word code for which a witness exists. It does *not* sum over witness tuples. A single word may have infinitely many arithmetic witnesses without changing its probability. Consequently this theorem does not assume a finite-fold or single-fold MRDP representation. Conversely, any effectively specified polynomial relation whose output-labeled projection is prefix-free defines a left-c.e. subprobability law by this same formula.

Corollary 4.5 (Ordinary strict thresholds). *For any effective output-law interpreter, the relation $b\mu_e(z) > a$, where $a \in \mathbb{N}$ and $b \geq 1$, is Diophantine and has a fixed quartic representation.*

Proof. A monotone rational approximation exceeds a/b at a finite stage exactly when the limit does. This is a c.e. relation in (e, z, a, b) . Apply MRDP and quartic lowering, including $b \geq 1$ in the decidable input guard. $\square$

Neither Theorem 4.3 nor Corollary 4.5 supplies the full expanded universal polynomial in the accompanying software. They give an existence construction through a fixed c.e. interpreter. The finite polynomial exported by the software is the substantially more explicit one constructed next.

5 The exact normalization budget

5.1 A mass-allocation problem

Let $p_0, p_1, \dots$ be strictly positive probability vectors on $\{1, \dots, m\}$, with $m \geq 2$. Suppose that by checkpoint s we have accumulated masses $a_{s,i}$ of disjoint returned outcomes. Those masses must be coordinatewise nondecreasing. Write

$$h_s = \sum_i a_{s,i}, \quad a_{s,i} = h_s p_{s,i}. \quad (5.1)$$

Here h_s is the probability already assigned to a successful return, also called the *heralded mass*. Fix an initial mass $h_0 = \delta > 0$ and a unit total budget $h_s \leq 1$.

The checkpoints are points in an effective enumeration of accumulated probability. They need not coincide with successive machine instructions or successive coin flips. This distinction is necessary: exact rational mass increments need not be realizable in a uniformly bounded number of fair coin flips.

Definition 5.1 (Normalization cost). For $s \geq 1$, define

$$R_s = \max_{1 \leq i \leq m} \frac{p_{s-1,i}}{p_{s,i}}, \quad G_0 = 1, \quad G_s = \prod_{j=1}^s R_j. \quad (5.2)$$

The path's total multiplicative cost is G_N for a finite horizon, and $G_\infty = \sup_s G_s$ for an infinite history.

Since both vectors sum to one, $R_s \geq 1$. Otherwise every old coordinate would be strictly smaller than the new coordinate, contradicting equal sums.

Theorem 5.2 (Optimal normalization budget). *A finite or infinite positive probability history admits masses satisfying (5.1), $h_0 = \delta$, $h_s \leq 1$, and $a_{s,i} \geq a_{s-1,i}$ if and only if*

$$\delta G_s \leq 1 \quad \text{for every checkpoint } s. \quad (5.3)$$

Its unique pointwise minimal feasible history is

$$h_s^* = \delta G_s, \quad a_{s,i}^* = \delta G_s p_{s,i}. \quad (5.4)$$

In particular, the largest feasible initial mass for a finite path is G_N^{-1} .

Proof. For a single transition, coordinatewise monotonicity is equivalent to

$$h_s p_{s,i} \geq h_{s-1} p_{s-1,i} \quad \text{for every } i,$$

and, because all quantities being divided by are positive, to $h_s \geq h_{s-1} R_s$. Iterating gives $h_s \geq \delta G_s$. This proves necessity and pointwise minimality of any possible candidate.

Conversely, the choice in (5.4) satisfies the single-step inequality with equality in the maximizing ratio and inequality in all other coordinates. Its coordinates are nondecreasing, its initial mass is δ , and the assumed budget condition gives total mass at most one. It is therefore feasible. Each of its total masses is forced for any pointwise minimal solution, and the prescribed vectors then force every coordinate. The argument applies to every finite prefix of an infinite history without any compactness or choice step. $\square$

Corollary 5.3 (Effective realization of a feasible history). *If δ and the $p_{s,i}$ are computable rationals, and the budget condition holds, the greedy masses define a computable nondecreasing rational approximation to an output subprobability law. A fair-coin program realizes that limiting law.*

Proof. Finite maxima and products of positive rationals are computable exactly. Theorem 5.2 supplies monotonicity and boundedness. Each coordinate therefore converges and is left-c.e.; the limiting sum is at most one. Apply Theorem 4.2. When necessary, round rational cumulative masses down to finer dyadic lower approximations before allocation. $\square$

Example 5.4 (An excursion costs mass). Consider

$$p_0 = (1/2, 1/2), \quad p_1 = (2/3, 1/3), \quad p_2 = (1/2, 1/2).$$

Then $R_1 = 3/2$, $R_2 = 4/3$, and $G_2 = 2$. Thus $\delta \leq 1/2$ is necessary and sufficient, although the initial and final conditional laws agree. At the maximum initial mass, the greedy cumulative masses are

$$(1/4, 1/4), \quad (1/2, 1/4), \quad (1/2, 1/2).$$

The first transition spends additional mass on the first output; the return transition spends it on the second. Previously assigned mass cannot be withdrawn.

5.2 An information-theoretic expression

For positive classical probability vectors, write

$$D_\infty(p||q) = \log \max_i \frac{p_i}{q_i},$$

with the natural logarithm throughout this article. Then

$$\log G_N = \sum_{s=1}^N D_\infty(p_{s-1}||p_s). \quad (5.5)$$

The divergence itself is standard, and is the commuting specialization of max-relative entropy [8]. Its role here is an exact additive budget for irreversible accumulation of terminal mass. The direction of the arguments is significant: the old law is compared with the new law. Reversing them generally changes the answer.

6 A single-fold quartic certificate with explicit size

6.1 Parameters and statement

Fix an external horizon $N \geq 0$ and an external number of labels $m \geq 2$. The free natural parameters are

$$(u_{s,i})_{0 \leq s \leq N, 1 \leq i \leq m}, \quad (v_s)_{0 \leq s \leq N}, \quad \alpha, \beta.$$

Call an input *valid* when

$$u_{s,i} \geq 1, \quad v_s = \sum_i u_{s,i}, \quad 1 \leq \alpha \leq \beta. \quad (6.1)$$

It specifies $p_{s,i} = u_{s,i}/v_s$ and $\delta = \alpha/\beta$. Denominators need not be reduced, and different rows may use independently rescaled denominators.

Theorem 6.1 (Explicit unique-witness quartic). *For each fixed N, m there is an explicitly constructible integer polynomial $F_{N,m}$ of total degree at most four with*

$$W(N, m) = 5Nm + m + 4N + 3 \quad (6.2)$$

$$E(N, m) = 5Nm + m + 6N + 4 \quad (6.3)$$

natural witnesses and squared residuals, respectively, such that:

- (i) *a valid input has a witness if and only if its prescribed history is feasible with initial mass α/β ;*
- (ii) *every valid feasible input has exactly one natural witness tuple;*
- (iii) *invalid inputs have no natural witness tuple.*

The residual system has an arithmetic-circuit implementation of size $O(Nm)$. This is not a claim that its fully expanded polynomial has only $O(Nm)$ monomials.

6.2 Validity, ratios, and selectors

Introduce positivity witnesses $g_{s,i}$ and g_α, g_β , with residuals

$$\begin{aligned} u_{s,i} - 1 - g_{s,i} &= 0, & v_s - \sum_i u_{s,i} &= 0, \\ \alpha - 1 - g_\alpha &= 0, & \beta - \alpha - g_\beta &= 0. \end{aligned} \quad (6.4)$$

For every transition $s = 1, \dots, N$ introduce $X_{s,i}, Y_{s,i}$ and enforce

$$X_{s,i} - u_{s-1,i}v_s = 0, \quad Y_{s,i} - v_{s-1}u_{s,i} = 0. \quad (6.5)$$

Their ratios are the candidates for R_s : $X_{s,i}/Y_{s,i} = p_{s-1,i}/p_{s,i}$, and their denominators are positive.

Introduce selectors $e_{s,i}$ satisfying

$$e_{s,i}(e_{s,i} - 1) = 0, \quad \sum_i e_{s,i} - 1 = 0. \quad (6.6)$$

They select exactly one coordinate. Define selected numerator and denominator witnesses A_s, B_s by

$$A_s - \sum_i e_{s,i}X_{s,i} = 0, \quad B_s - \sum_i e_{s,i}Y_{s,i} = 0. \quad (6.7)$$

Every residual displayed so far has total degree at most two.

6.3 Canonical earliest-maximum rule

Ordinary maximum constraints allow several selectors when two ratios tie. To eliminate this multiplicity, introduce natural slacks $t_{s,i}$ and use

$$A_s Y_{s,i} - B_s X_{s,i} - t_{s,i} - \sum_{j>i} e_{s,j} = 0. \quad (6.8)$$

If coordinate k is selected, the last sum is one for $i < k$ and zero for $i \geq k$. Thus the selected ratio must be *strictly* larger than every earlier ratio and at least as large as every later one. Integer strictness is encoded by subtraction of one, not by a new quantifier or a high-degree product.

Lemma 6.2 (Unique pivot). *For any positive integer lists $(X_i), (Y_i)$, the selector and comparison constraints (6.6)–(6.8) have exactly one solution in the selector, selected-ratio, and slack variables. Its selected index is the least index maximizing X_i/Y_i .*

Proof. The Boolean and sum constraints select an index k . Nonnegative slacks give $X_k Y_i - Y_k X_i \geq 1$ for $i < k$ and ≥ 0 for $i \geq k$. Positivity of all denominators makes these exactly the asserted strict and weak ratio comparisons. Hence k must be the least maximizing index. Conversely, that index satisfies those comparisons: a strictly positive integer cross difference is at least one. It determines all selectors, $A = X_k$, $B = Y_k$, and all slacks uniquely. $\square$

6.4 Canonical products and the terminal budget

Set $C_0 = \alpha$, $D_0 = \beta$ as expressions, not additional witnesses. For $s = 1, \dots, N$ introduce C_s, D_s and impose

$$C_s - C_{s-1} A_s = 0, \quad D_s - D_{s-1} B_s = 0. \quad (6.9)$$

Finally introduce a natural slack z and impose

$$D_N - C_N - z = 0. \quad (6.10)$$

For $N = 0$, this uses $D_0 = \beta, C_0 = \alpha$ directly. Products deliberately remain unreduced. There are no arbitrary common factors and no gcd certificates to create extra witnesses.

Let $\mathcal{R}_{N,m}$ be the complete residual list in (6.4)–(6.10), including every indicated index, and put

$$F_{N,m} = \sum_{r \in \mathcal{R}_{N,m}} r^2. \quad (6.11)$$

This is the promised explicit polynomial definition. It is a finite sum-of-squares specification, not an appeal to MRDP.

6.5 Proof of the compiler theorem

Proof of Theorem 6.1. Soundness. A natural zero of $F_{N,m}$ makes each residual zero. The validity residuals force (6.1). Equations (6.5) then force positive ratio denominators. Lemma 6.2 forces the least maximizing ratio at each stage, so $A_s/B_s = R_s$. The product residuals force

$$\frac{C_N}{D_N} = \frac{\alpha}{\beta} \prod_{s=1}^N R_s = \delta G_N.$$

The denominator is positive. The final natural slack forces $C_N \leq D_N$, hence $\delta G_N \leq 1$. Since G_s is nondecreasing, the budget condition holds for every earlier stage. Apply Theorem 5.2.

Completeness. For a valid feasible input, assign the positivity and input slacks their forced values, compute the X, Y lists, choose each least maximizing coordinate, and assign its uniquely determined nonnegative comparison slacks. Compute the unreduced products C_s, D_s . Feasibility makes $D_N - C_N$ nonnegative; assign it to z . Every residual is zero.

Uniqueness. The input fixes the validity slacks and all X, Y values. Lemma 6.2 fixes the selectors, A, B , and comparison slacks. The recurrence fixes the unreduced products, and the final equation fixes z . There are no other witnesses. The existence of other, nonminimal feasible mass histories is irrelevant: the certificate describes the canonical optimal budget, not an arbitrary feasible history.

Size and degree. The positivity slacks contribute $(N + 1)m$ witnesses, the two input slacks contribute two, and each transition contributes $4m + 4$ witnesses: X, Y, e, t for every coordinate and A, B, C, D . The terminal slack contributes one. This gives (6.2). Validity contributes $(N + 1)(m + 1) + 2$ residuals, each transition contributes $4m + 5$, and the terminal equation contributes one, giving (6.3). Each residual is quadratic or linear, so their squares have degree at most four. Shared suffix sums of the selectors evaluate all comparison residuals with $O(m)$ arithmetic operations per transition. $\square$

Remark 6.3 (The exact single-fold scope). The horizon N and label count m index a family of finite polynomials. They are not unbounded runtime variables inside one fixed-arity single-fold universal halting polynomial. This construction therefore does not resolve a general finite-fold MRDP question. Its uniqueness is over natural witnesses. Replacing each natural variable by an arbitrary four-square representation would in general destroy that uniqueness.

6.6 The exported $N = 2, m = 2$ polynomial

For Example 5.4, take rows $(u_{0,1}, u_{0,2}) = (1, 1)$, $(u_{1,1}, u_{1,2}) = (2, 1)$, $(u_{2,1}, u_{2,2}) = (1, 1)$ and $(\alpha, \beta) = (1, 2)$. The denominators are $(2, 3, 2)$. The selected coordinate is two at the first transition and one at the second. Thus

$$(A_1, B_1) = (3, 2), \quad (A_2, B_2) = (4, 3), \quad (C_1, D_1) = (3, 4), \quad (C_2, D_2) = (12, 12).$$

The terminal slack is zero. This instance has 11 free input coordinates, 33 witnesses, 38 residuals, total degree four, and 165 expanded monomials. The bundle supplies the residual list, variable ordering, full natural assignment, and expanded polynomial. These values are computed by the included script rather than inferred from a numerical optimizer.

7 Variation bounds and a sharp binary constant

7.1 A general bound

For probability vectors define $\text{TV}(p, q) = \frac{1}{2} \sum_i |p_i - q_i|$.

Proposition 7.1 (Variation consumes mass). *For the positive histories of Theorem 5.2,*

$$\sum_{s=1}^N \text{TV}(p_{s-1}, p_s) \leq \sum_{s=1}^N (1 - R_s^{-1}) \leq \log G_N. \quad (7.1)$$

Every history feasible from initial mass δ therefore has total variation at most $\log(1/\delta)$, including an infinite history.

Proof. If $R_s = 1$, then $p_s \geq p_{s-1}$ coordinatewise and equal total sums imply equality. If $R_s > 1$, coordinatewise domination gives a probability vector q_s such that

$$p_s = R_s^{-1} p_{s-1} + (1 - R_s^{-1}) q_s.$$

Hence $\text{TV}(p_{s-1}, p_s) \leq 1 - R_s^{-1}$. For $R \geq 1$, $1 - R^{-1} \leq \log R$, as follows by integrating $1/t$ on $[1, R]$ or differentiating their difference. Sum over s and apply the budget theorem. For an infinite path pass to the supremum of the nonnegative partial sums. $\square$

7.2 The binary path identity

The binary case contains a stronger symmetric bound. Put $p_s = (x_s, 1 - x_s)$ with $0 < x_s < 1$, and define

$$\Phi(x) = -\frac{1}{2} \log(x(1-x)). \quad (7.2)$$

Theorem 7.2 (Sharp balanced binary bound). *Every binary path satisfies*

$$2 \sum_{s=1}^N |x_s - x_{s-1}| + \Phi(x_N) - \Phi(x_0) \leq \log G_N. \quad (7.3)$$

In particular, a feasible path starting at $x_0 = 1/2$ and initial mass δ satisfies

$$\sum_{s=1}^N |x_s - x_{s-1}| \leq \frac{1}{2} \log(1/\delta). \quad (7.4)$$

For every $0 < \delta < 1$, the constant $1/2$ in this statement is sharp as a supremum over finite balanced-start paths.

Proof. If a step increases x from x to y , its largest ratio is $(1-x)/(1-y)$, so its logarithmic cost is $\int_x^y (1-u)^{-1} du$. If the step decreases x to y , the cost is $\int_y^x u^{-1} du$. Since

$$\Phi'(u) = \frac{2u-1}{2u(1-u)},$$

both cases have the exact common expression

$$\log R_s = \int_{\min(x_{s-1}, x_s)}^{\max(x_{s-1}, x_s)} \frac{du}{2u(1-u)} + \Phi(x_s) - \Phi(x_{s-1}). \quad (7.5)$$

Now $u(1-u) \leq 1/4$, so the integral is at least $2|x_s - x_{s-1}|$. Sum and telescope the potential to obtain (7.3). The potential is minimized at $1/2$, and $\log G_N \leq \log(1/\delta)$ for a feasible path, proving (7.4).

For sharpness, repeat the cycle $1/2 \rightarrow 1/2 + \varepsilon \rightarrow 1/2$ exactly n times. Its one-cycle multiplicative cost is $(1 + 2\varepsilon)/(1 - 2\varepsilon)$ and its variation is 2ε . Choose

$$r_n = (1/\delta)^{1/n}, \quad \varepsilon_n = \frac{r_n - 1}{2(r_n + 1)}.$$

The full cost is exactly $1/\delta$, while the full variation is $n(r_n - 1)/(r_n + 1)$, which tends to $\frac{1}{2} \log(1/\delta)$. Thus no smaller constant works. If rational paths are required, approximate each ε_n from below by rationals; feasibility is preserved and the same supremum is obtained. $\square$

This bound is a quantitative restriction on successive conditional estimates. Conditioning can hide noncomputable limiting information, but it cannot make a fixed positive amount of accumulated mass support an arbitrarily oscillatory normalized path. Small excursions near balance are the asymptotically least expensive way to accumulate binary variation.

8 Which real probabilities can normalization produce?

8.1 Weak computability and finite variation

A real is *weakly computable*, also called a difference of left-c.e. reals, when $c = A - B$ for finite left-c.e. reals A, B . The equivalence with computable rational approximation of bounded total variation is established in the literature [4, 5]. The elementary argument needed here is short.

Lemma 8.1 (Jordan decomposition of an approximation). *A real c is weakly computable if and only if there is a computable rational sequence (x_s) tending to c for which $\sum_s |x_{s+1} - x_s| < \infty$.*

Proof. If $x_s = A_s - B_s$ with monotone convergent computable rational approximations, then $|x_{s+1} - x_s| \leq (A_{s+1} - A_s) + (B_{s+1} - B_s)$, and the sum is finite. Conversely, set

$$P_s = \sum_{j < s} \max(x_{j+1} - x_j, 0), \quad Q_s = \sum_{j < s} \max(x_j - x_{j+1}, 0).$$

Both are computable increasing rational sequences with finite limits, and $x_s = x_0 + P_s - Q_s$. Their limits express c as a difference of left-c.e. reals. $\square$

8.2 An exact success-floor classification

Two disjoint successful terminal outcomes, A and B , have unconditional masses a, b and heralding probability $h = a + b$. Their conditional probability of A , given eventual successful return, is

$$c = \frac{a}{a + b}, \quad h > 0. \quad (8.1)$$

The denominator is an eventual-event probability. It is not assumed known or computable.

Theorem 8.2 (High-success normalization realizes exactly weak computability). *For a real $c \in [0, 1]$, the following are equivalent:*

- (i) *c is a conditional terminal-output probability of an effective process with positive heralding probability;*
- (ii) *c is weakly computable;*
- (iii) *for every rational $0 < \delta < 1$, there is such a process with heralding probability at least δ and conditional probability exactly c .*

The realization in (iii) may be chosen to have a computable rational accumulation checkpoint of total mass exactly δ .

Proof. (i) implies (ii). The masses a, b are left-c.e. by Lemma 3.2. If one is zero, c is an endpoint and is computable. Otherwise choose monotone rational approximations a_s, b_s and discard a finite initial segment until both are positive. Set $h_s = a_s + b_s$ and $x_s = a_s/h_s$. These give an effective positive binary history with initial mass $h_0 > 0$ and total budget at most one. Proposition 7.1 bounds its total variation by $\log(1/h_0)$. It converges to $a/(a + b)$, so Lemma 8.1 proves weak computability.

(ii) implies (iii). The endpoint cases are realized by a constant output. Suppose $0 < c < 1$, and take a computable rational bounded-variation approximation to c . Choose rational $\eta > 0$ with $c \in (\eta, 1 - \eta)$. Clamp the approximation to $[\eta, 1 - \eta]$; clamping is a contraction, preserves the limit, and does not increase total variation. Discard a sufficiently long finite initial segment so that the remaining variation is at most $\eta \log(1/\delta)$.

For a binary step confined to that interval, the integral formulas in the proof of Theorem 7.2 give $\log R_s \leq |x_s - x_{s-1}|/\eta$. Hence the remaining path has $\sum_s \log R_s \leq \log(1/\delta)$. Its greedy realization from initial mass δ is therefore feasible by Theorem 5.2. Its monotone rational coordinate masses converge to a, b with $a + b \geq \delta$. Because their normalized ratios are the chosen x_s , continuity at the positive denominator gives $a/(a + b) = c$. Apply Corollary 5.3 to realize the limiting law.

(iii) implies (i) is immediate. $\square$

Remark 8.3 (The nonuniform choice must not be hidden). The proof of (ii) $\Rightarrow$ (iii) chooses an interior interval and a late small-variation tail. It is an existence theorem for each real and each success floor; it is not an algorithm for discovering those choices from an arbitrary bounded-variation approximation index. The rational greedy compiler is uniform once a suitable tail and a valid budget bound are supplied. The finite-horizon quartic compiler is fully explicit and has no such unbounded tail choice.

Weakly computable reals are not all limit-computable reals; the distinction is established by computability-theoretic results such as those discussed by Ng, Stephan, and Wu [5]. Thus the theorem does not say that conditioning can realize an arbitrary Δ_2^0 real. It identifies an exact intermediate class using a concrete mass constraint.

Theorem 8.4 (The full-success boundary). *The conditional probabilities of effective two-outcome processes that return one of the two outcomes with probability one are exactly the computable reals in $[0, 1]$.*

Proof. Here $a + b = 1$. Increasing rational lower approximations give intervals $[a_s, 1 - b_s]$ containing a , whose widths tend to zero. To compute a to any required precision, search for an interval of sufficiently small width. The search terminates under the probability-one promise. Conversely, for a computable c , both c and $1 - c$ are uniformly left-c.e. Their two-label law has total mass one and is realized by Theorem 4.2. $\square$

There is therefore a genuine endpoint distinction: every fixed permitted failure probability greater than zero allows all weakly computable conditional probabilities; requiring failure probability exactly zero restricts the class to computable probabilities. This is an expressivity statement, not a physical procedure for obtaining a noncomputable real to certified arbitrary precision from experimental samples.

9 The arithmetic hierarchy of probability questions

9.1 Indexing and promises

A Σ_1^0 relation has an existential quantifier over a computable predicate; Σ_2^0 has an existential block followed by a universal block. The dual classes are Π_1^0 and Π_2^0 . Completeness below means computable many-one completeness. Inputs are program indices for a fixed effective universal model, with rational threshold $q \in (0, 1)$ fixed in advance.

When a success floor is required, use an effective family of wrappers that reserves a positive rational mass for guaranteed return before handing the remaining mass to an arbitrary source program. Restrict the syntax to wrappers whose reserved mass is at least the specified δ . All indices in that family satisfy the floor; no undecidable semantic validation is included in the language being classified. Equivalently, one may state the upper bounds as promise-problem bounds and ensure

that every hardness reduction lands inside the promise. We will do both explicitly rather than silently conjoin an arbitrary-code problem with an additional floor predicate.

These distinctions matter because almost-sure termination itself has high arithmetic complexity, as is known from prior work [6]. Our proofs below establish the particular threshold claims directly.

9.2 Unconditional probabilities

Theorem 9.1 (Unconditional comparison table). *For a fixed interior rational q and a universal effective output-law model, the following classifications hold:*

$p > q$	$p \leq q$	$p < q$	$p \geq q$	$p = q$	$p \neq q$
$\Sigma_1^0\text{-c.}$	$\Pi_1^0\text{-c.}$	$\Sigma_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Sigma_2^0\text{-c.}$

Here “-c.” means complete for the displayed class.

Proof. Let $p_s \uparrow p$ be computable rational lower bounds. Then $p > q \iff \exists s (p_s > q)$ and $p \leq q$ is its complement. Moreover,

$$p \geq q \iff \forall k \exists s (p_s > q - 2^{-k}),$$

so $p \geq q$ is Π_2^0 and $p < q$ is Σ_2^0 . Equality is the conjunction of $p \geq q$ and $p \leq q$, hence is Π_2^0 ; inequality is its complement.

For the first hardness claim, use a rational baseline q and add a small rational mass $\varepsilon > 0$ precisely when a given ordinary computation halts. The resulting mass exceeds q exactly in the halting case. Taking complements proves the Π_1^0 claim.

For the remaining claims, let φ_e be the e th ordinary partial computable function. A fair-coin program chooses $n \geq 0$ with probability $2^{-(n+1)}$, simulates $\varphi_e(n)$, and returns precisely if that simulation halts. Its return probability is

$$t_e = \sum_{n \geq 0} 2^{-(n+1)} \mathbf{1}_{\{\varphi_e(n) \downarrow\}}. \quad (9.1)$$

Thus $t_e = 1$ exactly when e belongs to the Π_2^0 -complete totality set. Set $p_e = qt_e$. Then $p_e < q$ and $p_e \neq q$ exactly for nontotal indices, whereas $p_e \geq q$ and $p_e = q$ exactly for total indices. These are the required reductions. $\square$

For completeness, totality has the form $\forall n \exists t$ “ $\varphi_e(n)$ halts by t ”, and every Π_2^0 predicate $\forall n \exists t R(e, n, t)$ with computable R reduces to it by the program that, on n , searches for such a t . No oracle is used by the probabilistic program in (9.1).

9.3 Conditioning defeats the existential threshold boundary

Theorem 9.2 (Conditional comparison at any subunit success floor). *Fix rationals $q \in (0, 1)$ and $\delta \in (0, 1)$. In a universal effective wrapper family guaranteeing $a + b \geq \delta$, set $c = a/(a + b)$. Then*

$c > q$	$c < q$	$c \neq q$	$c \geq q$	$c \leq q$	$c = q$
$\Sigma_2^0\text{-c.}$	$\Sigma_2^0\text{-c.}$	$\Sigma_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$

In particular, the strict conditional comparisons are not existential Diophantine relations in such a universal family.

Proof. Choose computable rational lower approximations a_s, b_s . Their normalized ratios $c_s = a_s/(a_s + b_s)$ are computable once the denominator is positive; use any fixed rational value before that point. They converge to c . Consequently

$$c > q \iff \exists k \exists n \forall s \geq n \quad c_s \geq q + 2^{-k}. \quad (9.2)$$

The analogous expression handles $c < q$. Thus these strict cuts are Σ_2^0 , and their complements are Π_2^0 . Equality has the more useful form

$$c = q \iff \forall k \forall n \exists s \geq n \quad |c_s - q| < 2^{-k}. \quad (9.3)$$

Convergence makes repeated arbitrary closeness equivalent to equality, so this is a Π_2^0 expression. Its complement is Σ_2^0 .

For hardness choose a positive rational $\varepsilon < \min(1 - \delta, q, 1 - q)$ and use t_e from (9.1). The first wrapper has masses

$$a_e = q, \quad b_e = 1 - q - \varepsilon + \varepsilon t_e. \quad (9.4)$$

It reserves mass $1 - \varepsilon$ for guaranteed return and assigns the remaining mass to the totality sampler. Its heralding probability is $h_e = 1 - \varepsilon + \varepsilon t_e \geq \delta$, and $c_e = q/h_e \geq q$. It follows that $c_e > q$ and $c_e \neq q$ hold exactly when e is nontotal, while $c_e \leq q$ and $c_e = q$ hold exactly when e is total.

For the other direction use

$$a_e = q - \varepsilon + \varepsilon t_e, \quad b_e = 1 - q. \quad (9.5)$$

The same denominator gives $c_e - q = -(1 - q)\varepsilon(1 - t_e)/h_e$. Thus $c_e < q$ exactly for nontotal indices and $c_e \geq q$ exactly for total indices. All masses in both wrappers are nonnegative and their sums are at most one. These reductions are effective, preserve the syntactic floor guarantee, and prove every hardness assertion. MRDP and the strict arithmetic hierarchy then exclude ordinary existential Diophantine representations for the complete Σ_2^0 cuts. $\square$

For example, the theorem applies with a success guarantee of 0.999999. The difficulty is not confined to rare postselection events: the unknown remaining denominator mass can encode a universal nontermination question. A finite lower bound for a does not certify that future additions to b will not reverse a conditional threshold comparison.

Remark 9.3 (A quantifier trap). The naive formula saying that c_s is eventually within every tolerance of q has the pattern $\forall k \exists n \forall s$ and only gives a Π_3^0 upper bound. Formula (9.3) is sharper because convergence is already known. Omitting that argument would obscure the exact boundary.

9.4 The probability-one promise

Proposition 9.4 (Comparisons under almost-sure return). *Under the promise $a + b = 1$, strict comparisons $c > q, c < q$ and inequality $c \neq q$ are Σ_1^0 -complete promise problems. Weak comparisons and equality are Π_1^0 -complete promise problems. The reductions can be chosen from a uniformly total family of samplers.*

Proof. Under the promise $c = a$ and $1 - c = b$, so $c > q$ and $c < q$ are witnessed by lower approximations to a and b , respectively. Their disjunction is inequality; taking complements within the promise gives the upper bounds.

Let $x_e = 2^{-t}$ if a designated computation halts at time t , and let $x_e = 0$ if it never halts. This is a uniformly computable real: after s steps, a still-undetected later halting time can contribute at most 2^{-s} . For a sufficiently small positive rational ε , the probabilities $q + \varepsilon x_e$ and $q - \varepsilon x_e$ lie in $[0, 1]$ and are uniformly computable. Their complementary two-label laws have uniform total samplers by Theorem 4.2. Strict comparison in the appropriate direction, or inequality with q , detects halting; equality and the opposite weak comparison detect nonhalting. $\square$

This is a promise classification. Deciding which arbitrary probabilistic programs satisfy the promise has not become easy. It is incorrect to erase that promise and infer a Π_1^0 algorithm for arbitrary exact-output questions.

9.5 Polynomial quantifier normal forms

The hierarchy can itself be expressed using quartic integer equations. For a computable relation $R(e, u, v)$, MRDP applies both to R and to its complement. Consequently the four normal forms are

Class	Quartic arithmetic form, with natural quantified tuples
Σ_1^0	$\exists \mathbf{w} P(e, \mathbf{w}) = 0$
Π_1^0	$\forall \mathbf{w} P(e, \mathbf{w}) \neq 0$
Π_2^0	$\forall u \exists v, \mathbf{w} P(e, u, v, \mathbf{w}) = 0$
Σ_2^0	$\exists u \forall v, \mathbf{w} Q(e, u, v, \mathbf{w}) \neq 0$

For the last line choose Q to represent $\neg R$, so that universal nonvanishing expresses $\forall v R(e, u, v)$. Apply Lemma 3.3 to each matrix. These are arithmetic formulas *over* Diophantine relations, not all existential Diophantine representations. The strict hierarchy is exactly why the distinction cannot generally be removed.

All tables assumed $0 < q < 1$. At endpoints some classifications collapse: $p > 0$ is Σ_1^0 , $p = 0$ is Π_1^0 , $p = 1$ can be Π_2^0 -complete, and $p > 1$ is impossible. Endpoint statements must not be read off the interior tables without checking them.

Part II

Quantum substrates (manuscripts 08 and 07)

10 [write] Two quantum models

Part II joins two independent treatments of quantum computation. Manuscript 08 (Sections 11–12) works with exact amplitudes of Clifford+ T circuits over $\mathbb{Z}[1/\sqrt{2}, i]$, with the terminal classical output laws of measured programs, and with an operator-valued analogue of Part I's normalization budget. Manuscript 07 (Sections 13–22) works with *rational real* instruments $\sum_j A_j^\top A_j = I$ with one Kraus operator per outcome, and with the exact zero-probability (mortality) problem for their

outcome words. The models overlap only in their common semantics of unnormalized branch states ($A_w \rho A_w^\dagger$ in 07, the unnormalized measured state in 08); there is no shared theorem. 08's corollary on quantum output laws in Section 11 makes every effective quantum program's terminal output law a fixed-quartic projected-root law, while 07's Theorem 18.2 shows that the *zero set* of the output probabilities of rational instruments is undecidable already in dimension four; the two statements are compatible, since a strict threshold $p > q$ is Σ_1^0 while $p = 0$ is Π_1^0 (Theorem 9.1).

11 An explicit quantum computational front end

11.1 Exact amplitudes rather than floating-point paths

The Clifford+ T gate family admits exact arithmetic in $\mathbb{Z}[1/\sqrt{2}, i]$; the exact synthesis literature gives a much stronger structural account of this ring and its realizable operators [7]. Here we only need a direct forward evaluator for the gates H, T, S, X , and CNOT.

After L elementary gates, represent each amplitude as

$$\psi_j = \frac{a_j + b_j\sqrt{2} + i(c_j + d_j\sqrt{2})}{2^L}, \quad a_j, b_j, c_j, d_j \in \mathbb{Z}. \quad (11.1)$$

The common denominator is intentionally unreduced: each gate increments L by one, even when it could be implemented without a new denominator. For $z = a + b\sqrt{2} + i(c + d\sqrt{2})$ the relevant integer-linear maps are

$$\begin{aligned} \sqrt{2}z &\longleftrightarrow (2b, a, 2d, c), \\ iz &\longleftrightarrow (-c, -d, a, b), \\ \sqrt{2}(1+i)z &\longleftrightarrow (2(b-d), a-c, 2(b+d), a+c). \end{aligned} \quad (11.2)$$

For an H gate on an amplitude pair (z_0, z_1) , the new numerators are $\sqrt{2}(z_0 + z_1)$ and $\sqrt{2}(z_0 - z_1)$. For T , multiply the bit-zero numerator by two and the bit-one numerator by $\sqrt{2}(1+i)$. For S , use two and $2i$ respectively. The X and CNOT gates permute amplitudes and multiply all numerators by two. Dividing by the new 2^{L+1} gives precisely the usual gate actions.

The squared norm of one numerator is

$$|z|^2 = A + B\sqrt{2}, \quad A = a^2 + 2b^2 + c^2 + 2d^2, \quad B = 2(ab + cd). \quad (11.3)$$

Thus for a selected set J of measurement outcomes,

$$\Pr(J) = \frac{\sum_{j \in J} A_j + (\sum_{j \in J} B_j)\sqrt{2}}{4^L}. \quad (11.4)$$

The symbolic coefficient B can be negative, although the final probability is nonnegative.

Proposition 11.1 (Decidable finite-circuit probabilities). *For a finite circuit over these gates with a computational-basis input, comparison of a terminal basis-event probability with a rational threshold is decidable by exact integer arithmetic. Its graph and its comparison relations are Diophantine, with quartic representations.*

Proof. The gate recurrences compute the four integer coordinates exactly. After clearing positive denominators, a rational comparison reduces to the sign of $A + B\sqrt{2}$ with $A, B \in \mathbb{Z}$. If either

coefficient is zero or they have the same sign, the sign is immediate. If they have opposite signs, compare A^2 and $2B^2$, reversing the resulting sign when $A < 0$. Equality $A^2 = 2B^2$ with both coefficients nonzero is impossible: cancelling common factors would give a reduced rational square root of two. This supplies a terminating integer algorithm and therefore a computable relation. Apply MRDP and Lemma 3.3.

For a direct bounded evaluator, use linear gate residuals and quadratic norm residuals. Signed intermediate coordinates can be encoded by their natural positive and negative parts; adding $z^+z^- = 0$ makes that split canonical. The finite comparison algorithm can be lowered to the same arithmetic-gate format. This last observation does not assert the particular witness count of Theorem 6.1 for the quantum compiler. $\square$

Example 11.2 (Cancellation and an irrational output probability). Starting in $|0\rangle$, the circuit H^2 returns outcome zero with probability one. If its intermediate basis alternatives were treated as independent classical alternatives without recombination, the cancellation in the outcome-one amplitude would be lost.

The circuit HTH , acting right to left on the input as usual, gives

$$\Pr(0) = \frac{2 + \sqrt{2}}{4}, \quad \Pr(1) = \frac{2 - \sqrt{2}}{4}.$$

The accompanying sequential evaluator applies H , then T , then H , which is the same palindromic circuit. Both probabilities are represented exactly, not by decimal approximations. A finite fair-coin decision tree with a uniform depth bound has only dyadic output probabilities, so no such tree can realize this law exactly. The unbounded sampler in Theorem 4.2 can realize it and, because the two masses sum to one, returns almost surely.

11.2 Unbounded measured programs

Consider an unbounded classical controller with finitely described quantum gates and projective measurements, allowing only finitely many qubits at each finite execution stage. A finite measurement history has a weight computed by evolving an *unnormalized* state through its gates and measurement projections. Its squared norm is the history probability. Keeping the state unnormalized avoids division by a possibly zero branch probability.

Distinct recorded measurement histories are mutually exclusive classical events. Summing the weights of first-return histories therefore gives a presentation in Definition 3.1. Lemma 3.2 and Theorem 4.3 apply. The same argument works for uniformly computable gate entries when valid evolution and an effective finite history presentation are part of the model assumptions; exact algebraic zero tests are unnecessary for obtaining lower bounds.

Corollary 11.3 (Quantum output-law Diophantine representation). *Every such effective quantum program's discrete classical terminal output law has the fixed-quartic projected-root representation (4.2). Its ordinary strict output-probability thresholds are existential Diophantine. Its conditional eventual-return thresholds are subject to the distinctions in Section 9.*

Proof. The measurement-history presentation gives a left-c.e. law; apply the corresponding representation and threshold theorems. Classical fair-coin control is a special case, so the general hardness obstructions cannot be removed by admitting quantum operations. $\square$

This is an exact statement about a terminal *classical probability law*. It is not an efficient classical simulation of quantum computation, not a replacement for quantum states in interactive protocols, and not a claim that all quantum behavior is captured by classical random variables. The reference state-vector implementation uses 2^n amplitudes for n qubits and is deliberately a small-instance exact checker. It does not implement the unbounded universal MRDP compiler.

12 An operator-valued normalization budget

12.1 Accumulated ensembles

A quantum analogue of coordinatewise accumulated output mass is a sequence of positive semidefinite operators ρ_s , with $\rho_s - \rho_{s-1} \succeq 0$ and $\text{Tr } \rho_s \leq 1$. For example, these may represent sums of subnormalized density operators over disjoint returned histories. Write

$$\rho_s = h_s \sigma_s, \quad \text{Tr } \sigma_s = 1.$$

This is an accumulation process, not the state trajectory of a single system undergoing arbitrary unitary evolution.

For positive definite density matrices define

$$R(\sigma, \tau) = \inf\{r > 0 : \sigma \preceq r\tau\} = \lambda_{\max}(\tau^{-1/2} \sigma \tau^{-1/2}). \quad (12.1)$$

The logarithm of this quantity is the established max-relative entropy $D_{\max}(\sigma \| \tau)$, with logarithm-base conventions adjusted to our natural logarithm [8]. The equality in (12.1) follows by conjugating the operator inequality by $\tau^{-1/2}$.

Theorem 12.1 (Exact operator accumulation cost). *For a finite or infinite sequence of positive definite density matrices $\sigma_0, \sigma_1, \dots$, put*

$$R_s = R(\sigma_{s-1}, \sigma_s), \quad G_s = \prod_{j=1}^s R_j.$$

There are accumulated ensembles $\rho_s = h_s \sigma_s$ with $h_0 = \delta > 0$, $h_s \leq 1$, and $\rho_s - \rho_{s-1} \succeq 0$ if and only if $\delta G_s \leq 1$ for every s . Their unique pointwise minimal total masses are $h_s^ = \delta G_s$. Equivalently, the logarithmic cost is*

$$\log G_N = \sum_{s=1}^N D_{\max}(\sigma_{s-1} \| \sigma_s). \quad (12.2)$$

Proof. A single monotonicity condition is $h_s \sigma_s \succeq h_{s-1} \sigma_{s-1}$, equivalent to $h_s/h_{s-1} \geq R_s$. Taking traces shows $R_s \geq 1$. The proof of Theorem 5.2 now applies verbatim to the scalar factors: iteration gives necessity, and choosing equality at every step gives positive semidefinite increments and sufficiency. $\square$

For singular density matrices, the same result holds with $R_s = +\infty$ when $\text{supp}(\sigma_{s-1})$ is not contained in $\text{supp}(\sigma_s)$, and with the inverse taken on the latter support otherwise. To see the obstruction, a vector in the new kernel must also lie in the old kernel whenever the old operator is bounded above by a multiple of the new one. Under support inclusion, the finite-dimensional restriction gives the same formula.

Corollary 12.2 (Data processing and trace variation). *For a completely positive trace-preserving map Λ ,*

$$R(\Lambda\sigma, \Lambda\tau) \leq R(\sigma, \tau).$$

Thus applying the same channel at each checkpoint cannot increase the optimal accumulation cost. Moreover, any feasible operator path satisfies

$$\sum_{s=1}^N \frac{1}{2} \|\sigma_s - \sigma_{s-1}\|_1 \leq \log(1/\delta).$$

Proof. A positive map preserves $\sigma \preceq r\tau$, so every feasible r before the map is feasible afterward. For the variation claim, if $R_s > 1$ then

$$\sigma_s = R_s^{-1} \sigma_{s-1} + (1 - R_s^{-1}) \omega_s$$

for a positive trace-one operator ω_s . The triangle inequality and $\|\omega_s\|_1 = \|\sigma_{s-1}\|_1 = 1$ give trace distance at most $1 - R_s^{-1}$. Handle $R_s = 1$ by equality, sum, and use the scalar logarithmic inequality from Proposition 7.1. $\square$

The data-processing property of max-relative entropy is known [8]. Here its immediate consequence is monotonicity of an entire accumulation budget, rather than a new entropy monotonicity theorem.

12.2 Two exact matrix examples

Example 12.3 (A coherent excursion erased by dephasing). Let

$$\sigma_0 = \sigma_2 = I/2, \quad \sigma_1 = \begin{pmatrix} 1/2 & t \\ t & 1/2 \end{pmatrix}, \quad 0 < t < 1/2.$$

The eigenvalues of σ_1 are $1/2 \pm t$. Thus

$$R_1 = \frac{1}{1-2t}, \quad R_2 = 1+2t, \quad \delta_{\max} = \frac{1-2t}{1+2t}.$$

At $t = 1/6$ the cost is two and the maximum initial mass is $1/2$. Dephasing in the displayed basis sends every matrix to $I/2$ and reduces the cost to one. These matrices commute because the endpoints are scalar; the example demonstrates a coherence-sensitive cost, not noncommutation.

Example 12.4 (A genuinely noncommuting cycle). Set

$$\rho = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad \sigma = \begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/2 \end{pmatrix}.$$

Their commutator is nonzero. Both have determinant $2/9$, and

$$\det(r\sigma - \rho) = \det(r\rho - \sigma) = \frac{4r^2 - 9r + 4}{18}.$$

The least admissible domination factor in either direction is $R = (9 + \sqrt{17})/8$. Indeed, the determinant vanishes there and the Hermitian difference has positive trace, hence is positive semidefinite; the smaller root is below one and cannot dominate a trace-one operator. For the cycle $\rho \rightarrow \sigma \rightarrow \rho$,

$$G = R^2 = \frac{49 + 9\sqrt{17}}{32}, \quad \delta_{\max} = \frac{49 - 9\sqrt{17}}{32}.$$

Dephasing turns the middle state into $I/2$ and gives the strictly smaller cycle cost $(4/3)(3/2) = 2$. The symbolic checker verifies both determinant identities and the reciprocal cost exactly.

The operator theorem is a finite-dimensional order optimization theorem. It does not by itself compile every prescribed computable matrix sequence into a circuit over the discrete gate set of Section 11. Exact state preparation and a small Diophantine certificate for generalized eigenvalue maxima are separate problems.

13 Research target, provenance, and the precise contribution

The suggested research direction is particularly promising when “computation” is not identified with a discrete execution log. A quantum outcome word carries an ordered product of matrices. Its possibility depends on exact cancellation among amplitudes. Encoding only the existence of combinatorial paths can therefore give the wrong answer. This manuscript studies that cancellation directly and connects it to the natural-number polynomial interfaces already present in ProveIt.

The quantum measurement occurrence problem is already known to be undecidable. Eisert, Müller, and Gogolin established this for nine rational Kraus operators in dimension fifteen [10]. The research question pursued here is more specific: can the normalization of an arbitrary matrix-mortality instance be made much smaller while retaining *all* outcome words and a transparent relation between witnesses? The answer proved below is affirmative. This is a quantitative refinement of a known undecidability mechanism, not a new discovery of that mechanism.

13.1 Dependency and originality ledger

The logical dependencies should be kept separate from the proposed contribution.

Component	Status in this manuscript
Matrix mortality hardness	Imported from Neary’s published six-generator, 3×3 result; its proof is not reproduced [11].
Rational quadratic forms	Meyer’s theorem and Lagrange’s four-square theorem are classical inputs; the consequences needed here are proved explicitly [12].
Packed Kraus normalization	Full proof given here, including the erasure and one-step witness bounds. This is the principal contribution proposed for review.
Fixed-length quartics	Explicit formulas, correctness proof, exact counts, executable compiler, and sample expanded polynomial. The underlying arithmetic-circuit technique is standard.
Unbounded Diophantine form	Consequence of MRDP, not a new MRDP theorem and not a single-fold result [14, 13].
Validation	Exact finite computations; no machine-checked proof of the general theorems and no independent priority certification.

The literature check used the published STACS 2015 version of Neary’s result. In particular, the numerical input used throughout is *six* 3×3 matrices, as in its Corollary 12 [11]; no stronger bound from a differently worded early preprint is needed. The comparison with dimension fifteen is with the explicit theorem in [10]. A targeted literature search is not sufficient to declare the bound below the best previously unknown bound in every formulation of quantum mortality.

14 Words, mortality, and exact quantum outcomes

Throughout, $\mathbb{N}$ includes zero. Matrices have finite dimension, and an input rational number is specified exactly by integers. All products are ordered chronologically: for a word $w = i_1 \cdots i_n$,

$$M_w = M_{i_n} \cdots M_{i_1}, \quad M_\varepsilon = I_d. \quad (14.1)$$

This convention will also be used in the code. It eliminates the common ambiguity about whether appending a letter multiplies on the left or on the right.

Definition 14.1. A finite list $\mathcal{M} = (M_1, \dots, M_k)$ of $d \times d$ matrices is *mortal* if there is a nonempty word w with $M_w = 0$. Let $\mu(\mathcal{M})$ be the smallest such length, and put $\mu(\mathcal{M}) = \infty$ when there is no such word.

Repeated generators are allowed: labels are outcomes, not necessarily distinct matrices. Repeating a generator does not change whether a list is mortal. The empty word is never a witness because $d \geq 1$.

A rational, real, single-Kraus-per-outcome instrument on $\mathbb{C}^D$ is a list

$$\mathcal{A} = (A_0, \dots, A_k), \quad A_j \in \mathbb{Q}^{D \times D}, \quad \sum_{j=0}^k A_j^\top A_j = I_D. \quad (14.2)$$

The use of real entries is a restriction within ordinary complex quantum mechanics, not a different theory. For a density matrix ρ , the unnormalized state after the outcome word w is $A_w \rho A_w^\dagger$. Thus its probability is

$$p_\rho(w) = \text{tr}(A_w \rho A_w^\dagger). \quad (14.3)$$

Using unnormalized states avoids division by a zero probability when describing an impossible word. These are the instrument semantics used in the known occurrence problem [10].

Lemma 14.2 (Full-rank zero test). *If ρ is positive definite, then $p_\rho(w) = 0$ if and only if $A_w = 0$. In particular, for $\rho = I_D/D$,*

$$p(w) = \frac{\|A_w\|_F^2}{D}. \quad (14.4)$$

Proof. The expression in (14.3) equals $\|A_w \rho^{1/2}\|_F^2$. A sum of squared absolute values vanishes exactly when every entry vanishes. Since $\rho^{1/2}$ is invertible, this is equivalent to $A_w = 0$. The stated formula follows by substituting $\rho = I_D/D$. $\square$

The full-rank condition matters. For example, a nonzero projection can annihilate a particular pure state. Without full rank, an impossible word need not represent a zero matrix product. All hardness statements below use a full-rank state, and the explicit probability calculations use the maximally mixed state.

15 Rational Gram factors and their row budget

Let Q be a symmetric positive definite rational $d \times d$ matrix. We need a rational rectangular factor

$$Q = B^\top B, \quad B \in \mathbb{Q}^{s \times d}. \quad (15.1)$$

Over the reals one can take $s = d$, but the entries of a square factor need not be rational. Replacing an irrational square root by a decimal approximation would destroy exact normalization. The number of rational rows is therefore an actual resource in the construction.

15.1 The elementary four-square construction

Proposition 15.1 (An explicit $4d$ -row factor). *Every positive definite rational $d \times d$ matrix has a rational Gram factor with $4d$ rows. Such a factor can be found using rational elimination and finite integer searches.*

Proof. Rational elimination gives

$$Q = L \operatorname{diag}(q_1, \dots, q_d) L^\top,$$

where L is unit lower triangular over $\mathbb{Q}$ and every q_j is positive rational. This is the usual $LDL^\top$ factorization; positivity ensures that every pivot is positive and nonzero. Write $q_j = a_j/b_j$ with positive integers a_j, b_j . Lagrange’s four-square theorem gives integers $t_{j1}, \dots, t_{j4}$ such that

$$a_j b_j = \sum_{\ell=1}^4 t_{j\ell}^2, \quad q_j = \sum_{\ell=1}^4 (t_{j\ell}/b_j)^2.$$

For each (j, ℓ) , put a row $(t_{j\ell}/b_j) e_j^\top L^\top$ into B . Summing its outer products gives (15.1). The four-square search is finite: it suffices to enumerate integer coordinates of absolute value at most $\sqrt{a_j b_j}$. No efficient bit-complexity bound is being asserted. $\square$

Zero rows can be omitted in a concrete example. The uniform $4d$ bound deliberately does not rely on such simplification. This construction is implemented directly in the package, using exact fractions rather than floating-point Cholesky factorization.

15.2 A classical improvement to $d + 3$ rows

The stronger row count follows from the following standard rational quadratic-form theorem [12].

Theorem 15.2 (Meyer; classical input). *A nondegenerate indefinite quadratic form over $\mathbb{Q}$ in at least five variables has a nonzero rational isotropic vector.*

For clarity, indefinite means that the form takes both positive and negative values over $\mathbb{R}$. Theorem 15.2 is an imported theorem, not proved from elementary number theory here. It is a consequence of the Hasse–Minkowski theorem and the local isotropy theorem for rational quadratic forms of rank at least five. The weaker bound in Proposition 15.1 does not need this input.

Lemma 15.3 (Representation by a positive four-dimensional space). *Let V be a positive definite rational quadratic space of dimension at least four. Every positive rational number a is represented by its quadratic form.*

Proof. On $V \oplus \mathbb{Q}$, consider the form $q(v) - at^2$. It is nondegenerate, indefinite, and has dimension at least five. Meyer gives a nonzero rational solution to $q(v) = at^2$. If $t = 0$, positive definiteness forces $v = 0$, a contradiction. Therefore $q(v/t) = a$. $\square$

Proposition 15.4 (A $(d + 3)$ -row factor). *Every positive definite rational $d \times d$ matrix admits a rational Gram factor with $d + 3$ rows. There is a total search algorithm for such a factor.*

Proof. As above, write $Q = L \operatorname{diag}(q_1, \dots, q_d) L^\top$. In the standard positive quadratic space $\mathbb{Q}^{d+3}$, choose successively rational vectors $v_1, \dots, v_d$ satisfying

$$v_i^\top v_j = 0 \ (i \neq j), \quad v_j^\top v_j = q_j.$$

Suppose $v_1, \dots, v_{j-1}$ have been chosen. They are linearly independent because their norms are positive and they are orthogonal. Their rational orthogonal complement has dimension

$$d + 3 - (j - 1) \geq 4.$$

The restriction of the standard form is positive definite. Lemma 15.3 therefore supplies v_j . If V is the matrix whose columns are these vectors, then $V^\top V = \operatorname{diag}(q_1, \dots, q_d)$, so $B = VL^\top$ works.

For effectivity one does not need an efficient implementation of the local theory. Enumerate rational $(d + 3) \times d$ matrices by increasing common denominator and numerator height, and test $B^\top B = Q$ exactly. Existence guarantees termination. This is an algorithmic existence statement, not a claim that the search is practical on large inputs. $\square$

15.3 The general rational row bound is sharp

The next fact is included to identify a genuine arithmetic limitation of the method. It is not offered as a new theorem about quadratic forms.

Proposition 15.5 (Sharpness for arbitrary positive rational forms). *For every $d \geq 1$, the matrix $Q_d = \operatorname{diag}(1, \dots, 1, 7)$ has no rational Gram factor with $d + 2$ rows. Consequently $d + 3$ is the smallest uniform row bound valid for all positive definite rational $d \times d$ matrices.*

Proof. First, 7 is not a sum of three rational squares. Otherwise clearing denominators gives

$$x^2 + y^2 + z^2 = 7t^2,$$

with integers $t > 0$ and with no common factor among x, y, z, t . If t is even, reduction modulo 4 forces x, y, z all even, contradicting primitivity. If t is odd, the right side is 7 modulo 8, whereas a sum of three residues from $\{0, 1, 4\}$ cannot be 7 modulo 8.

Now suppose $B^\top B = Q_d$ with $B \in \mathbb{Q}^{(d+2) \times d}$. The first $d - 1$ columns are rational orthonormal vectors. Any rational unit vector u can be carried to e_1 by a rational orthogonal matrix: if $u \neq e_1$, use the Householder reflection

$$H = I - \frac{2(u - e_1)(u - e_1)^\top}{(u - e_1)^\top (u - e_1)}.$$

It is rational and orthogonal and satisfies $Hu = e_1$. If $u = e_1$, use the identity. Applying this procedure successively in orthogonal complements sends the first $d - 1$ columns to the first $d - 1$ coordinate vectors. The last column is then supported on the remaining three coordinates and has squared norm 7, a contradiction. The case $d = 1$ is exactly the first argument. A factor with fewer rows could be padded to $d + 2$ rows, so it is also impossible. $\square$

This does *not* prove that our final quantum dimension is optimal. The residual forms used below are special forms $c^2 I - \sum_i M_i^\top M_i$, and c is under our control. A better choice of c , a different embedding, or a different reduction could outperform the general-purpose bound.

16 The packed normalization theorem

We now state the main algebraic result. The row packing, rather than the existence of some normalization, is the important operation.

Theorem 16.1 (Mortality-preserving rational Gram packing). *Let $\mathcal{M} = (M_1, \dots, M_k)$ be rational $d \times d$ matrices, with $d, k \geq 1$. Choose a positive rational c such that*

$$Q = c^2 I_d - \sum_{i=1}^k M_i^\top M_i \quad \text{is positive definite.} \quad (16.1)$$

Suppose $Q = B^\top B$ with s rational rows. Put

$$r = \lceil s/k \rceil, \quad D = d + r.$$

Pad B with zero rows to obtain kr rows, and split it into k consecutive $r \times d$ blocks $B_1, \dots, B_k$. Define

$$A_i = \alpha \begin{pmatrix} M_i & 0_{d \times r} \\ B_i & 0_{r \times r} \end{pmatrix} \quad (1 \leq i \leq k), \quad A_0 = \begin{pmatrix} \beta I_d & 0 \\ 0 & I_r \end{pmatrix}, \quad \alpha = \frac{4}{5c}, \quad \beta = \frac{3}{5}. \quad (16.2)$$

Then $\mathcal{A} = (A_0, \dots, A_k)$ is an exactly normalized rational instrument, and

$$\text{Mort}(\mathcal{A}) \iff \text{Mort}(\mathcal{M}). \quad (16.3)$$

If mortal, the shortest witness lengths satisfy

$$\mu(\mathcal{M}) \leq \mu(\mathcal{A}) \leq \mu(\mathcal{M}) + 1. \quad (16.4)$$

Proof. We separate the normalization, soundness, and completeness arguments.

Normalization. Since the right block columns of A_i vanish,

$$\sum_{i=1}^k A_i^\top A_i = \alpha^2 \begin{pmatrix} \sum_i M_i^\top M_i + \sum_i B_i^\top B_i & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} (16/25)I_d & 0 \\ 0 & 0 \end{pmatrix}.$$

On the other hand, $A_0^\top A_0 = \text{diag}((9/25)I_d, I_r)$. Their sum is I_D . All entries are rational. The constants $3/5$ and $4/5$ are simply a rational point on the unit circle with both coordinates nonzero.

No false mortal word. Every generator is block lower triangular. Taking its upper-left $d \times d$ block is multiplicative within this block-triangular algebra. If w has m active letters from $\{1, \dots, k\}$ and z idle letters 0, and u is obtained by deleting all zeros, then

$$(A_w)_{\text{top}} = \alpha^m \beta^z M_u. \quad (16.5)$$

If $m = 0$, then $A_w = A_0^z$ is invertible and cannot vanish. If $A_w = 0$ and $m > 0$, the nonzero scalar in (16.5) implies $M_u = 0$. Thus every target witness yields a nonempty source witness, of length no greater than $|w|$.

Every source witness has a target extension. If u is nonempty and $M_u = 0$, the product of the active matrices has the form

$$A_u = \begin{pmatrix} 0 & 0 \\ R & 0 \end{pmatrix}$$

for some $r \times d$ matrix R . Multiplication on the left by any active matrix kills this residual:

$$A_i A_u = \alpha \begin{pmatrix} M_i & 0 \\ B_i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ R & 0 \end{pmatrix} = 0 \quad (1 \leq i \leq k). \quad (16.6)$$

Therefore ui is a target witness. Applying the preceding two directions to shortest words gives (16.4). $\square$

16.1 Effective choice of the scalar

Let $S_{\mathcal{M}} = \sum_i M_i^T M_i$ and $t_{\mathcal{M}} = \text{tr}(S_{\mathcal{M}}) \geq 0$. Since $S_{\mathcal{M}}$ is positive semidefinite, $\lambda_{\max}(S_{\mathcal{M}}) \leq t_{\mathcal{M}}$. Hence any integer $c > \sqrt{t_{\mathcal{M}}}$ satisfies (16.1). An exact choice is

$$c = \left\lfloor \sqrt{\lfloor t_{\mathcal{M}} \rfloor} \right\rfloor + 1.$$

This can be computed with integer square root and rational arithmetic. No numerical eigenvalue calculation is required. A smaller user-supplied rational c is also allowed, provided positive definiteness of Q is checked exactly, for example by LDL^T pivots.

16.2 Why packing saves dimension

The identity $B^T B = Q$ constrains the *sum* of the contributions of its rows, not which Kraus operator receives each row. There is no requirement that every generator receive an entire $d \times d$ correction matrix. The s correction rows can instead be distributed among the k pre-existing active outcomes. Each added state-space coordinate provides k such row slots. Consequently $\lceil s/k \rceil$ additional coordinates suffice.

This allocation does not introduce a separate potentially dangerous “correction generator.” Each correction block remains attached to its original M_i . Soundness is then protected by the upper-left projection, while the zero right columns make one additional active outcome sufficient to erase the lower-left residual.

Corollary 16.2 (General dimension bounds). *An input list of k rational $d \times d$ matrices admits a mortality-preserving rational instrument with $k + 1$ outcomes and dimension*

$$D_{\text{Meyer}} = d + \left\lceil \frac{d + 3}{k} \right\rceil.$$

Using only the elementary construction of Proposition 15.1, one obtains

$$D_{\text{elem}} = d + \left\lceil \frac{4d}{k} \right\rceil.$$

Both constructions are effective, and both satisfy the witness-length bound (16.4).

Proof. Apply Theorem 16.1 with the factors from Propositions 15.4 and 15.1, respectively. Zero-row padding leaves the Gram identity unchanged. $\square$

17 A stronger word-level invariant

The projection argument is enough for undecidability, but a more precise formula is useful both for arithmetic certificates and for debugging an implementation.

Proposition 17.1 (Idle erasure and the final active letter). *Suppose a target word w contains the active subsequence $i_1 \cdots i_m$, with $m \geq 1$. Let z be its total number of zeros and z_- the number of zeros before its final active letter. Put*

$$P = M_{i_{m-1}} \cdots M_{i_1},$$

where $P = I_d$ if $m = 1$. Then

$$A_w = \begin{pmatrix} \alpha^m \beta^z M_{i_m} P & 0 \\ \alpha^m \beta^{z-} B_{i_m} P & 0 \end{pmatrix}. \quad (17.1)$$

In particular,

$$A_w = 0 \iff \begin{pmatrix} M_{i_m} \\ B_{i_m} \end{pmatrix} P = 0. \quad (17.2)$$

Arbitrary insertion or deletion of idle letters does not change whether a word with an active letter has zero product.

Proof. Before the final active letter, the upper-left block is $\alpha^{m-1} \beta^{z-} P$. Left multiplication by A_{i_m} replaces the lower-left block by αB_{i_m} times this upper-left block: the old lower-left block contributes nothing because the right block column of A_{i_m} is zero. Each remaining idle letter multiplies the upper-left block by β and leaves the lower-left block unchanged. This proves (17.1). The two scalar factors are nonzero, so they do not affect vanishing. $\square$

The proposition should not be misread as asserting that idle insertion multiplies the *whole* product by one scalar. It generally scales the two block rows differently. Only the zero/nonzero predicate is invariant. Keeping this distinction explicit prevents an otherwise tempting false identity.

17.1 Sharpness of the one-step interval

Both endpoints of (16.4) occur. For $d = k = 1$, take $M_1 = 0$, $c = 1$, and $B_1 = (1)$. The active matrix is

$$A_1 = \frac{4}{5} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

It is nonzero but its square is zero. Hence the source minimum is 1 and the target minimum is 2.

For equality, take $d = 1$, $M_1 = 0$, $M_2 = 1$, and $c = 2$. Put $r = 3$, $B_1 = (0, 0, 0)^\top$, and $B_2 = (1, 1, 1)^\top$. Their combined Gram matrix is $3 = c^2 - (0^2 + 1^2)$, while $A_1 = 0$. Both minima are 1. These are sharpness examples for the interval, not hard mortality instances.

18 Four-dimensional, seven-outcome undecidability

We import the following published result [11, Corollary 12].

Theorem 18.1 (Six-generator mortality; external input). *There is no algorithm deciding mortality for arbitrary lists of six integer 3×3 matrices. The proof proceeds through an effective reduction from halting, via binary tag systems and Post correspondence.*

Theorem 18.2 (Dimension four and seven outcomes). *The following decision problem is undecidable: given seven rational real 4×4 matrices $A_0, \dots, A_6$ satisfying $\sum_{j=0}^6 A_j^T A_j = I_4$, decide whether there exists a nonempty outcome word with probability zero for the maximally mixed input state.*

Undecidability already holds on the subfamily in which

$$A_0 = \text{diag}(3/5, 3/5, 3/5, 1),$$

all other matrices have a zero fourth column, and their upper-left 3×3 blocks are a common nonzero rational scalar times six integer matrices.

Proof. Given the six integer matrices, choose c as above and a six-row rational Gram factor of Q . Proposition 15.4 supplies exactly the needed row budget because $d + 3 = 6$. Thus $r = \lceil 6/6 \rceil = 1$ and $D = 4$.

The construction is a total computable transformation: a rational Gram factor may be found by exhaustive search, whose termination is guaranteed. By Theorem 16.1, the source is mortal if and only if the target is mortal. By Lemma 14.2, target mortality is exactly the existence of a zero-probability word. A decision algorithm for the stated target family would therefore decide the source problem, contradicting Theorem 18.1. $\square$

Corollary 18.3 (A more elementary resource bound). *The same occurrence problem is undecidable for seven rational outcomes in dimension five, using Lagrange’s four-square theorem in place of Meyer.*

Proof. Use $s = 4d = 12$ correction rows. Then $r = \lceil 12/6 \rceil = 2$ and $D = 5$. $\square$

Construction	Source (d, k)	Quantum dimension	Outcomes
Eisert–Müller–Gogolin [10]	(3, 8)	15	9
Packed, elementary	(3, 8)	5	9
Packed, Meyer	(3, 8)	4	9
Packed, elementary with [11]	(3, 6)	5	7
Packed, Meyer with [11]	(3, 6)	4	7

The table compares proved constructions, not known optimal lower bounds. The first two packed rows do not require the improved six-generator source theorem. A separate tradeoff is available by duplicating the source generators: twelve active outcomes suffice to fit an elementary 12-row factor into one extra dimension, yielding dimension four with thirteen outcomes without Meyer.

18.1 Computational completeness and the lack of a stopping bound

Finite products can be evaluated exactly, so the positive instances of the target problem are recursively enumerable. The effective halting reductions underlying Theorem 18.1, followed by Theorem 16.1, make this set recursively-enumerable complete under computable many-one reductions. No polynomial-time bound for the Gram-search portion of this reduction is asserted.

Corollary 18.4 (No total computable witness-length bound). *There is no total computable function of a valid rational four-dimensional, seven-outcome instrument description that bounds the shortest mortal-word length whenever the instrument is mortal.*

Proof. If such a bound were available, calculate it and test every nonempty word up to that length. There are finitely many, and every product is exactly computable. The bounded search would decide the problem in Theorem 18.2. $\square$

In particular, no total computable bound depending only on the bit length of the instrument description can serve this purpose. This concerns a bound guaranteed for every positive instance. It does not exclude useful bounds for restricted matrix classes or for individual examples with separately established invariants.

19 A fully explicit interference example

An example makes the distinction between a source zero product and its one-step target extension visible. It also provides a small exact regression test in the same dimension and outcome count as Theorem 18.2.

Let

$$\begin{aligned} M_1 &= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & M_2 &= \begin{pmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & M_3 &= \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ M_4 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & M_5 &= M_4, & M_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \end{aligned}$$

Direct calculation gives $\sum_i M_i^\top M_i = 4I_3$. Choose $c = 3$, so $Q = 5I_3$, and take the six correction rows

$$B_1 = (1, 0, 0), B_2 = (2, 0, 0), B_3 = (0, 1, 0), B_4 = (0, 2, 0), B_5 = (0, 0, 1), B_6 = (0, 0, 2). \quad (19.1)$$

Their outer products sum to $5I_3$. Equation (16.2) now gives seven explicit rational 4×4 operators, with $\alpha = 4/15$.

The chronological source word $(2, 1)$ gives

$$M_{(2,1)} = M_1 M_2 = 0.$$

Its upper-left entry vanishes by the cancellation $1 - 1 = 0$. Taking absolute values of entries destroys this cancellation: $|M_1||M_2| \neq 0$.

The corresponding target product does *not* vanish yet. Its only nonzero block is the bottom-left row

$$\alpha^2 B_1 M_2 = \alpha^2 (1, 0, 0).$$

Consequently

$$p(2, 1) = \frac{\alpha^4}{4} = \frac{64}{50625} > 0, \quad p(2, 1, 1) = 0. \quad (19.2)$$

The appended active outcome removes exactly the lower-left residual predicted by the proof. This example disproves the stronger, generally false assertion $M_w = 0 \Rightarrow A_w = 0$ for the *same* word.

All operators have a common denominator 15. Writing $H_j = 15A_j$, the exact normalization becomes

$$\sum_{j=0}^6 H_j^\top H_j = 225I_4, \quad H_0 = \text{diag}(9, 9, 9, 15), \quad H_i = 4 \begin{pmatrix} M_i & 0 \\ B_i & 0 \end{pmatrix}. \quad (19.3)$$

The artifact `data/07-quantum-mortality-four_dimensional_instrument.json` contains these integer matrices. The example is not presented as a universal instance; it is a transparent test of the general compiler.

20 Exact probability gaps and their limits

Let $\mathcal{A}$ be any rational real instrument, not necessarily one produced by the packing construction. Choose a positive common denominator N and put $H_j = NA_j \in \mathbb{Z}^{D \times D}$.

Proposition 20.1 (Arithmetic separation at a fixed length). *For a word of length n and the maximally mixed state,*

$$p(w) = \frac{\sum_{a,b} (H_w)_{ab}^2}{DN^{2n}}. \quad (20.1)$$

Hence

$$p(w) = 0 \quad \text{or} \quad p(w) \geq \delta_n := \frac{1}{DN^{2n}}. \quad (20.2)$$

Proof. The common denominator factors out of the ordered product, giving $A_w = N^{-n}H_w$. Substitute this identity into (14.4). A nonzero integer matrix has at least one nonzero integer entry, so its squared Frobenius norm is a positive integer, at least one. $\square$

The denominator argument is the same basic mechanism behind length-dependent probability separation in the earlier quantum mortality work [10]. Here the formula is stated for the maximally mixed state so that its normalization factor and lower bound are explicit. For a different positive definite ρ , one obtains the lower bound $\lambda_{\min}(\rho)N^{-2n}$ instead. No rationality of ρ is required for that inequality.

Proposition 20.2 (A finite-length perturbation estimate). *Suppose A_j and $\tilde{A}_j$ are corresponding Kraus operators of two valid instruments of the same dimension, and $\|A_j - \tilde{A}_j\|_{\text{op}} \leq \varepsilon$. For every length- n word,*

$$|p(w) - \tilde{p}(w)| \leq 2n\varepsilon$$

for maximally mixed inputs.

Proof. Normalization implies $A_j^\dagger A_j \leq I$ for each j , so every Kraus operator is a contraction. The same is true of the tilded operators. Telescoping the difference of two ordered products yields $\|A_w - \tilde{A}_w\|_{\text{op}} \leq n\varepsilon$, and hence $\|A_w - \tilde{A}_w\|_{\text{F}} \leq \sqrt{D}n\varepsilon$. Each product has Frobenius norm at most $\sqrt{D}$. Therefore

$$\frac{|\|A_w\|_{\text{F}}^2 - \|\tilde{A}_w\|_{\text{F}}^2|}{D} \leq \frac{(2\sqrt{D})(\sqrt{D}n\varepsilon)}{D} = 2n\varepsilon.$$

$\square$

For a fixed exact rational reference instrument, a threshold of $\delta_n/2$ correctly distinguishes a zero reference probability from a nonzero one when the perturbed probability is known exactly and $\varepsilon < \delta_n/(4n)$. Any additional numerical or statistical error must also fit inside the remaining margin.

This is not a uniform experimental solution to the unbounded problem. The required error tolerance can decay exponentially with n , and Corollary 18.4 precludes a total computable stopping length in general. Exact rational input is a finite mathematical description; it is not an assertion that arbitrary real-world calibration achieves exact zero amplitudes.

21 A canonical quartic compiler for bounded histories

We now give an explicit Diophantine representation, rather than merely appeal to MRDP. Fix $D, K, n \geq 1$. Let

$$H_0, \dots, H_{K-1} \in \mathbb{Z}^{D \times D}$$

be the integer numerator matrices of an instrument. The polynomial construction also works for arbitrary integer matrix lists; normalization is only needed for the quantum interpretation.

21.1 Natural coordinates for signed entries

An integer x is encoded canonically by natural numbers x^+, x^- with

$$x = x^+ - x^-, \quad x^+ x^- = 0. \quad (21.1)$$

The pair is unique: $(x^+, x^-) = (\max(x, 0), \max(-x, 0))$. The complementarity equation is essential. Without it, a single integer would have infinitely many representations $(x^+ + t, x^- + t)$, destroying the witness-counting statement below.

Introduce natural selectors e_{tj} for $0 \leq t < n$ and $0 \leq j < K$, subject to

$$e_{tj}(e_{tj} - 1) = 0, \quad \sum_{j=0}^{K-1} e_{tj} = 1. \quad (21.2)$$

Thus exactly one outcome is selected at every time.

For each time introduce signed matrices Y_t and X_t , represented by canonical natural pairs. The selected one-step matrix and its accumulated product are enforced by

$$Y_{t,ab} = \sum_{j=0}^{K-1} e_{tj} H_{j,ab}, \quad 0 \leq t < n, \quad (21.3)$$

$$X_{t+1,ab} = \sum_{c=1}^D Y_{t,ac} X_{t,cb}, \quad 0 \leq t < n, \quad (21.4)$$

$$X_{0,ab} = \delta_{ab}, \quad X_{n,ab} = 0. \quad (21.5)$$

Every occurrence of X , Y , or H in these formulas denotes a difference of its positive and negative coordinates. The signed pairs for H are free *input* coordinates. They may be fixed canonically before constructing the witness; the representation is also valid for any fixed pair encoding the same integer inputs.

The intermediate variables Y_t are not optional if the matrix entries are treated as polynomial input variables and degree four is to be retained by this simple construction. Direct substitution of (21.3) into (21.4) would produce terms $e_{tj}H_{j,ac}X_{t,cb}$ of degree three. The separate Y_t gates keep every residual quadratic.

21.2 The polynomial and its exact count

Let $\mathcal{R}_{D,K,n}$ consist of the following residuals:

- all equations in (21.2), moved to the left;
- the complementarity products X^+X^- and Y^+Y^- for every entry;
- all equations in (21.3)–(21.5), moved to the left.

Define the integer polynomial

$$\mathcal{Q}_{D,K,n}(H^+, H^-, \mathbf{w}) = \sum_{R \in \mathcal{R}_{D,K,n}} R^2. \quad (21.6)$$

This is an explicit finite formula, not an appeal to an unspecified encoding theorem.

Theorem 21.1 (Parsimonious bounded-history representation). *The polynomial (21.6) has total degree at most four, including its free matrix input coordinates. For each fixed integer matrix list H and fixed encoding of that input, its natural-number solution tuples are in bijection with the length- n words w satisfying $H_w = 0$.*

It has

$$W(D, K, n) = nK + (4n + 2)D^2 \quad (21.7)$$

natural witness coordinates and is the sum of squares of

$$E(D, K, n) = n(K + 1) + (4n + 3)D^2 \quad (21.8)$$

quadratic-or-lower residuals. The $2KD^2$ natural coordinates describing H are free inputs and are not included in W .

Proof. Every residual has degree at most two, so its square has degree at most four. A sum of integer squares is zero if and only if each summand is zero.

Suppose $\mathbf{w}$ is a natural solution. The selector equations yield a unique label j_t at each time. Equation (21.3) forces $Y_t = H_{j_t}$. Starting with $X_0 = I_D$, induction in (21.4) gives

$$X_t = H_{j_{t-1}} \cdots H_{j_0}.$$

The final equations give $H_w = 0$ for $w = j_0 \cdots j_{n-1}$.

Conversely, for such a word choose the corresponding one-hot selectors, set $Y_t = H_{j_t}$, and let X_t be the actual prefix products. Encode every signed entry by its unique canonical pair. All residuals vanish. These choices are forced by the word: selectors are fixed, signed Y_t and X_t are determined successively, and (21.1) fixes their natural pairs. Thus the correspondence is a bijection, not merely a surjection.

For the count, there are nK selectors, $2nD^2$ coordinates for $Y_0, \dots, Y_{n-1}$, and $2(n+1)D^2$ for $X_0, \dots, X_n$. This gives (21.7). The residual count is displayed explicitly below and sums to (21.8). $\square$

Residual family	Number
Selector Boolean equations	nK
One-hot sums	n
Complementarity for all X_t	$(n+1)D^2$
Complementarity for all Y_t	nD^2
Matrix-selection equations	nD^2
Matrix-multiplication equations	nD^2
Initial identity	D^2
Final zero	D^2
Total	$n(K+1) + (4n+3)D^2$

Corollary 21.2 (Four-dimensional specialization). *For seven outcomes in dimension four, the explicit length- n certificate uses*

$$W(4, 7, n) = 71n + 32, \quad E(4, 7, n) = 72n + 48.$$

For $n = 3$, these numbers are 245 natural witnesses and 264 residuals.

Because the solutions are in bijection with words, a fixed input has at most K^n solutions to this particular bounded-history polynomial. This finite-fold property belongs to the fixed-length family. It is not a statement about the fixed-arity unbounded MRDP representation in Section 22.

21.3 Physical validity can be certified separately

If one also wants the polynomial input to certify a rational instrument, include a positive denominator $N = u + 1$ and the residuals

$$\sum_{j=0}^{K-1} \sum_{c=1}^D H_{j,ca} H_{j,cb} - N^2 \delta_{ab} = 0 \quad (1 \leq a, b \leq D). \quad (21.9)$$

After replacing signed entries by differences, these are quadratic. Adding their squares keeps the total degree at most four. The denominator does not enter the zero-product equations: multiplying every generator by the same nonzero scalar does not change mortality. The counts in Theorem 21.1 exclude these optional validity residuals and the denominator coordinate.

21.4 Fixed words and the companion nonzero certificate

A prescribed word can simply be substituted into the selectors. Alternatively, retain the selectors and add natural prefix variables $z_0, \dots, z_n$ with

$$z_0 = 0, \quad z_{t+1} = Kz_t + \sum_{j=0}^{K-1} j e_{tj}, \quad z_n = h.$$

For a fixed length, these equations encode the word as its base- K integer h , including leading zero labels. They add $n + 1$ natural witnesses and $n + 2$ linear residuals, and the extra witnesses are uniquely determined.

Proposition 21.3 (A bounded nonzero companion). *Replace the D^2 final-zero residuals by one residual*

$$\sum_{a,b} X_{n,ab}^2 - (v+1) = 0, \quad v \in \mathbb{N}. \quad (21.10)$$

Then the resulting sum of squares is still quartic, and its natural solutions are in bijection with the length- n words having $H_w \neq 0$.

Proof. The trace equations still force $X_n = H_w$. Its squared Frobenius norm is a nonnegative integer. It is positive exactly when it is $v+1$ for a natural v , and that v is unique. The new residual is quadratic in the signed-pair coordinates, so its square is quartic. $\square$

For a fixed word length, both possibility and impossibility therefore have completely explicit arithmetic certificates. This is consistent with the undecidability theorem: the difficult property is the existential search over *all lengths*. The assertion that every word of every length is possible has an unbounded universal quantifier.

22 Unbounded Diophantine representation and the computational substrate

22.1 The semidecider

Choose an effective encoding of finite rational matrix descriptions by naturals. Invalid encodings can be rejected. Define $\mathbf{QMort}_{4,7}$ to be the set of codes of valid rational four-dimensional, seven-outcome instruments for which some nonempty word has zero product. A semidecider is straightforward: first check exact normalization; then enumerate words in increasing length and evaluate their integer numerator products until a zero product is found.

The validity test is decidable, and every individual word test is decidable. If the input is mortal, its shortest witness is eventually reached. If it is not mortal, the word search continues indefinitely. The hard part is not evaluating a product but deciding that this unbounded search will never succeed.

Theorem 22.1 (A fixed-arity unbounded polynomial). *There exist a fixed $m \in \mathbb{N}$ and a fixed polynomial $U \in \mathbb{Z}[e, y_1, \dots, y_m]$ such that*

$$e \in \mathbf{QMort}_{4,7} \iff \exists \mathbf{y} \in \mathbb{N}^m U(e, \mathbf{y}) = 0. \quad (22.1)$$

The represented set is recursively-enumerable complete. Its complement is not Diophantine over $\mathbb{N}$.

Proof. The semidecider establishes recursive enumerability. MRDP supplies (22.1); this is exactly the form of the natural-input finite-polynomial interface described by ProveIt [14]. Completeness follows by composing the effective halting reduction into six-generator matrix mortality with the total computable construction of Theorem 18.2.

If the complement were also Diophantine, it too would be recursively enumerable: enumerate all natural witness tuples and evaluate its polynomial. Running the two enumerators in parallel would decide membership in $\mathbf{QMort}_{4,7}$, contradicting Theorem 18.2. $\square$

This theorem does not print an expanded universal polynomial U , nor claim a numerical witness bound for it. The explicit polynomials supplied in this package are $\mathcal{Q}_{D,K,n}$, a length-indexed family. One cannot obtain a fixed-arity polynomial merely by writing $\exists n \mathcal{Q}_{D,K,n} = 0$, because the number of coordinates in that notation grows with n . MRDP, or a genuine coded-trace construction, is the additional step that removes this changing arity.

22.2 What “Turing-complete substrate” means here

The chain of effective reductions is

$$\begin{aligned} \text{Turing-machine halting} &\longrightarrow \text{binary tag-system halting} \longrightarrow \text{Post correspondence} \\ &\longrightarrow \text{six integer } 3 \times 3 \text{ matrices} \longrightarrow \text{seven rational } 4 \times 4 \text{ Kraus operators} \\ &\longrightarrow \text{a natural-number Diophantine equation.} \end{aligned} \tag{22.2}$$

The first matrix-hardness portion is imported from the published reduction [11]. The new quantitative normalization is Theorem 16.1; the final unbounded polynomial step is MRDP. A binary tag system is a finite rewriting program acting on an unbounded word, whereas the quantum description has fixed matrix size but permits arbitrarily long outcome histories. These are different locations for the unbounded resource.

In particular, Theorem 18.2 does not say that one fixed four-dimensional device with a fixed finite set of coefficients is a universal programmable quantum computer. Its input is the *description of the instrument*: the rational coefficients vary with the encoded computation. Nor does the theorem show how to decide halting by physically waiting for an impossible outcome. Such an outcome never occurs. The decision problem concerns the exact mathematical description of the entire family of histories.

The substantive role of interference is that signed matrix products retain cancellation. Replacing the signed substrate by entrywise supports changes its computational behavior, as both the example and the following finite-state argument show.

Remark 22.2 ([write] What ProveIt formalizes of this chain). Of the chain (22.2), ProveIt formalizes in Lean, for its own purposes and independently of this report, a different route from Turing machines to *binary tag systems*: three-counter programs, single-register machines, a 30-tag system and Neary’s track-and-shift binary layer, ending in the undecidability of an encoded binary tag family (Jones1980.tag91_undecidable; Section 1.3). It does not formalize the Post-correspondence or matrix-mortality steps of Neary’s Corollary 12, nor Theorem 16.1. The source’s sentence that the first matrix-hardness portion is imported from the published reduction remains accurate for the matrix steps.

22.3 A decidable boundary: nonnegative entries

Proposition 22.3 (Nonnegative mortality is decidable). *Mortality for any finite family of entrywise nonnegative rational $D \times D$ matrices is decidable. A breadth-first search through at most 2^{D^2} Boolean support matrices suffices.*

Proof. For a nonnegative matrix M , let $\text{supp}(M)$ have entry 1 where M is positive and 0 otherwise. Nonnegativity implies

$$\text{supp}(MN) = \text{supp}(M) \odot \text{supp}(N),$$

where $\odot$ is Boolean matrix multiplication: an entry is 1 exactly when some intermediate index makes both factors positive. There is no cancellation. Thus reachable supports form a finite semigroup with at most 2^{D^2} elements. Starting with the identity support, multiply by the generator supports and explore every newly reached support. Mortality holds exactly when the zero support is reached. $\square$

This decidable contrast is already part of the quantum occurrence literature [10]; the proof is included to make the boundary explicit. It applies equally to real rational Kraus operators whose entries happen all to be nonnegative in the chosen representation. It does not apply merely because probabilities themselves are nonnegative: their underlying amplitude matrices can have signs.

22.4 A scalar matrix-product observable

There is also a precise tensor version of the same arithmetic invariant. For $M \in \mathbb{Q}^{d \times d}$ and $\Omega = \sum_{a=1}^d e_a \otimes e_a$, one has

$$\Omega^\top (M \otimes M) \Omega = \sum_{a,b} M_{ab}^2. \quad (22.3)$$

Consequently, with local matrices $T_i = M_i \otimes M_i$,

$$\Omega^\top T_{i_n} \cdots T_{i_1} \Omega = 0 \iff M_w = 0.$$

The proof is direct expansion and the identity $(M \otimes M)(N \otimes N) = (MN) \otimes (MN)$. This gives a scalar observable of a matrix-product representation that vanishes exactly on mortal words. It is an algebraic representation; additional normalization or boundary-state choices are required before identifying it with a particular normalized physical tensor-network state. No new matrix-product-state undecidability theorem is claimed here.

Part III

Continuous computation (manuscript 09)

23 The question beyond ordinary MRDP

The bare assertion that a Turing-complete substrate has a Diophantine halting set is not a new result. Once finite successful executions can be enumerated, MRDP applies. A useful research contribution must identify a property whose finite-witness structure is not already the finite execution itself, expose that structure, and delimit where it fails.

This article does that for *infinite-time robust avoidance* in a compact continuous-state computer. The finite witness is a cut in a rational graph, not a completed execution. Soundness rules out *all* finite perturbed executions, of arbitrarily large length. Completeness applies exactly when a strictly positive noise margin exists.

Three questions organize the development. Can an infinite-time safety property of a continuous universal machine be compiled into an ordinary integer polynomial? Can one fixed small-dimensional machine realize arbitrary independent enumerable acceptance and robust-rejection behaviors? Can its exact noise threshold be a computable real even though the sign of that threshold is undecidable? We answer all three under explicit hypotheses. These are the questions posed in this manuscript; we do not assert that they were previously advertised as named open problems.

[write] Reading guide for Part III

For the main realization theorem, read Sections 27 and 28; the crucial mechanisms are the enlarged rejecting slab and the expanding emergency coordinate. For finite certificates and their arithmetic meaning, read Sections 25–26, followed by Section 29. The fixed MRDP polynomial must be distinguished from the directly generated finite-grid quartics. For executable material, start with Section 32 and the report README. Section 35.3 describes the proof-assistant work still needed, and Appendix E records dependencies and non-claims. (The source’s own reading guide, of which this is an adaptation, preceded its first section; its repository baseline, Section 1.1 there, is printed in Section 1.)

23.1 Relation to existing research

Moore’s generalized shifts already connect symbolic computation with continuous dynamical systems and establish undecidability phenomena [16]. Our two tape coordinates are a direct relative of this classical Cantor encoding. Packing the finite control into one of those coordinates is an elementary space-saving step, not a claim that continuous universality begins here.

Bournez, Graça, and Hainry study adversarial perturbations and show that appropriate infinitesimal robustness forces decidability. Their bounded-domain arguments use finite spatial abstractions [17]. We credit that qualitative framework. The present proofs isolate a numerical bottleneck radius, its two-sided horizon-free approximation, an explicit quartic cut checker, and a simultaneous three-outcome realizer with time–radius bounds. Whether this exact package, or a stronger equivalent formulation, already appears elsewhere requires a wider priority audit.

Coarse chain recurrence studies stepwise perturbations, filtrations, and finite-error dynamical information [18]; the critical radius is naturally related to that viewpoint. The distinction between approximate numerical information and exact truth also appears in δ -decision procedures [19]. We do not identify our unbounded-time graph algorithm with bounded first-order δ -decision.

A recent preprint by Platzer and Qian treats axiomatization of robust safety for polynomial differential equations, with a bounded-horizon topological robustness notion [20]. Our model instead has discrete time, a continuous compact state space, and a uniform bound on adversarial error at *every* step. Neither theorem is automatically a theorem about the other noise model.

23.2 Principal statements

The quantitative core is

$$\rho \leq r_N \leq \rho + \frac{L+1}{2N}, \quad (23.1)$$

where r_N is a finite rational graph bottleneck. A Boolean cut is checked by an explicit degree-four polynomial. The universal core is a single three-dimensional map realizing, for every disjoint computably enumerable pair (A, B) ,

$e \in A$	$e \in B$	$e \notin A \cup B$
exact acceptance	positive noise radius	fragile divergence.

The radius of a rejecting input, when rejection first occurs at time T , obeys

$$\frac{1}{12m S_T(9m+6)} \leq \rho \leq \min\{1, 2 \cdot 3^{-(T-1)}\}, \quad S_T(L) = \sum_{j=0}^{T-1} L^j, \quad (23.2)$$

where m is the fixed number of interpreter states. The following sections prove these statements and their arithmetic consequences.

24 Noise semantics and the three outcomes

Fix $K = [0, 1]^d$ with the sup norm. Unless otherwise stated, $f : K \rightarrow K$ is a continuous rational piecewise-affine map, specified by finitely many rational polyhedra and rational affine formulas agreeing on overlaps. A rational number $L \geq 0$ is supplied with a proof that

$$\|f(u) - f(v)\|_\infty \leq L \|u - v\|_\infty \quad (u, v \in K). \quad (24.1)$$

For example, the largest row-sum norm of the affine linear parts is a valid bound: subdivide the segment from u to v at the finitely many cell faces and add the local estimates. Continuity is essential in this argument.

Let $H \subseteq K$ be a nonempty finite union of closed rational axis-aligned boxes. Degenerate boxes are permitted. Initially take $x \in K \cap \mathbb{Q}^d$ with $x \notin H$.

Definition 24.1 (Critical radius). For a finite path $\gamma = (x_0, \dots, x_T)$, where $T \geq 1$, $x_0 = x$, all $x_t \in K$, and $x_T \in H$, set

$$c_f(\gamma) = \max_{0 \leq t < T} \|x_{t+1} - f(x_t)\|_\infty.$$

The *critical noise radius* is

$$\rho_f(x, H) = \inf_{\gamma} c_f(\gamma). \quad (24.2)$$

If $x \in H$, define $\rho_f(x, H) = 0$ to include a time-zero hit.

The infimum is over real paths, not rational paths or symbolic encodings. It is finite because a one-step jump from $f(x)$ to any point of H costs at most one. In particular $0 \leq \rho_f(x, H) \leq 1$. An ε -pseudo-orbit is a path whose cost is at most ε . Errors may change at every step and may be selected adversarially using the previous states. The path must stay inside K ; there is no implicit extension outside the cube.

Definition 24.2 (Outcomes). Write

$$\begin{aligned} \text{Hit}(x) &\iff \exists t \geq 0 \quad f^t(x) \in H, \\ \text{Safe}(x) &\iff \rho_f(x, H) > 0, \\ \text{Frag}(x) &\iff \neg \text{Hit}(x) \text{ and } \rho_f(x, H) = 0. \end{aligned}$$

We call these *exact acceptance*, *robust avoidance*, and *fragile avoidance*, respectively. In the universal simulation, fragile avoidance will coincide with divergence of the interpreter.

An exact accepting orbit gives a zero-cost path, so **Hit** and **Safe** are disjoint. Every initial state has exactly one of these three outcomes. Furthermore,

$$\text{Safe}(x) \iff \exists \varepsilon > 0 \text{ no finite } \varepsilon\text{-pseudo-orbit from } x \text{ hits } H. \quad (24.3)$$

For the forward implication choose $0 < \varepsilon < \rho$. Conversely, if $\rho = 0$, paths of cost below any positive ε exist by the definition of infimum.

Remark 24.3 (The threshold itself). Neither $\rho = 0 \Rightarrow \text{Hit}$ nor attainment of the infimum is valid in general. A lower bound $\rho \geq \delta > 0$ certifies safety for all errors *strictly below* δ ; without additional information it does not certify safety at the endpoint δ . All later certificate statements respect this distinction.

Example 24.4 (An infinite horizon changes the answer). For $f(t) = t$, $x = 0$, and $H = \{1\}$, the exact orbit never moves. Nevertheless, $0, 1/T, \dots, 1$ is a path of cost $1/T$, so $\rho = 0$. No zero-cost path hits H . Thus an exactly nonaccepting state can be arbitrarily noise-sensitive.

25 Computing the radius without bounding time

Choose a positive integer D clearing all denominators of all endpoints of the boxes making up H . For $n \geq 0$, let

$$N = D2^n, \quad G_N = \{0, 1/N, \dots, 1\}^d, \quad V_N = G_N \cup \{x\}, \quad T_N = G_N \cap H.$$

The union defining V_N removes duplicates. In particular, a nongrid initial point is kept *exactly*; it is not silently rounded. The target set T_N is nonempty because all box endpoints are grid points.

On the complete directed graph with vertex set V_N , assign edge weight

$$w_N(u, v) = \|v - f(u)\|_\infty \in \mathbb{Q}_{\geq 0}. \quad (25.1)$$

Define r_N to be the minimum, over graph paths from x to T_N , of their largest edge weight. A repeated vertex can be removed together with the intervening cycle without increasing the largest weight. Thus a minimizing path exists and may be chosen with at most $|V_N| - 1$ edges.

Theorem 25.1 (Horizon-free grid approximation). *For every aligned mesh N , put $h_N = 1/(2N)$ and $C = L + 1$. Then*

$$\rho_f(x, H) \leq r_N \leq \rho_f(x, H) + Ch_N. \quad (25.2)$$

The numbers r_{D2^n} form a nonincreasing rational sequence. Consequently

$$[\max\{0, r_N - Ch_N\}, r_N] \quad (25.3)$$

is a computable interval containing $\rho_f(x, H)$ and of length at most Ch_N . The estimate does not involve the length of an accepting pseudo-orbit.

Proof. Every graph path is also a path of real states in K with the same cost. Taking infima gives $\rho \leq r_N$.

Conversely, take any real path $x_0 = x, \dots, x_T \in H$ of cost c . Keep $y_0 = x$ exactly. Round each coordinate of x_t for $t \geq 1$ to a nearest grid coordinate, obtaining $y_t \in G_N$ with $\|x_t - y_t\|_\infty \leq h_N$. The final state can be rounded inside H : choose a box containing x_T ; since both endpoints of each coordinate interval are on the grid, nearest-grid rounding stays in that interval. Thus $y_T \in T_N$.

For $1 \leq t < T$, the triangle inequality and (24.1) give

$$\|y_{t+1} - f(y_t)\|_\infty \leq \|y_{t+1} - x_{t+1}\|_\infty + \|x_{t+1} - f(x_t)\|_\infty + \|f(x_t) - f(y_t)\|_\infty \leq c + h_N + Lh_N.$$

The first edge has the better bound $c + h_N$, because $y_0 = x_0$. Therefore $r_N \leq c + Ch_N$. Taking the infimum over all real paths proves the upper bound even when that infimum is not attained.

The grid at mesh N is contained in the grid at mesh $2N$; the source and old targets remain available, with the same weights. Hence $r_{2N} \leq r_N$. All graph data are rational and finite, so the interval is effectively computable. No error was accumulated over time: each edge was estimated against the corresponding edge of the original path independently. $\square$

25.1 Algorithm, complexity, and what remains undecidable

A bottleneck version of Dijkstra's algorithm starts with distance zero at the source and relaxes an edge by

$$d(v) \leftarrow \min\{d(v), \max(d(u), w_N(u, v))\}.$$

For $M = |V_N| \leq (N + 1)^d + 1$, a simple implementation takes $O(M^2)$ rational comparisons and stores $O(M^2)$ edge weights. The companion implementation uses precisely this finite algorithm. The arithmetic bit costs also depend on the rational input and map descriptions; no unit-cost claim for arbitrarily large integers is intended.

Given an additive precision 2^{-p} , choose $N = D2^n$ with $C/(2N) \leq 2^{-p}$. The resulting interval proves that the critical radius is a *computable real*, uniformly from the rational data and a valid L . In fixed dimension, the explicit graph size is exponential in the number of precision bits. A four-dimensional or three-dimensional grid can already be impractical at moderate precision.

Corollary 25.2 (A promised threshold comparison). *Given rational ε and $\delta > 0$, under the promise $|\rho - \varepsilon| \geq \delta$, one can decide whether $\rho < \varepsilon$ or $\rho > \varepsilon$ by using an interval of width less than δ . No time cutoff is needed.*

This does not decide whether $\rho = 0$. Every interval in (25.3) may continue to contain zero. Later we construct a fixed map for which positivity is as hard as the halting problem.

25.2 Stability of the value, not of its zero test

Proposition 25.3 (Lipschitz dependence of the critical value). *For maps $f, g : K \rightarrow K$ and the same source outside H ,*

$$|\rho_f(x, H) - \rho_g(x, H)| \leq \sup_{u \in K} \|f(u) - g(u)\|_\infty.$$

If f is L -Lipschitz and $x, y \notin H$, then

$$|\rho_f(x, H) - \rho_f(y, H)| \leq L \|x - y\|_\infty.$$

Proof. For any fixed path, changing the map changes every edge cost, and therefore the maximum cost, by at most the uniform map distance. Take infima in both directions. To change the source, keep every vertex of a path except its first vertex. Only the first edge cost changes, by at most $\|f(x) - f(y)\|_\infty \leq L \|x - y\|_\infty$. Again take infima in both directions. $\square$

The exclusion of sources in H in the second statement is deliberate: our time-zero convention overrides the positive-length variational definition. Even on a region where the radius is Lipschitz and computable, equality to zero can be undecidable. Numerical regularity is not a decision procedure for exact membership in a zero set.

25.3 An effective-continuity extension

Proposition 25.4 (Beyond rational affine formulas). *Suppose instead that f is computable and has a known effective uniform modulus ω , with $\omega(h) \rightarrow 0$ and $\|f(u) - f(v)\|_\infty \leq \omega(\|u - v\|_\infty)$. At*

each graph vertex, compute a rational approximation $\tilde{f}(u)$ within $\sigma > 0$, and let $\tilde{r}_N$ be the bottleneck formed from $\|v - \tilde{f}(u)\|_\infty$. Then

$$\max\{0, \tilde{r}_N - \sigma - h_N - \omega(h_N)\} \leq \rho_f(x, H) \leq \tilde{r}_N + \sigma. \quad (25.4)$$

Thus the critical radius remains a computable real in this setting.

Proof. The rounding proof replaces Lh_N by $\omega(h_N)$. At fixed vertices, each approximate edge weight differs from the true edge weight by at most σ . A maximum along a path and then a minimum over paths preserve that additive error bound. Combine the two estimates. $\square$

This extension does *not* preserve every exact-acceptance assertion below. For a simple uniform counterexample, let $a_e = 0$ when machine e never halts, and $a_e = 2^{-(T+1)}$ when it halts at time T . The real a_e is uniformly computable: simulate for enough steps to make every later possible value smaller than the requested precision. For the computable constant map $f_e(t) = a_e$, source 1, and target $\{0\}$, exact acceptance is precisely nonhalting. Rational piecewise-affine formulas avoid this closed-target equality obstruction because every finite iterate is exactly rational.

26 Finite cuts as infinite-time safety proofs

For a finite complete weighted directed graph with source s and nonempty target set T disjoint from s , an admissible cut is a set S with $s \in S$ and $S \cap T = \emptyset$. Its bottleneck is

$$\kappa(S) = \min_{u \in S, v \notin S} w(u, v).$$

The minimum exists because S and its complement are nonempty.

Theorem 26.1 (Bottleneck/cut duality). *If r is the minimum path bottleneck from s to T , then*

$$r = \max_{S \text{ admissible}} \kappa(S). \quad (26.1)$$

An optimal cut is the set of vertices reachable from s using only edges of weight strictly less than r .

Proof. Every source-to-target path crosses every admissible cut. Its largest edge weight is therefore at least $\kappa(S)$, giving $r \geq \max_S \kappa(S)$. Let S be the set reachable below r . It cannot meet T , since that would produce a path with bottleneck strictly below r . Every edge from S to its complement has weight at least r , or its endpoint would also be reachable. Consequently $\kappa(S) \geq r$, which proves equality. This reasoning also covers $r = 0$, when there are no negative-weight edges. $\square$

Applying [theorems 25.1](#) and [26.1](#) gives the certificate interpretation

$$\kappa(S) \geq \eta > Ch_N \implies \rho_f(x, H) \geq \eta - Ch_N > 0. \quad (26.2)$$

The cut excludes paths of *every* length. This is the step that replaces an infinite-time universal assertion by a finite object.

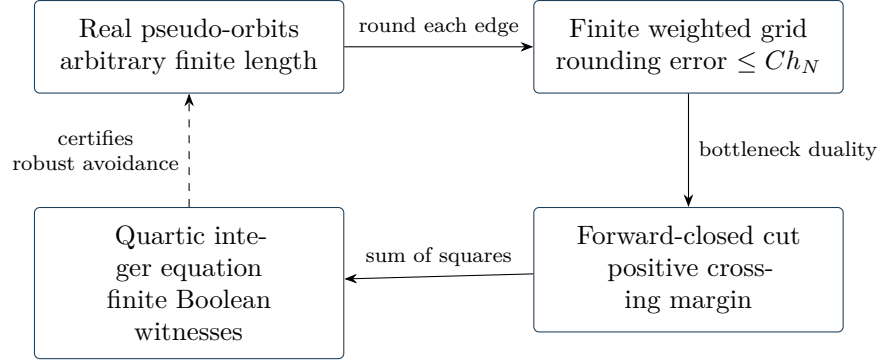


Figure 1: The certificate is a finite obstruction to all perturbed paths, not a single successful computation trace. The polynomial variables encode the cut; they are not the real state coordinates.

26.1 An explicit quartic polynomial

Set

$$\lambda_N = \frac{L+1}{N} = 2Ch_N, \quad E_N = \{(u, v) \in V_N^2 : w_N(u, v) < \lambda_N\}.$$

Introduce one variable b_v for each vertex. Define

$$\begin{aligned} Q_N(b) = & (b_x - 1)^2 + \sum_{t \in T_N} b_t^2 + \sum_{v \in V_N} (b_v(b_v - 1))^2 \\ & + \sum_{(u,v) \in E_N} (b_u(1 - b_v))^2. \end{aligned} \quad (26.3)$$

After the finite graph is computed, this is an ordinary integer polynomial of degree at most four. Its coefficients contain no real parameters. All rational comparisons have already been performed while constructing E_N .

Theorem 26.2 (Complete quartic safety certificates). *For each mesh N ,*

$$\exists b \in \mathbb{N}^{V_N} \ Q_N(b) = 0 \iff r_N \geq \lambda_N. \quad (26.4)$$

Every such zero certifies $\rho_f(x, H) \geq (L+1)/(2N) > 0$. Conversely,

$$\rho_f(x, H) > 0 \iff \exists n \geq 0 \ \exists b \in \mathbb{N}^{V_{D^{2n}}} \ Q_{D^{2n}}(b) = 0. \quad (26.5)$$

The same zero-set assertions hold if witnesses are allowed to range over $\mathbb{Z}$.

Proof. An integer sum of squares is zero exactly when every square is zero. The third group of terms forces $b_v \in \{0, 1\}$. The first two groups put the source in the set $S = \{v : b_v = 1\}$ and all targets outside it. The final group says that no edge of weight below λ_N leaves S . Thus S is an admissible cut with crossing weights at least λ_N , and [theorem 26.1](#) gives $r_N \geq \lambda_N$.

Conversely, if $r_N \geq \lambda_N$, let S consist of vertices reachable using edges below λ_N . No target is in S , and no such edge leaves S . Its indicator vector makes every term of Q_N vanish.

The lower bound follows from $\rho \geq r_N - Ch_N \geq \lambda_N - Ch_N = Ch_N$. If $\rho > 0$, choose n so large that $\lambda_{D^{2n}} \leq \rho$. Then $r_N \geq \rho \geq \lambda_N$, and a certificate exists. This proves completeness. $\square$

There are $M = |V_N|$ variables and at most $1 + |T_N| + M + M^2$ square terms. These are transparent size bounds, not optimized ones. A bit vector, together with the mesh, is a finite witness for an unbounded-time statement. The strict inequality defining E_N is part of the exact equivalence in (26.4); equality-weight edges must not be silently inserted.

26.2 From the explicit family to a fixed Diophantine equation

Let $e \mapsto x_e$ be a primitive-recursive rational input encoding for a fixed f, H, L, D . The number of vertices, the rational evaluations, the comparisons forming E_N , and the Boolean cut test are primitive recursive in their natural-number codes. For clarity, one can encode a cut by a bit mask less than 2^M . All graph loops are bounded by primitive-recursive integers, even though those integers grow very quickly.

Theorem 26.3 (Fixed-polynomial compilation). *The set $\{e : \rho_f(x_e, H) > 0\}$ is computably enumerable. Consequently there are fixed k and $P_{\text{Safe}} \in \mathbb{Z}[X, Y_1, \dots, Y_k]$ with*

$$\rho_f(x_e, H) > 0 \iff \exists w \in \mathbb{N}^k P_{\text{Safe}}(e, w) = 0. \quad (26.6)$$

For rational piecewise-affine f and rational boxes H , exact acceptance also has a fixed Diophantine representation P_{Hit} .

Proof. Enumerate meshes and check whether the finite below-threshold graph has a source-to-target path. Halt when it does not. By theorem 26.2, this semidecides exactly positive radius. Alternatively, enumerate mesh/bit-mask pairs and run the finite certificate checker. Apply (1.1) to this computably enumerable predicate. For exact acceptance, iterate the rational map exactly and test the rational target inequalities at every time; this is a semidecision procedure. Apply MRDP again. $\square$

There are two different levels of explicitness here. Formula (26.3) is a fully specified polynomial for each finite grid, with varying arity. Formula (26.6) is a fixed-arity existence theorem obtained from the finite-checker semidecider and MRDP. This article does not expand that second polynomial, establish its numerical degree or witness count, or claim single-fold or finite-fold representability. In particular, one may not transfer the word “quartic” and the count M directly from the first level to the second.

27 A three-dimensional continuous computer

We now construct the map used in the sharpness results. The construction is generic: a finite binary Turing-machine transition table determines a finite rational piecewise-affine map. Choosing a single universal interpreter table produces one fixed map. An expanded universal transition table is not needed for the existence proof, but producing and checking one is a separate implementation milestone.

27.1 The binary machine and its rational tape coordinates

Take a deterministic binary Turing machine with $m \geq 3$ states, numbered $0, \dots, m-1$. Its start state s , accepting state h , and rejecting state r are distinct. Every other state has an instruction

$$(j, a) \mapsto (j', b, \text{move}), \quad a, b \in \{0, 1\}, \quad \text{move} \in \{L, R, S\}.$$

The tape is two-sided, is indexed relative to the current head, and has blank symbol zero. Let a_i be the bit at relative location i . Define

$$u = \sum_{k \geq 1} \frac{2a_{-k}}{3^k}, \quad v = \sum_{k \geq 0} \frac{2a_k}{3^{k+1}}. \quad (27.1)$$

Both coordinates lie in the middle-third Cantor set. A finitely supported tape gives rational coordinates. On encoded tapes, the leading bit of a coordinate is identified by the disjoint closed intervals

$$J_0 = [0, 1/3], \quad J_1 = [2/3, 1].$$

There is no ambiguity between these two intervals, including at endpoints.

Let $c = a_{-1}$ be the bit immediately to the left of the head and $a = a_0$ the scanned bit. Writing b and moving gives

$$R: \quad u' = (2b + u)/3, \quad v' = 3v - 2a, \quad (27.2)$$

$$L: \quad u' = 3u - 2c, \quad v' = v/3 + 2c/3 + 2(b - a)/9, \quad (27.3)$$

$$S: \quad u' = u, \quad v' = v + 2(b - a)/3. \quad (27.4)$$

These are affine formulas on each digit rectangle $J_c \times J_a$.

Lemma 27.1 (Tape update on complete rectangles). *Equations (27.2)–(27.4) simulate the indicated tape operations on encoded tapes. In addition, each maps the entire closed rectangle $J_c \times J_a$ into $[0, 1]^2$, not merely its Cantor subset.*

Proof. Write $u = (2c + u_1)/3$ and $v = (2a + v_1)/3$, where $u_1, v_1 \in [0, 1]$. For a right move, the new left stack begins with b and then the old left stack, while the new right stack is the old right tail: this gives $u' = (2b + u)/3$ and $v' = v_1$.

For a left move, the left tail becomes u_1 . The new right stack begins with c , then b , then the old right tail. Its value is $2c/3 + 2b/9 + v_1/9$, which is precisely (27.3). The range is contained in $[0, 2/3 + 2/9 + 1/9] = [0, 1]$. For a stationary move, replacing the first right digit gives $v' = (2b + v_1)/3$. The remaining range assertions follow at once. $\square$

27.2 Packing the control state into the left coordinate

Set

$$\beta = \frac{1}{3m}, \quad \alpha_j = (3j + 1)\beta, \quad I_j = [\alpha_j, \alpha_j + \beta]. \quad (27.5)$$

Instead of retaining j and u as separate real coordinates, use

$$\xi = \alpha_j + \beta u. \quad (27.6)$$

The state intervals are separated: the gap between consecutive I_j is 2β . Consequently the encoded state can be read from ξ without confusing two different control states.

The three coordinates of the continuous machine are now $(\xi, v, z) \in [0, 1]^3$. The last coordinate z is an *emergency coordinate*; the exact simulation starts with $z = 0$ and keeps it zero until a terminal state is reached. Tiny positive values of z , however, can eventually trigger acceptance while the interpreter is still running.

Let

$$I_j^+ = [\alpha_j - \beta/2, \alpha_j + 3\beta/2], \quad p_j = (\alpha_j + \beta/2, 1/2, 1/2).$$

The enlarged terminal intervals I_h^+, I_r^+ remain inside $(0, 1)$ and are disjoint from each other and from all other state intervals. Define the accepting target box

$$H = I_h \times [0, 1] \times [0, 1]. \quad (27.7)$$

It is fixed, rational, closed, and full-dimensional. The accepting sink p_h lies in its interior relative to the cube.

27.3 Prescribed boxes and the emergency mechanism

Specify the map on the following finitely many closed rational boxes.

Terminal slabs. For $j = h, r$, set

$$F(\xi, v, z) = p_j \quad \text{whenever} \quad \xi \in I_j^+. \quad (27.8)$$

Here v, z range over the whole unit interval. The enlarged rejecting slab, not merely a single rejecting point, will provide robust rejection.

High emergency coordinate. For each active state $j \notin \{h, r\}$, set

$$F(\xi, v, z) = p_h \quad \text{on} \quad I_j \times [0, 1] \times [2/3, 1]. \quad (27.9)$$

Low emergency coordinate and valid leading digits. For each active j , each $c, a \in \{0, 1\}$, and the instruction $(j, a) \mapsto (j', b, \text{move})$, prescribe

$$F(\alpha_j + \beta u, v, z) = (\alpha_{j'} + \beta u', v', 3z) \quad \text{on} \quad u \in J_c, v \in J_a, z \in [0, 1/3], \quad (27.10)$$

where (u', v') is given by (27.2), (27.3), or (27.4). Since $u = (\xi - \alpha_j)/\beta$, this is affine in (ξ, v, z) on the prescribed box.

Every prescribed box maps into the cube by lemma 27.1. No two distinct prescriptions conflict. The terminal slabs are separated from active-state bands; low and high emergency intervals are separated by a gap; and the two digit intervals are separated by a gap. In particular, there is room to fill in all unprescribed regions continuously.

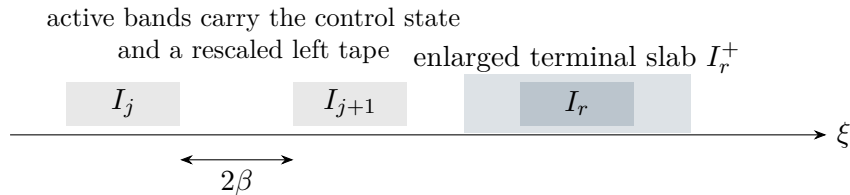


Figure 2: Schematic state packing. Each base interval has width β ; the terminal interval is enlarged by $\beta/2$ on either side. The drawing is schematic rather than a complete partition for a particular interpreter.

27.4 A constructive global extension

Lemma 27.2 (Continuous rational extension with an explicit bound). *The prescriptions (27.8)–(27.10) extend to a continuous rational piecewise-affine map $F : [0, 1]^3 \rightarrow [0, 1]^3$. One such extension has global sup-norm Lipschitz bound*

$$L = 9m + 6. \quad (27.11)$$

The extension can be computed from the finite transition table using rational arithmetic alone.

Proof. For the ξ axis, use all points $0, 1$ and $\alpha_j + k\beta/3$ for $j = 0, \dots, m-1$ and $k = 0, 1, 2, 3$, together with the endpoints of I_h^+ and I_r^+ . For each of the v, z axes use $0, 1/3, 2/3, 1$. Every prescribed box is a union of cells of this rectilinear grid.

At each grid vertex belonging to a prescribed box, assign the prescribed rational image. The disjointness just proved makes the assignment unambiguous. Assign p_r to every remaining vertex. Subdivide each rectangular cell by the same coordinate-ordered Kuhn triangulation: after normalizing its coordinates to a unit cube, use the six regions specified by an ordering of those coordinates. Interpolate the vertex values affinely on each simplex. Adjacent simplices and adjacent rectangular cells agree on their common faces, so the result is globally continuous. Convexity of the cube keeps every interpolated value in the cube. On any prescribed box, affine interpolation reproduces its prescribed affine formula.

Every interval in the ξ grid has length at least $\beta/3 = 1/(9m)$; the other two axis intervals have length $1/3$. Along the chain of vertices of a Kuhn simplex, successive vertices differ in exactly one coordinate. Hence each column of its affine linear part is an image difference divided by that coordinate's interval length. Each scalar image difference has absolute value at most one. Every row-sum norm is therefore at most $9m + 3 + 3 = 9m + 6$. The segment-subdivision argument following (24.1) makes this a global bound. $\square$

This extension is not claimed to be differentiable, reversible, injective, volume-preserving, or a polynomial map. Its continuity on the entire cube, including off-encoding states and guard boundaries, is part of the proved statement. Merely writing a discontinuous guarded update would not be enough for the robustness argument.

27.5 Exact simulation and finite input encoding

If the initial state is $(\alpha_s + \beta u, v, 0)$ with a finite encoded tape, then every active exact iterate satisfies (27.10), and its third coordinate remains zero. The first two coordinates perform exactly the binary machine computation. Entering state h places the image in H ; entering state r places it in the rejecting band, whose next image is p_r .

For a concrete primitive-recursive input coding, let $b_1 \dots b_k$ be the ordinary binary expansion of n (use $k = 1, b_1 = 0$ for zero), and take the self-delimiting word

$$\text{word}(n) = 1^k 0 b_1 \dots b_k 1. \quad (27.12)$$

These words are prefix-free and end in a one. Place the word starting under the head, put blanks elsewhere, and define

$$\iota(n) = \left(\alpha_s, \sum_{i=0}^{\ell(n)-1} \frac{2 \text{word}(n)_i}{3^{i+1}}, 0 \right). \quad (27.13)$$

The encoding is injective and primitive recursive as a rational map. Its values lie outside H . No uncomputable real parameter is present.

28 The universal trichotomy and its quantitative proof

Fix an effective enumeration $(W_a)_{a \in \mathbb{N}}$ of the computably enumerable subsets of $\mathbb{N}$. Fix also a primitive-recursive pairing bijection on natural numbers, with primitive-recursive decoding; the usual Cantor pairing suffices. We use $\langle a, b, e \rangle = \langle a, \langle b, e \rangle \rangle$.

Choose a *single* deterministic binary universal interpreter that, on input $\langle a, b, e \rangle$, dovetails the two semidecidings for $e \in W_a$ and $e \in W_b$. It accepts if the first succeeds first, rejects if the second succeeds first, and uses a fixed tie convention. When W_a and W_b are disjoint the tie convention is immaterial. A binary, one-tape, two-sided interpreter exists by the usual finite-alphabet and multitape simulations; those changes only replace one finite table by another. Its finite number m of states is fixed from now on.

Apply [section 27](#) to this table, and fix its map F , target H , input encoding ι , and bound $L = 9m + 6$ once and for all.

Theorem 28.1 (Universal three-dimensional trichotomy). *For every disjoint computably enumerable pair $(A, B) = (W_a, W_b)$ and every $e \in \mathbb{N}$, put $x_e = \iota(\langle a, b, e \rangle)$. Then*

$$e \in A \iff \text{Hit}(x_e), \quad (28.1)$$

$$e \in B \iff \rho_F(x_e, H) > 0, \quad (28.2)$$

$$e \notin A \cup B \iff \text{Frag}(x_e). \quad (28.3)$$

The map and target do not depend on a, b, e ; only the rational initial point changes. Moreover, on every $e \in B$, all sufficiently small perturbed trajectories enter the enlarged rejecting slab after finitely many steps and remain there.

We prove a quantitative rejection estimate and the emergency mechanism first.

Lemma 28.2 (Positive margin for a rejecting computation). *Suppose the exact interpreter first enters rejecting state r at time $T \geq 1$. Put $S_T(L) = \sum_{j=0}^{T-1} L^j$. No pseudo-orbit whose per-step error is at most*

$$\varepsilon_T = \frac{\beta}{4S_T(L)} \quad (28.4)$$

can hit H . Every such pseudo-orbit is in the enlarged rejecting slab at time T and stays there afterward. In particular, $\rho_F(x, H) \geq \varepsilon_T$.

Proof. Let $y_t = F^t(x)$ be the exact orbit and let x_t be a pseudo-orbit from the same source. Up to time T , global Lipschitz continuity gives

$$\|x_{t+1} - y_{t+1}\|_\infty \leq \varepsilon_T + L \|x_t - y_t\|_\infty.$$

Induction yields $\|x_t - y_t\|_\infty \leq \varepsilon_T S_t(L) \leq \beta/4$ for $1 \leq t \leq T$.

Before and at rejection, every exact packed coordinate is in a state interval other than I_h . Its distance from I_h is at least 2β . Thus no x_t for $t \leq T$ is in H . At time T , the exact packed coordinate belongs to I_r , so the perturbed one is in I_r^+ , because the enlargement has margin $\beta/2$ and the error is at most $\beta/4$.

On the entire rejecting slab, F is the constant p_r . Since $\varepsilon_T \leq \beta/4$, the next perturbed packed coordinate remains within $\beta/4$ of the center of I_r and hence within I_r^+ . This argument repeats forever and the slab is disjoint from H . It is important here that the terminal rule is constant for all values of the other two coordinates. $\square$

Lemma 28.3 (Arbitrarily small emergency acceptance). *If the exact interpreter never enters either terminal state, then $\rho_F(x, H) = 0$. If it first rejects at time $T \geq 2$, then*

$$\rho_F(x, H) \leq 2 \cdot 3^{-(T-1)}. \quad (28.5)$$

Proof. In an infinite active computation, take the first exact step, but change only the new emergency coordinate from zero to $2 \cdot 3^{-N}$, where $N \geq 1$. The cost of this single perturbation is $2 \cdot 3^{-N}$. Leave every subsequent step unperturbed. While $z \leq 1/3$, the first two coordinates continue the exact simulation and z is multiplied by three. At time N the emergency coordinate is $2/3$; the next step uses (27.9) and reaches p_h . Thus arbitrarily small costs suffice.

For a computation first rejecting at time $T \geq 2$, use the same construction with $N = T - 1$. At time $T - 1$ the interpreter is still active and $z = 2/3$. At time T , the emergency rule accepts instead of taking the rejecting instruction. The total path cost is the single initial perturbation $2 \cdot 3^{-(T-1)}$. $\square$

Proof of theorem 28.1. On exact encoded states the interpreter simulation is correct. If $e \in A$, the interpreter accepts, so $\text{Hit}(x_e)$ holds and the radius is zero. If $e \in B$, it rejects after a finite time, and lemma 28.2 gives positive radius and robust entry into the rejecting slab. If e belongs to neither set, the interpreter remains active forever. It never exactly accepts, while lemma 28.3 gives radius zero. These mutually exclusive cases exhaust all inputs and prove all three equivalences. $\square$

Corollary 28.4 (The time–radius sandwich). *For a first rejection at time $T \geq 1$,*

$$\frac{\beta}{4S_T(L)} \leq \rho_F(x, H) \leq \min\{1, 2 \cdot 3^{-(T-1)}\}, \quad \beta = \frac{1}{3m}, \quad L = 9m + 6. \quad (28.6)$$

Proof. The lower bound is lemma 28.2. For $T \geq 2$, combine lemma 28.3 with the universal one-step upper bound $\rho \leq 1$. For $T = 1$, that one-step bound gives the stated right side directly. $\square$

The constants are deliberately conservative. Their significance is the explicit dependence on the actual rejection time, not numerical optimality. A known lower bound $\rho \geq \delta > 0$ implies, for $T \geq 2$,

$$T \leq 1 + \log_3(2/\delta). \quad (28.7)$$

Thus a certified positive margin carries hidden information about how soon rejection must have occurred. It is not surprising that no computable inputwise positive margin can cover every rejecting input.

29 Arithmetic classification and limits of finite certificates

For the one fixed universal map, define sets of natural codes

$$\mathcal{H} = \{n : \text{Hit}(\iota(n))\}, \quad \mathcal{S} = \{n : \text{Safe}(\iota(n))\}, \quad \mathcal{F} = \{n : \text{Frag}(\iota(n))\}.$$

These definitions apply to all codes, even those decoding to overlapping pairs of enumerable sets. Their trichotomy follows from the dynamics, not from the disjoint-pair promise used for the universal realization theorem.

Theorem 29.1 (Exact computability classification). *The sets $\mathcal{H}$ and $\mathcal{S}$ are disjoint Σ_1^0 -complete sets. Their complement $\mathcal{F} = \mathbb{N} \setminus (\mathcal{H} \cup \mathcal{S})$ is Π_1^0 -complete. Completeness holds under primitive-recursive many-one reductions.*

Proof. Both positive sets are computably enumerable by [theorem 26.3](#). Let K_0 be a standard computably enumerable-complete halting set, with index k , and let o index the empty set. The map $e \mapsto \langle k, o, e \rangle$ reduces K_0 to $\mathcal{H}$, while $e \mapsto \langle o, k, e \rangle$ reduces K_0 to $\mathcal{S}$.

The complement of a union of two computably enumerable sets is co-computably enumerable. On the first slice, $e \notin K_0$ is equivalent to $\langle k, o, e \rangle \in \mathcal{F}$, giving Π_1^0 -hardness. All displayed coding maps are primitive recursive. $\square$

29.1 Two existential polynomials and one unavoidable negation

Corollary 29.2 (The Diophantine trichotomy). *There are fixed finite integer polynomials $P_{\mathcal{H}}, P_{\mathcal{S}}$ such that*

$$\begin{aligned} n \in \mathcal{H} &\iff \exists u \in \mathbb{N}^k P_{\mathcal{H}}(n, u) = 0, \\ n \in \mathcal{S} &\iff \exists v \in \mathbb{N}^\ell P_{\mathcal{S}}(n, v) = 0. \end{aligned}$$

There is no ordinary existential Diophantine representation of $\mathcal{F}$ over natural witnesses. It does, however, have the fixed universal nonvanishing representation

$$n \in \mathcal{F} \iff \forall u \in \mathbb{N}^k \forall v \in \mathbb{N}^\ell P_{\mathcal{H}}(n, u)P_{\mathcal{S}}(n, v) \neq 0. \quad (29.1)$$

Proof. Apply MRDP to the two computably enumerable sets. If $\mathcal{F}$ had an existential Diophantine representation, enumerating natural witness tuples would make it computably enumerable. Since it is already co-computably enumerable and is Π_1^0 -complete, this would decide the halting problem. The product in (29.1) is nonzero for every pair of tuples exactly when neither factor has a zero; the sets of tuples are nonempty, including when a witness dimension is zero. $\square$

The universal quantifier in (29.1) cannot be removed by the methods here. It expresses a genuine computability obstruction, not simply a missing clever encoding. This is a precise boundary for the proposed Diophantine treatment of unconventional computation: robust rejection has finite witnesses, but fragile divergence does not.

29.2 A stronger obstruction: no total separator

The simultaneous realization of arbitrary disjoint computably enumerable pairs gives more than separate completeness of the two positive outcomes.

Theorem 29.3 (Recursive inseparability). *There is no total computable Boolean classifier that outputs 1 on every $n \in \mathcal{H}$ and 0 on every $n \in \mathcal{S}$, even when either answer is allowed on $\mathcal{F}$.*

Proof. We first give the standard diagonal pair explicitly. Let (φ_e) be an effective enumeration of partial computable functions and set

$$A_* = \{e : \varphi_e(e) \downarrow = 0\}, \quad B_* = \{e : \varphi_e(e) \downarrow = 1\}.$$

These sets are disjoint and computably enumerable. Suppose a computable set C contained A_* and avoided B_* . Let d be an index for its total characteristic function χ_C . If $\chi_C(d) = 0$, then $d \in A_* \subseteq C$, a contradiction. If $\chi_C(d) = 1$, then $d \in B_*$ and hence $d \notin C$, again a contradiction. Thus no such C exists.

Use [theorem 28.1](#) for (A_*, B_*) . Composing the proposed total classifier with its primitive-recursive input coding would give exactly such a separator C . Therefore the classifier cannot exist. $\square$

An algorithm may still run the two semidecision procedures in parallel and halt on every input promised to be in $\mathcal{H} \cup \mathcal{S}$. The theorem says that this partial promise procedure has no total computable extension which remains correct on both positive outcomes.

29.3 Where decidability returns

Proposition 29.4 (A family without fragile cases). *For a computable input family of a fixed rational piecewise-affine map, suppose every input is either exactly accepting or robustly avoiding. Then exact acceptance on that family is decidable.*

Proof. Dovetail exact orbit simulation with enumeration of finite safety certificates. The two outcomes cannot both occur, and the assumption ensures that one semidecision halts. Its identity decides the answer. $\square$

This is a specialized form of the known robustness/decidability phenomenon [17]. The new universal construction shows how the missing third case can have full co-enumerable complexity, rather than being a negligible numerical inconvenience.

30 How the uncomputability hides in the noise scale

Restrict the universal map to the slice $(A, B) = (\emptyset, K_0)$, and write $\rho(e) = \rho_F(\iota(\langle o, k, e \rangle), H)$. Then $\rho(e)$ is a uniformly computable real in $[0, 1]$ and

$$\rho(e) > 0 \iff e \in K_0.$$

For $e \in K_0$, let $T(e)$ be the first rejection time of the fixed interpreter on that encoded input. These are interpreter times, not the step counts of the original machine whose halting is being tested.

Theorem 30.1 (No computable positive gap). *There is no total computable $g : \mathbb{N} \rightarrow \mathbb{N}$ such that*

$$e \in K_0 \implies \rho(e) \geq 2^{-g(e)}. \quad (30.1)$$

Indeed, for every total computable g , infinitely many $e \in K_0$ violate (30.1).

Proof. If such a bound existed, compute an interval for $\rho(e)$ of length less than $2^{-g(e)}/3$. Under the dichotomy $\rho(e) = 0$ or $\rho(e) \geq 2^{-g(e)}$, the interval distinguishes the two cases and decides K_0 . This contradicts undecidability.

If there were only finitely many violations, hardcode their membership answers and use the same decision procedure on all other inputs. A finite set is computable, irrespective of whether we have effectively found its list. This would again contradict undecidability. $\square$

30.1 A Busy Beaver-type envelope

For $n \in \mathbb{N}$, define two finite extrema with default values:

$$B(n) = \max(\{T(e) : e \leq n, e \in K_0\} \cup \{1\}), \quad (30.2)$$

$$\mu(n) = \min(\{\rho(e) : e \leq n, e \in K_0\} \cup \{1\}). \quad (30.3)$$

Thus $B(n)$ is finite and $\mu(n) > 0$ for every n , even when no relevant machine halts.

Proposition 30.2 (Quantitative finite-input envelope). *For every n ,*

$$\frac{\beta}{4S_{B(n)}(L)} \leq \mu(n) \leq \min\{1, 2 \cdot 3^{-(B(n)-1)}\}. \quad (30.4)$$

The function B has no total computable upper bound. The positive function μ has no total computable strictly positive rational lower bound.

Proof. The lower bound follows from [corollary 28.4](#), the monotonicity of $S_T(L)$ in T , and the added default value one. If $B(n) > 1$, an input attaining that rejection time exists, and its radius gives the upper bound. If $B(n) = 1$, the upper bound is simply $\mu(n) \leq 1$.

A computable upper bound for $B(e)$ would let us simulate the interpreter long enough to decide whether input e is in K_0 . A positive computable rational lower bound for $\mu(e)$ would similarly separate $\rho(e) = 0$ from $\rho(e) > 0$ using its computable approximation. Both conclusions are impossible. $\square$

The phrase “Busy Beaver-type” refers to the absence of any computable upper bound on a finite maximum of halting times. We do not claim eventual domination of every computable function for an arbitrary numbering convention. The precise proved assertion is the one in [proposition 30.2](#).

30.2 Consequences for certificate engineering

Corollary 30.3 (No uniform resolution or witness bound). *No total computable function of the input bounds the first mesh index at which a positive-radius input receives a certificate of [theorem 26.2](#). More generally, no sound, complete decidable finite-certificate system for $\mathcal{S}$ has a total computable inputwise certificate-length bound. For any fixed Diophantine representation of $\mathcal{S}$, the height of its smallest natural witness tuple has no total computable inputwise bound on all positive instances.*

Proof. A computable mesh bound would permit checking all meshes up to that bound, thereby deciding $\mathcal{S}$. A computable certificate-length bound would permit checking all finitely many certificate strings up to that length. A computable natural-coordinate bound for a fixed polynomial would permit checking all witness tuples in a finite box. Each is a decision procedure for a Σ_1^0 -complete set, which is impossible. $\square$

This result does not say that certificates are usually large, or that useful subclasses cannot have excellent bounds. It says that a complete universal pipeline must expose the possibility of unbounded search rather than promise a computable worst-case stopping criterion.

31 Transferring the result to other substrates

A simulation of exact trajectories need not preserve robustness. Extra state coordinates can introduce shortcuts to acceptance which were absent in the original model. The following sufficient condition makes the noise issue explicit rather than hiding it behind the word “simulation.”

Theorem 31.1 (Noise-faithful retract transfer). *Let $f : K \rightarrow K$ and $g : M \rightarrow M$ be maps of metric state spaces. Suppose $E : K \rightarrow M$ and $\pi : M \rightarrow K$ are Lipschitz maps with positive bounds A, B and*

$$\pi E = \text{id}_K, \quad gE = Ef, \quad \pi g = f\pi. \quad (31.1)$$

For $H \subseteq K$, take the target $J = \pi^{-1}(H)$ in M . Then

$$\frac{1}{B} \rho_f(x, H) \leq \rho_g(E(x), J) \leq A \rho_f(x, H). \quad (31.2)$$

Exact acceptance, positive-radius avoidance, and fragile avoidance coincide on corresponding encoded inputs. Thus the three-way arithmetic classification transfers whenever the input encoding into M is effective.

Proof. Lift a path $x_0, \dots, x_T$ in K by E . Its final point belongs to J , and

$$d_M(E(x_{t+1}), g(E(x_t))) = d_M(E(x_{t+1}), E(f(x_t))) \leq A d_K(x_{t+1}, f(x_t)).$$

Taking infima gives the upper bound. Conversely, project any path $y_0 = E(x), \dots, y_T \in J$ by π . Its edge costs are bounded by

$$d_K(\pi(y_{t+1}), f(\pi(y_t))) = d_K(\pi(y_{t+1}), \pi(g(y_t))) \leq B d_M(y_{t+1}, g(y_t)).$$

Taking infima gives $\rho_f \leq B \rho_g$, proving the lower bound. The exact conjugacy on the embedded copy and the definition of J give exact acceptance equivalence. The two positive constants in (31.2) preserve the sign of the radius, and therefore the third outcome as well. $\square$

For example, adding auxiliary coordinates by $g(x, z) = (F(x), q(x, z))$, with $q(x, z_0) = z_0$, the product sup norm, and target $H \times Z$ preserves the radius *exactly*: use $E(x) = (x, z_0)$ and $\pi(x, z) = x$, both with bound one. Invertible affine changes of coordinates preserve the three outcomes and change the quantitative radius by at most the associated norm condition factors. These are concrete transfer results, not claims about arbitrary physical implementations.

31.1 A rational recurrent ReLU realization

There is also a direct neural-circuit interpretation. Let $\text{ReLU}(t) = \max\{0, t\}$. A standard max–min representation theorem states that a continuous piecewise-affine scalar function on a closed convex domain is a finite maximum of finite minima of its affine components [21]. Applying this theorem coordinatewise to F gives

$$F_i(x) = \max_{S \in \mathcal{A}_i} \min_{j \in S} \ell_{ij}(x), \quad i = 1, 2, 3,$$

where all ℓ_{ij} are rational affine functions. This cited representation theorem is an additional external dependency of the neural corollary, not of the main dynamical construction.

Corollary 31.2 (A fixed rational neural update block). *There exists a finite feedforward ReLU network with rational weights and biases and three outputs whose restriction to $[0, 1]^3$ is exactly the fixed map F . Repeatedly applying this one block to its three-dimensional state register realizes the trichotomy of [theorem 28.1](#), with exactly the same critical radius for adversarial perturbations of that state register after each complete network evaluation.*

Proof. Replace each maximum and minimum in the displayed representation by the identities

$$\max(a, b) = b + \text{ReLU}(a - b), \quad \min(a, b) = a - \text{ReLU}(a - b).$$

This produces a finite acyclic circuit of rational affine operations and ReLU gates. Signed intermediate wires can be carried through ordinary ReLU layers by the two nonnegative channels $\text{ReLU}(t)$ and $\text{ReLU}(-t)$, whose difference is t . Layering the finite circuit and carrying necessary wires forward therefore gives a standard finite feedforward network with rational parameters and an affine output layer. Its update function is F on the cube, so both the exact and the specified perturbed state trajectories are identical. $\square$

This is a three-dimensional *recurrent state*, not a claim of three neurons, three layers, or a small hidden network. The construction gives no optimized size bound and the network's full weight list is not extracted in the companion code. Noise at every hidden neuron, fixed parameter errors, stochastic noise, finite floating-point hardware, and continuous-time neural circuits are different models. Their equivalence would require additional estimates or a transfer theorem such as [theorem 31.1](#).

32 A worked example: an unattained critical threshold

32.1 A contraction with an unattained critical threshold

Proposition 32.1 (An explicit radius). *For $f(t) = at$ on $[0, 1]$, $0 \leq a < 1$, source zero, and target $[b, 1]$ with $0 < b \leq 1$,*

$$\rho_f(0, [b, 1]) = (1 - a)b. \tag{32.1}$$

If $0 < a < 1$, no pseudo-orbit with error at most this exact critical value hits the target in finite time. If $a = 0$, the critical value is attained in one step.

Proof. Any error- ε path satisfies $x_{t+1} \leq ax_t + \varepsilon$, so

$$x_t \leq \varepsilon(1 + a + \cdots + a^{t-1}) = \frac{\varepsilon(1 - a^t)}{1 - a}.$$

When $0 < a < 1$ and $\varepsilon \leq (1 - a)b$, this is strictly less than b for finite t . Conversely, when $\varepsilon > (1 - a)b$, the positive-error recursion reaches or passes b after finitely many steps. One can stop at b on that step, or clip at one, without exceeding the allowed error. Thus the infimum is $(1 - a)b$ but is not attained. For $a = 0$, every step is within ε of zero, and the jump to b has exactly cost b . $\square$

Mesh N	Vertices	r_N	Certified interval for ρ	Quartic zero found
4	5	$1/2$	$[5/16, 1/2]$	yes
8	9	$7/16$	$[11/32, 7/16]$	yes
16	17	$13/32$	$[23/64, 13/32]$	yes
32	33	$25/64$	$[47/128, 25/64]$	yes
64	65	$49/128$	$[95/256, 49/128]$	yes

Table 1: A computable critical radius need not be realized by a critical-cost finite path. All intervals contain $3/8$.

For $a = 1/2$, $b = 3/4$, the exact value is $3/8$. The reference implementation gives the following rational enclosures using $L = 1/2$. Every entry was computed with integer fractions, without floating-point tolerances.

At mesh $N = 4$, the vertices are $0, 1/4, 1/2, 3/4, 1$, and $\lambda_N = 3/8$. The cut indicator

$$(b_0, b_1, b_2, b_3, b_4) = (1, 1, 0, 0, 0)$$

is a zero of (26.3). Its least outgoing edge has weight $3/8$, so the cut alone certifies $\rho \geq 3/16$. Optimizing the cut to obtain the actual bottleneck $r_4 = 1/2$ gives the stronger lower bound $5/16$ in the table. The full factored quartic and its witness are supplied as `data/09-continuous-barriers-small_quartic.txt` and `data/09-continuous-barriers-small_certificate.json`.

Part IV

Other substrates, implementation, formalization and questions

33 Other unconventional substrates

The decisive property for the output-law theorem is not a conventional instruction set. It is an effective, nonovercounting presentation of finite terminal histories with computable weights. Universality is needed for the completeness lower bounds, but not for the positive representation theorem. The following applications state their semantics explicitly.

Randomized combinator reduction. A lambda, SK, SKI, or Iota reducer with a specified effective reduction strategy and a fair-coin branching primitive has finite syntactic histories and dyadic history weights. Its first-return law therefore has (4.2). The random primitive and reduction strategy are part of the proposed extension; this is not a claim that the inspected repository already implements that probabilistic language. Adding probabilities does not excuse ambiguities between deterministic evaluation strategy and nondeterministic rewrite choice.

Stochastic fractional programs. One can give a FRACTRAN-style system an explicit rational random dispatcher between finite deterministic fractional programs, preserving each program's ordered divisibility semantics. A transition is then decidable arithmetic, with a rational branch

probability. Finite histories fit our interface. This construction deliberately avoids saying “choose an applicable fraction randomly” without specifying how conflicts, priorities, and normalization are resolved. Deterministic FRACTRAN-to-Diophantine reductions are already present in the literature [3]; the additional object represented here is the probability law and its conditional boundary.

Random local rewriting and cellular systems. A finite description of each current active configuration, an effective finite set of next local choices, and rational transition weights are enough for an effective first-return presentation. Infinite configurations may also be admitted when a finite-history simulation and observable can be computed from their effective descriptions. Merely declaring an arbitrary infinite initial configuration does not meet that requirement.

Continuous-time jump systems. For a countable-state jump process with finitely many outgoing rational rates q_{ij} at each effectively represented state, the next-jump probability is $q_{ij} / \sum_k q_{ik}$ when the denominator is positive. A zero-rate state is treated as absorbing. Events that reach a designated terminal state after finitely many jumps therefore have an effective rational-weight history presentation. This application uses the embedded jump chain. It neither assigns a terminal value to an infinite explosion history nor claims a result about arbitrary deadline-constrained events. Those questions require additional semantics and analysis.

Robust observations of computable continuous trajectories. Suppose a trajectory is supplied with an algorithm for certified evaluation at rational times and is continuous. Reaching an effectively open target at some finite time is semidecidable: enumerate rational times and certified interior balls, and accept once an evaluation is certified inside one. Continuity ensures that an actual open-target visit contains an appropriate rational time, including a one-sided interval at time zero. This yields a Diophantine reachability predicate by MRDP. It says nothing about exact zero crossings with no robust margin, or about differential equations whose solutions are not uniformly computable from the supplied data.

The effective boundary is indispensable. An arbitrary real oracle parameter can encode a non-c.e. set, and an observation that asks for an exact infinite-time limit can move beyond existential arithmetic. Such a model is not covered merely because it is called Turing-complete. The theorem concerns effective computational substrates, not every imaginable extension that contains ordinary Turing machines as a submodel.

Remark 33.1 ([write] Relation to Part III). The paragraph on robust observations of computable continuous trajectories concerns reaching an effectively *open* target in continuous time, which is semidecidable. Part III treats the complementary discrete-time question for *closed* rational targets under adversarial per-step noise: positive critical radius is semidecidable and has complete quartic certificates (Theorem 26.2), exact acceptance is semidecidable, and the remaining noise-fragile outcome is Π_1^0 -complete (Theorem 29.1). Theorem 31.1 states when such a trichotomy transfers to another substrate.

34 Implementation and validation

34.1 Manuscript 08: what the software actually verifies

The bundle contains an exact rational budget evaluator, a symbolic single-fold quartic compiler, a canonical prefix allocator, and an exact small quantum-circuit evaluator. The script `code/08-probabilistic-laws-verify.py` uses a fixed random seed, 20260930, for reproducible sampled tests. It also runs exhaustive tests that do not rely on randomness.

Check	Observed result
Exhaustive binary histories	780 histories, 3,120 initial-mass cases, and 22,220 forced-pivot candidates; exactly one valid pivot candidate precisely in each of the 1,330 feasible cases.
Multi-outcome and large-integer tests	192 sampled schedules, including 32 additional rescalings by 10^{100} ; all exact invariance and feasibility checks pass.
Symbolic quartic	$N = 2, m = 2$: 11 parameters, 33 witnesses, 38 residuals, degree four, 165 expanded monomials. All residuals vanish at the exported witness.
Mutation checks	Increasing each of the 33 witness coordinates separately by one is rejected. This is a finite robustness check, not a search of all wrong tuples.
Quadratic-field signs	25,921 small integer pairs cross-checked with high-precision decimal arithmetic; 200 near-cancelling Pell cases checked by exact identities.
Quantum circuits	100 sampled circuits of 30 gates; 3,000 exact norm checks pass, together with H^2 , HTH , and a Bell-state example.
Prefix allocation	12 dyadic approximation stages of the HTH output law; disjoint allocated mass is 4095/4096.
Operator budgets	Four exact coherent examples and one noncommuting cycle pass symbolic matrix checks.

The two allocated masses at the twelfth HTH stage are 437/512 and 599/4096. Their sum is 4095/4096, leaving exactly 1/4096 unallocated at that finite stage. No claim is made that twelve stages already represent the full irrational law. The script also includes floating-point diagnostics for the sharpness limit in Theorem 7.2; those diagnostics are explicitly labeled illustrative and do not support the exact proof.

The detailed results and software versions are in `data/08-probabilistic-laws-verification.json`. The full finite quartic is supplied both as a residual specification and as an expanded expression. The code uses Python integers and `Fraction` for exact rational arithmetic; SymPy is used for symbolic expansion and the matrix checks.

34.2 Manuscript 07: what the delivered programs do

The package uses only the Python standard library. Matrix entries are `fractions.Fraction`; all certificate computations are integer computations. The principal functions in `code/07-quantum-mortality-quantum-diophantine.py` are:

Function	Contract
<code>elementary_gram</code>	Rational LDL^T and four-square search; returns a $4d$ -row factor and checks its Gram identity.
<code>gram_search</code>	Height-ordered rational search, with an optional cap; succeeds on the included small examples with $d + 3$ rows. A cap failure is not an impossibility result.
<code>compile_instrument</code>	Checks the factor, packs its rows, builds all Kraus operators, and verifies normalization exactly.
<code>build_certificate</code>	Constructs the quadratic residuals with free signed-pair matrix inputs and asserts both symbolic counts.
<code>canonical_assignment</code>	Generates the uniquely determined natural assignment associated with a specified word.
<code>Certificate.expanded</code>	Expands the sum of residual squares into a sparse integer polynomial. Expansion is optional and can be large.

The test command, from the root of the delivered package layout (the README explains how to run it on a copy), is

```
python code/run_checks.py
```

It writes the example matrices, a canonical zero-word witness, the expanded small quartic, and a machine-readable result report.

34.3 Manuscript 07: observed test results

The delivered run passed 32,418 explicit checks. The total is a count of individual assertions, not a count of statistically independent experiments. It includes twelve seeded small random input lists and several deterministic examples. In particular, 3,460 outcome words were checked for the projection formula, exact integer probability formula, and arithmetic gap. The one-step flush implication was checked 12,449 times, using all active final letters when its premise held.

The suite verifies the four-square routine on every integer from 0 through 300, exercises the $(d + 3)$ -row rational search on three small positive forms, and tests the counterexample 7 to a three-square search within a specified finite height cap. The latter computational cap is only a regression test; the global impossibility is proved in Proposition 15.5.

For the uniform certificate with $(D, K, n) = (2, 2, 2)$, symbolic expansion yields 647 monomials and total degree four, with 44 witnesses and 50 residuals. For the dimension-four example and word $(2, 1, 1)$, the generated assignment satisfies all 264 residuals with 245 natural witnesses. Increasing any one of those witness coordinates by one makes the certificate nonzero. These mutation tests supplement, but do not replace, the inductive uniqueness proof.

The complete category counts are in `data/07-quantum-mortality-check_results.json`. A local run took about five seconds; this is not a benchmark for the general rational Gram search. The reference implementation is intentionally transparent rather than optimized.

34.4 Manuscript 09: what was tested

The script `code/09-continuous-barriers-run_checks.py` reproduces 5,660 passing assertions. Numerical calculations use exact rational arithmetic through Python's `fractions.Fraction`. The

deterministic sampling seed is 20260930. The checks include the following groups.

Check group	Assertions
Finite-tape update formulas against independently encoded tape operations	768
Images of whole digit rectangles, including non-Cantor points	600
Bottleneck/cut duality and optimal strict-threshold cuts	240
All simplex Lipschitz bounds in the five-state sample compiler	1,350
Prescribed low boxes, emergency boxes, and terminal slabs	495
Quartic zero test on all assignments in $\{-1, 0, 1, 2\}^5$	1,024
Rejection-time upper schedules and lower-noise samples	656
Other enclosure, cube, emergency, and certificate checks	527
Total	5,660

Table 2: Implementation checks, not a formal verification of the theorems.

The sample machine for the exhaustive simplex check has five states and uses left, right, and stationary instructions. All 1,350 of its simplices satisfy the advertised rational row-sum bound. The tests also include rejecting countdowns of lengths one through nine and emergency perturbations with exponents one through fifteen.

The universal interpreter table itself is not supplied or tested, and finite tests cannot establish universality or an infinite-time theorem. Their role is narrower: detecting formula, indexing, boundary, and implementation errors in the constructive ingredients. The general results rest on the proofs in [sections 25 to 28](#).

Reproducing the artifact. [write] The source’s reproduction instructions (run `python code/run_checks.py results`, with Python 3.10 or newer and no third-party packages, which rewrites the JSON verification report, example certificate, factored quartic and contraction table in the directory named; then build the article twice with pdfLaTeX) apply to the delivered layout. The source uses standard T_EX packages, and no downloaded font files or third-party papers were included in its archive.

34.5 Manuscript 08: a trust-boundary summary

The ordinary proofs establish the general theorems. The finite tests provide evidence about the supplied implementations, including arithmetic signs, indexing, tie resolution, and degree accounting. They cannot replace an inductive proof of the compiler theorem. Conversely, the MRDP existence proof does not certify that the software has generated a universal polynomial: it has not. The delivered archive made this separation explicit in both its README and its machine-readable report; the report README repeats it.

34.6 [write] Rerunning the suites in this report

The programs are shipped byte-identical to the delivery but under prefixed names (`code/07-quantum-mortality-*.py`, `code/08-probabilistic-laws-*.py`, `code/09-continuous-barriers-*.py`), and they import one another by their delivered module names and write to delivered directories (`examples/` and `verification/` for 07, `artifacts/` for 08, the named directory or `results/` for

09). They must therefore be run on a copy with the delivered layout, never in place; the README gives the commands. Manuscripts 07 and 09 need only the Python standard library; 08 needs SymPy (`data/08-probabilistic-laws-requirements.txt` pins `sympy==1.14.0`). Rerun on copies on 30 September 2026 (07 and 09 with Python 3.14.4, 08 with Python 3.13.5 and SymPy 1.14.0), all three suites passed with the recorded counts, and every rewritten output equals the shipped file apart from line endings (the programs write CRLF on Windows) and 07's recorded runtime. 08's report records the interpreter version it runs under, so another interpreter changes that field too.

35 Formalization

[write] What ProveIt has formalized, and what it has not, is stated in Section 1.3. The three sources propose the following plans; none of them has been carried out.

35.1 Manuscript 08: proof obligations suitable for Lean or Rocq

A direct formalization should keep the finite arithmetic layer separate from the analytic probability layer. The following dependency chain avoids requiring a large quantum semantics development before obtaining a useful kernel-checked result.

Finite order and ratio layer. Formalize positive rational rows, least maximizing indices, and the single-step equivalence $h'q_i \geq hp_i (\forall i) \iff h' \geq h \max_i(p_i/q_i)$. Induct on the horizon to prove the optimal product theorem. This portion requires finite sums, ordered fields, and finite maxima, not MRDP.

Certificate layer. Define a finite index type for parameters and witnesses, an explicit residual family of multivariate integer polynomials, and its sum of squares. Prove that natural zeros force every residual to vanish. Prove the selector lemma, the product invariant, and witness uniqueness. Derive the cardinality formulas from the index constructors rather than from an external count. Prove a compiler correctness theorem and use the Python example only as an untrusted certificate generator.

Computability and MRDP layer. Encode the allocation interpreter and the threshold semidecision procedure on naturals. The inspected `Diophantine.mrdp` interface can then produce the finite polynomial existence theorem. A separate circuit lowering theorem establishes degree at most four. This layer should not claim witness uniqueness for a generic MRDP construction.

Measure and limit layer. Prove the prefix-cylinder allocation invariant and its measure identity; then identify limits of cumulative masses with output laws. Prove the variation bounds and the weak-computability realization with its explicit nonuniform existential tail choice. Separate program extraction claims from existence in classical mathematics.

Quantum layer. First verify the integer-linear gate recurrences and the quadratic norm formula, with the state-vector dimension explicit. Then develop measured first-return semantics using unnormalized branch states. The operator-budget theorem can be formalized independently in a finite Hermitian matrix library.

This is a plan, not delivered Lean source or a report of successful Lean compilation. In particular, no guessed theorem names or placeholder `sorry` proofs are included as though they were checked results.

35.2 Manuscript 07: a realistic formalization decomposition

A useful implementation plan separates five proof obligations.

Matrix block algebra. Formalize the block matrices in (16.2), the sum-of-Grams identity, and the projection homomorphism. These are finite algebraic identities. The normalization and word-transfer theorem can first be proved under an explicit hypothesis $B^\top B = Q$; no number theory is needed at that stage.

Rational factor existence. Provide a constructive $4d$ factor from a rational $LDL^\top$ decomposition and four-square certificates. A separate theorem can improve this to $d+3$ rows when the appropriate rational quadratic-form theorem is available. This isolates the classical dependency rather than concealing it inside a tactic call.

Exact word evaluator. Encode an integer matrix by finitely many signed natural coordinates. Matrix multiplication and lookup in a finite generator list are computable operations. Their graphs can be handled directly by polynomial equations or through primitive-recursive graph interfaces. One should explicitly state the chronological multiplication convention in the evaluator theorem.

Quartic certificate correctness. Prove that the residual family is zero exactly when the selected matrices and products are correct. Then prove uniqueness of the natural sign split and count the coordinate families. A sum-of-squares theorem produces the single polynomial. This is a direct finite proof and need not invoke the entire MRDP construction.

Unbounded bridge. Prove that the exact enumerator semidecides $\text{QMort}_{4,7}$ and apply the public MRDP endpoint. An alternative route uses the existing `exactIter_dioph` and `existsExactIter_dioph` interfaces after representing the matrix-update relation [15]. These routes establish the same semantic property but need not preserve the same resource bounds or witness multiplicities.

No Lean source is supplied as “verified” in this package. A schematic statement with missing proofs would not add evidential value. The executable artifacts verify concrete instances and polynomial generation, while the general theorems are proved in the text.

35.3 Manuscript 09: a concrete formalization plan for ProveIt

The highest-value first formalization target is the finite checker and its soundness, not a large universal machine table. The mathematical architecture has clean trust boundaries.

35.3.1 Finite graph and polynomial layer

Represent a finite vertex type, a source, a finite target set, and rational weights. Prove bottleneck/cut duality with finite paths and cycle deletion. Then define (26.3) as an `MvPolynomial` over $\mathbb{Z}$. Its

zero-set proof requires only finite sums, nonnegativity of squares, and the identity $b(b - 1) = 0 \Rightarrow b = 0 \vee b = 1$ over integers.

An external graph search may propose a bit mask. A proved finite checker verifies source membership, target exclusion, and every below-threshold edge. The proof assistant need not trust the search algorithm, its heuristics, or its report of an optimal bottleneck. When a claimed numerical lower radius is used, the checker must also verify the stated rational crossing margin.

35.3.2 Real rounding and geometric layer

Formalize nearest-grid rounding in the sup norm and its compatibility with closed rational boxes. Keep the nongrid source as a separate vertex. The critical-radius infimum requires explicit nonemptiness and bounded-below proofs; do not smuggle in the existence of an optimal real path. The rounded path argument then yields [theorem 25.1](#) independently of its length.

For the map compiler, certify a rational triangulation, agreeing affine restrictions on shared faces, cube-valued vertex data, and row-sum bounds. The critical global fact is continuity across faces. A source file with correct guarded formulas but an unspecified interpolation is not a proof of the required map properties.

35.3.3 Computability and MRDP layer

Choose one canonical coding for rationals, finite tuples, finite tables, and bit masks. Prove the finite graph/cut predicate primitive recursive in those codes. A semidecider then enumerates mesh/certificate pairs. At that point the existing theorem `Diophantine.mrdp` is the correct public interface. Its returned fixed polynomial certifies the encoded predicate, not a different informal notion of safety.

No modification to the MRDP core is mathematically necessary. Possible new modules in a `RobustReachability` namespace would be `Grid`, `CutCertificate`, and `Diophantine`. These names are proposals, not existing or compiled repository declarations.

35.3.4 Universal interpreter layer

The final layer needs a concrete universal binary transition table, a verified finite-tape semantics, and a proof that the chosen input coding matches the interpreter. One must separately establish exact simulation, terminal-slab robustness, emergency timing, and the global map's Lipschitz bound. The universal trichotomy then follows from ordinary dovetailing.

The present archive contained no Lean files, no claimed Lean build receipt, and no axiom audit for these new results. Its value for formalization is a small set of explicit intermediate statements rather than a large untested proof script.

36 Further research questions

The following questions are proposed research directions, not claims that their full priority or open-status history has been settled by the present literature search. They are ordered to put concrete extensions ahead of broader conjectural targets.

Layer	Artifact that can be generated	Fact that must be proved
Finite certificate	Vertex list, edge list, bit mask	The checked cut blocks all low-cost graph paths
Polynomial	Factored sum of squares	Its integer zeros are exactly admissible Boolean cuts
Geometry	Rational simplex table	Continuity, cube invariance, and a global error bound
Semidecider	Mesh/certificate enumeration	Acceptance iff the real critical radius is positive
Universal compiler	Interpreter table and its PWA image	Correct simulation and the three-outcome correspondence

Table 3: Generated data and proved semantics must remain separate.

[write] The questions are printed by source: 08’s twelve in Section 36.1, 07’s ten in Section 36.2 (which the source posed as subsections and are numbered here), and 09’s ten, with its recommended order, in Section 36.3. How they overlap, and which one another source partly answers, is recorded in Section 36.4.

36.1 Manuscript 08: probabilistic laws and normalization

Research question 1 (Kernel-checked unique quartic compiler). Can Theorem 6.1, including invalid-input rejection and the exact count $5Nm + m + 4N + 3$, be formalized and connected to an executable polynomial exporter in ProveIt? A decisive deliverable would be a checked reflection theorem accepting the supplied $N = 2, m = 2$ residual certificate.

The finite positivity and tie-breaking lemmas make this the most sharply specified implementation target. It need not wait for quantum semantics or infinite probability theory.

Research question 2 (Smaller canonical arithmetic certificates). Can the X, Y cross-product witnesses or the selectors be eliminated while retaining degree at most four, natural-witness uniqueness, and a linear-size arithmetic circuit? What trade-off between witness count, degree, and coefficient size is optimal for this specific normalization predicate?

This is a finite compiler optimization problem. A mere MRDP existence argument cannot answer its size or uniqueness requirements.

Research question 3 (Sharp multi-outcome variation budgets). For a specified initial vector p_0 , determine the sharp supremum of $\sum_s \text{TV}(p_{s-1}, p_s)$ under $\sum_s D_\infty(p_{s-1} \| p_s) \leq B$. Does a useful potential formula extend the balanced binary constant $1/2$ to $m \geq 3$, and what paths are asymptotically extremal?

Proposition 7.1 gives the universal upper bound B but Theorem 7.2 shows that it can be substantially nonsharp when the starting law is fixed.

Research question 4 (Uniform high-success realization). What effective additional data are necessary and sufficient to turn the existential high-success realization in Theorem 8.2 into a uniform compiler? Natural input formats include certified tail-variation bounds, an interior interval, or a computable bound on the remaining logarithmic cost.

The present theorem explicitly leaves tail selection nonuniform. A solution should specify its representation of real numbers and its promised inputs; otherwise an apparent uniformization may hide an oracle.

Research question 5 (Exact symbolic infinite histories). For which algorithmically described rational paths can G_∞ be computed, or bounded sharply enough to synthesize a desired success floor? Examples include periodic transformations, linear recurrences, and piecewise-rational paths with summable perturbations.

Finite prefix feasibility is decidable and uniquely certified here. Certifying every prefix with one finite object is an additional mathematical problem, not an automatic consequence of those certificates.

Research question 6 (Operator certificates with controlled multiplicity). Can an explicit quartic natural-witness system certify the optimal factors R_s for algebraic density matrices, including support changes, without uncontrolled multiplicity from eigenvector choices? Can canonical principal minor or characteristic-polynomial data replace arbitrary spectral bases?

The scalar earliest-maximum rule has no immediate analogue for degenerate eigenspaces. Any proposed uniqueness theorem must address that issue rather than merely add spectral witnesses.

Research question 7 (Quantum front ends with structural compression). Which circuit classes admit polynomial-size exact Diophantine certificates that avoid the full 2^n state vector? Candidate interfaces include stabilizer structure, bounded tensor-network width, and a bounded number of non-Clifford gates, with all compression assumptions stated explicitly.

The current state-vector checker is exponential. A useful result would preserve exact zero tests and Born probabilities while making the source of any speedup mathematically visible.

Research question 8 (Interactive and compositional semantics). How much of the projected-root representation survives for interactive processes with adaptive inputs, correlated outputs, or sequential conditioning? Can a compositional representation preserve joint laws rather than only a terminal one-output marginal?

Independent samplers for marginals are not enough. A solution must preserve the common probability space or the appropriate quantum instrument.

Research question 9 (Direct universal substrate polynomials). Can one produce a fully expanded, audited quartic polynomial for a small measured quantum controller or a stochastic fractional language, avoiding a large generic interpreter? Which operation counts can be proved rather than estimated from informal circuit descriptions?

Theorem 4.3 gives a fixed polynomial in principle, not its coefficients or a competitive universal witness count.

Research question 10 (Counting executions without counting arithmetic witnesses). Can canonical finite source certificates be composed into useful resource-bounded counting reductions while preserving projected-event weights? Under what explicit restrictions can one obtain a genuinely parsimonious unbounded representation?

The finite single-fold construction here is not a general unbounded single-fold theorem. Any extension must confront that distinction directly.

Research question 11 (Continuous-time and robust physical semantics). Develop certified first-hit and deadline probabilities for effective jump systems and computable flows, with explicit treatments of explosion, Zeno behavior, exact equality observations, and robustness margins. Which observables remain c.e. threshold events, and which cross the normalization boundary proved here?

The finite-first-return interface identifies a tractable starting point; it does not prescribe a physically preferred semantics for limit events.

Research question 12 (An algebra of probability specifications). Build a specification language that tracks whether a quantity is left-c.e., right-c.e., weakly computable, or computable, and automatically assigns a sound arithmetic quantifier class to each comparison. Can successful normalization-budget certificates improve the inferred class for useful restricted program families?

Such a system would prevent the common mistake of compiling every probabilistic specification to a single existential equation merely because its source language is Turing-complete.

36.2 Manuscript 07: quantum mortality and its certificates

The following are proposed investigations, not assertions that every formulation is a previously catalogued open problem. Establishing priority and the exact state of the literature is itself part of the first stage of each project.

Research question 13 (Smaller rational instruments). Can the same unrestricted zero-word decision problem be shown hard in dimension three, or with fewer than seven outcomes in dimension four? A successful construction must avoid spurious zero products from any added normalization operators. It is not enough to simulate a computation only on a regular language of “well-formed” histories while leaving other outcome words unchecked. The projection-and-idle structure here gives a useful baseline for such an audit.

Research question 14 (Residual-specific Gram compression). For $Q(c) = c^2I - \sum_i M_i^T M_i$, can one always choose rational c so that $Q(c)$ has fewer than $d + 3$ rational Gram rows? Proposition 15.5 does not answer this because it concerns arbitrary positive forms and does not respect the freedom to vary c . An arithmetic analysis of this one-parameter family could improve the quantum resource bound without changing the block argument.

Research question 15 (Height bounds and practical rational factors). The existence proof for a $d + 3$ factor is effective by search, but the search can be enormous. Develop height bounds in terms of the bit length of Q , and compare algorithms based on rational isotropy, lattice reduction, and specialized four-square manipulations. The relevant output complexity is not just row count: the numerator and denominator heights of the Kraus operators affect both file size and the gap in (20.2).

Research question 16 (Choosing the lower blocks to preserve a shortest word). Theorem 16.1 confines the shortest length to a two-point interval. For a known source witness ending in i , equation (17.2) says exactly when no extra letter is needed: B_i must annihilate the preceding product. Can one choose a Gram factor with prescribed annihilation subspaces while retaining the same row budget? For unknown shortest witnesses this interacts with an undecidable search; for specified witnesses it is a concrete constrained quadratic-form problem.

Research question 17 (Smaller explicit Diophantine certificates). The formulas $71n + 32$ and $72n + 48$ are transparent upper bounds, not claimed optima. Eliminating known-zero matrix coordinates, exploiting the block structure, and using the final-active-letter criterion could reduce constants substantially. Such reductions should preserve the distinction between free matrix inputs and existential witnesses and should state whether the word-to-witness bijection survives.

Research question 18 (A complete proof-producing compiler). Implement and certify the entire chain from a binary tag program and input word through integer matrix mortality, rational Gram packing, and polynomial generation. The first useful milestone is an independently checked rational Gram certificate plus a checked block-algebra theorem. A practical artifact should include exact source and target semantics, not merely display a polynomial with the desired informal interpretation.

Research question 19 (The fixed-arity uniqueness frontier). Can a useful restricted class of these instruments admit a fixed-arity Diophantine representation whose witnesses retain a controlled multiplicity? The bounded family already has unique arithmetic witnesses per outcome word. Unbounded trace compression can introduce arbitrarily many choices, and the general single-fold or finite-fold questions must not be claimed solved by the bounded construction. A sensible first target is a clearly specified instrument class with additional growth or normal-form constraints.

Research question 20 (Promise problems robust under perturbation). Study promises with a fixed separation from zero, bounded history length, or restricted coefficient heights. The estimate in Proposition 20.2 supplies an exact baseline for a finite-length promise, but not a length-independent robust undecidability result. A meaningful physical formulation must specify both the norm of the implementation error and what information about the exact reference model is available.

Research question 21 (Decidable signed subclasses). The nonnegative support argument gives one decidable boundary. Other restrictions worth investigating include simultaneous triangularization with arithmetic constraints, low-rank generated semigroups, or common invariant cones with an effectively checkable description. These proposals require care: positivity of operators in the Loewner order is not the same as entrywise nonnegativity, and either condition may fail to survive a change of basis.

Research question 22 (Other exact scalar fields and tensor observables). For a fixed algebraic number field with a chosen embedding into $\mathbb{C}$, exact field arithmetic can be represented by rational coordinate tensors. Determine how the Gram packing changes when conjugation and norms are expressed in those coordinates. The tensor identity (22.3) gives an elementary entry point for scalar matrix-product observables, but physically normalized tensor networks and general quantum channels with several Kraus operators per recorded outcome require separate semantic proofs.

36.3 Manuscript 09: continuous computation

The following questions are concrete continuations of the results. Some are implementation or optimization problems; others may require new mathematical ideas. Their inclusion does not certify that no answer exists in the wider literature. A focused prior-art audit should precede any priority claim.

Research question 23 (Extract a fixed safety polynomial). Can the finite cut checker be compiled through ProveIt's existing machinery into a fully displayed polynomial with an audited degree, variable count, coefficient height, and witness-decoding procedure?

This is the most direct repository contribution. Start with a deliberately large but complete representation, then optimize. The milestone is an end-to-end proof that a zero decodes to a valid real safety statement. A small polynomial for just one grid does not meet the milestone. No single-fold or finite-fold conclusion should be assumed during optimization.

Research question 24 (Reduce the three-dimensional realizer). Can one fixed continuous rational piecewise-affine map of a square realize every disjoint computably enumerable pair with the same exact-acceptance/positive-radius trichotomy? What additional restrictions make such a realization impossible?

The current third coordinate provides an independently adjustable expanding emergency channel. Encoding that channel into the tape without allowing spurious robust rejections is the central challenge. Any lower-dimensional claim must retain a positive margin on every rejecting input and must specify a full continuous extension, not only a symbolic invariant subset.

Research question 25 (Smooth and polynomial realizers). Which regularity classes support the same simultaneous realization theorem on a compact state space: C^∞ maps, analytic maps, or rational polynomial maps?

Smooth generalized-shift constructions suggest an accessible route for the first class. The polynomial and analytic cases cannot simply reuse the constant terminal slabs: a real analytic map constant on a nonempty open set of a connected domain is constant throughout. One would need contracting trapping regions instead of literal plateaus, with a new proof of the positive rejection margin and a computably recognizable accepting target.

Research question 26 (Extract and certify the rational neural circuit). Can the max–min representation in [corollary 31.2](#) be generated as a concrete rational network, with a checked equivalence to the simplex table and useful size bounds?

The elementary max/min-to-ReLU identities supply a baseline. More important than reducing depth initially is keeping a machine-readable correspondence between the network, the PWA dynamics, and the certificate checker. Noise at hidden units should be treated as a separate theorem, not identified with noise at the recurrent state register.

Research question 27 (Quantitative transfers to genuinely different dynamics). Can discrete symbolic, chemical, mechanical, optical, or hybrid simulators admit effective noisy projection/lifting estimates analogous to [\(31.2\)](#)?

Exact Turing completeness alone is insufficient. A useful positive result would establish both directions of perturbation transfer, perhaps with a controlled time rescaling, and specify which target is observed. A useful negative result would identify a substrate for which an exact simulator necessarily destroys every positive rejection margin.

Research question 28 (Replace dense grids by small inductive regions). Can positive-radius inputs be certified by adaptive polyhedral partitions, interval boxes, or algebraic invariants with substantially smaller witnesses than the complete grid?

The current grid proves completeness but is deliberately expensive. A natural certified search starts with a coarse cut, refines only ambiguous boundary cells, and carries a local Lipschitz bound for each relevant cell. The theorem to preserve is completeness for every positive margin, not merely success on benchmark examples. The no-bound theorem excludes a computable universal worst-case guarantee, but permits powerful subclass-specific bounds.

Research question 29 (Sharp radius–time tradeoffs). Can the lower and upper exponential estimates in (28.6) be made comparable up to fixed constants or a fixed exponent for a universal realizer?

The present lower bound uses a coarse global Lipschitz constant while the upper bound uses the emergency multiplier three. A construction whose simulation and emergency expansion scales are matched could make the smallest robustness radius a quantitatively sharper geometric encoding of halting-time complexity. Any such statement must specify the interpreter and input-length convention.

Research question 30 (Classify exact critical-level reachability). For which subclasses is reachability at error exactly ρ decidable or characterizable, assuming ρ is supplied exactly?

The contraction example shows that equality of the error budget to the critical value need not yield a finite accepting path. Compactness of the state space does not bound the time variable. Useful hypotheses might force a bounded witness length or turn the relevant limiting chain into a finite one. A classification here would complement, rather than repeat, the computability of the value itself.

Research question 31 (Different noise semantics). What becomes of the Diophantine trichotomy for a single initial perturbation, a fixed parameter perturbation, stochastic errors, total-error budgets, or errors confined to specified coordinate directions?

Our fragile acceptance mechanism already uses only one perturbation in one coordinate, but the rejection theorem controls all coordinates at every step. This asymmetry provides a useful starting point. Probabilistic eventual acceptance, almost-sure avoidance, and existential noisy reachability must not be conflated; their quantifier structures can differ substantially.

Research question 32 (Geometry of the fragile boundary). What can be said about the topology, dimension, or effective descriptive complexity of the full set $\{x : \neg \text{Hit}(x), \rho_F(x, H) = 0\}$, beyond the Π_1^0 -complete classification on the encoded inputs?

On regions disjoint from H , the critical radius is Lipschitz by [proposition 25.3](#), so its positive set is open. The coded-input classification does not by itself determine the measure or fractal geometry of the full zero set. Studying that geometry could connect certificate complexity with dynamical notions of chain accessibility.

Recommended order. First formalize the finite graph and quartic checker, then the rounding soundness theorem, and only then extract a fixed MRDP polynomial. In parallel, make one concrete interpreter table and the neural representation explicit. Dimension reduction and polynomial realizations are better treated as separate mathematical projects after these certified baselines exist.

36.4 [write] How the questions overlap

- *Partly answered.* Question 11 (08) asks which observables of computable flows and jump systems remain c.e. threshold events once robustness margins are treated explicitly. For discrete-time rational piecewise-affine maps under adversarial per-step noise, Part III answers the robustness part: positive margin is Σ_1^0 -complete with complete quartic certificates, and noise-fragile nonacceptance is Π_1^0 -complete, hence not existential Diophantine (Theorems 26.2

and 29.1, Corollary 29.2). The question stays open for continuous time, jump processes, deadlines, explosion and stochastic noise, which is also 09's Question 31.

- *One frontier, two forms.* 08's Question 10 and 07's Question 19 both ask for a fixed-arity representation with controlled witness multiplicity; 09 (Section 26) likewise declines to claim it. The sibling report `canonical-diophantine-certificates` of this category shows that a single-fold (finite-fold) universal halting polynomial exists exactly when every computably enumerable set has such a representation, an open problem.
- *Extraction of a fixed polynomial.* 08's Question 9, 07's Question 18 and 09's Question 23 ask for explicit, audited forms of the three different fixed polynomials of Remark 2.1.
- *Smaller explicit certificates.* 08's Question 2 and 07's Question 17 ask the same kind of question for the two different explicit quartic families $F_{N,m}$ and $\mathcal{Q}_{D,K,n}$.
- *Formalization first.* 08's Question 1, 07's milestone in Question 18 and 09's recommended order all put a kernel-checked finite certificate layer before any unbounded statement.

37 Conclusions

37.1 Manuscript 08

The useful extension of Diophantine computation is not just another unconventional universal instruction set. It is a semantics-preserving account of which quantitative assertions admit finite arithmetic witnesses. Effective output laws admit a common projected-root representation, but normalization can raise a strict threshold from Σ_1^0 to Σ_2^0 , even with a success guarantee arbitrarily close to one.

The exact normalization budget supplies a constructive organizing theorem: accumulated mass can follow a prescribed positive conditional path exactly when its product of maximal coordinate ratios fits the available budget. That theorem yields a direct, unique-witness quartic compiler, a sharp binary oscillation bound, an exact weak-computability characterization, and a noncommutative operator-order extension. The explicit finite compiler and tests are ready to serve as a concrete formalization target; the broader universal and quantum results retain their stated existence and semantic boundaries.

37.2 Manuscript 07

The key quantitative step is to distribute rational Gram rows among the original active Kraus operators instead of allocating a large correction block to each one. Exact normalization then costs only $\lceil s/k \rceil$ additional state-space coordinates for an s -row factor. Classical rational quadratic-form theory supplies $s = d + 3$, while an elementary implementation supplies $s = 4d$.

This produces a proof of quantum zero-word undecidability in dimension four with seven rational outcomes, based on the published six-generator matrix theorem. Its witness transformation is unusually tight: erase idle outcomes in one direction, and append one active outcome in the other. The polynomial compiler subsequently records exact signed products, not merely legal paths, with quartic degree and explicit linear-in-length witness counts.

The work therefore extends the repository's Diophantine program toward an interference-based computational substrate without conflating exact mathematics, finite experimental precision, and

formal proof status. The row-packing theorem is ready for independent mathematical review; the elementary branch and the bounded quartic compiler are ready for focused formalization.

37.3 Manuscript 09

The main contribution is not another bare invocation of MRDP on a halting set. It is a finite certificate architecture for a genuinely infinite-time property, together with a universal dynamical construction that proves the architecture’s exact expressive boundary.

Positive critical radius is witnessed by a finite graph cut and hence by an explicit quartic equation at some mesh. A fixed MRDP polynomial exists for the resulting enumerable predicate. One fixed three-dimensional continuous rational map realizes arbitrary disjoint enumerable acceptance and robust rejection sets, while the remaining inputs diverge without any positive noise margin. Its radius is numerically computable with an error bound that does not depend on trajectory length, yet deciding whether the value is positive is maximally hard among enumerable problems. An exact rational neural update block inherits the same statement under the specified output noise semantics.

The result is therefore both constructive and limiting: finite arithmetic witnesses can certify robust nonacceptance of a continuous universal computer, but they cannot provide a complete existential certification of its fragile divergence. The distinction survives even when the numerical value separating the outcomes is a computable Lipschitz function.

A Reproduction notes and algorithmic contracts

A.1 Manuscript 08: using the delivered files

[write] Manuscript 08’s appendix listed the files of its archive. In this report they are:

Shipped path (delivered path)	Purpose
<code>article.tex</code> (08’s <code>article.tex</code> , now merged)	Source of this report; 08’s own PDF is not shipped.
<code>code/08-probabilistic-laws-normalization.py</code> (<code>code/normalization.py</code>)	Exact budget evaluator and residual compiler.
<code>code/08-probabilistic-laws-quantum_exact.py</code> (<code>code/quantum_exact.py</code>)	Exact ring arithmetic and small state vectors.
<code>code/08-probabilistic-laws-prefix_allocator.py</code> (<code>code/prefix_allocator.py</code>)	Canonical labeled prefix allocation.
<code>code/08-probabilistic-laws-verify.py</code> (<code>code/verify.py</code>)	Reproducible test and artifact generator.
<code>data/08-probabilistic-laws-verification.json</code> (<code>artifacts/verification.json</code>)	Test counts, versions, and limitations.
<code>README.md</code>	The report README (08’s delivered README is replaced).

The two files beginning `data/08-probabilistic-laws-normalization_quartic_N2_m2` contain the residual system with example assignment and the full expanded polynomial. The file `data/08-probabilistic-laws-HTH_prefix_allocation.json` records the finite prefix-code example. The source rebuilt its artifacts with `python code/verify.py` from its archive root; the README gives the equivalent command on a copy with the delivered layout. The Python scripts require

Python 3.10 or newer and SymPy. Tests were run with Python 3.13.5 and SymPy 1.14.0. A newly generated report records the versions actually used. The source code uses zero-based coordinate indices; the mathematical exposition uses indices $1, \dots, m$. The earliest-maximum rule is the same ordering in both conventions.

A.2 Manuscript 07: algorithmic contracts and reproduction notes

A.2.1 Rational Gram search

The search routine uses a common-denominator presentation. At height h , it considers denominators $1 \leq q \leq h$ and integer column vectors in $[-h, h]^s$. Candidate columns are first filtered by the equations

$$\|v_j\|^2 = q^2 Q_{jj},$$

and then by the mutual inner products $v_i^\top v_j = q^2 Q_{ij}$. A successful tuple gives $B = q^{-1}(v_1, \dots, v_d)$. Every rational matrix has such a representation for some finite h , so Proposition 15.4 proves termination when $s = d + 3$ and no height cap is imposed. Repeating earlier candidates is inefficient but harmless.

The implementation checks positive definiteness first. A finite height cap raises a timeout-style exception and is explicitly not interpreted as nonexistence. The same routine may be called with a smaller s , but then the general termination guarantee no longer applies.

A.2.2 Polynomial serialization

The expanded sample polynomial is stored as a list of integer coefficients and monomials. A monomial is a sorted list of variable names, with a name repeated according to its exponent. Thus

```
{"coefficient": -2, "monomial": ["x", "x", "y"]}
```

means $-2x^2y$. The file also lists the free input names and existential witness names. The symbolic checker includes the free H^+ , H^- coordinates in the degree computation, which verifies the uniform degree claim rather than only a specialization to constant matrix entries.

A.2.3 Independent checks worth repeating

The most failure-sensitive identities are the completeness equation, the order of multiplication in the word evaluator, the fact that a zero source word may need one extra target letter, and the uniqueness of the natural sign split. These are all tested in the package and proved in the text. A useful independent review should also inspect that the inactive right block columns are zero and that no restriction on the allowable target outcome language is silently being imposed.

[write] The primary artifacts of manuscript 07 were the manuscript source and PDF, the two Python files, and three example JSON files. Here the manuscript is merged into `article.tex`, its PDF is not shipped, and the Python and JSON files are shipped under the prefixed names of the README. The source compiled with a standard pdfL^AT_EX installation, and its test command is `python code/run_checks.py` in the delivered layout. The build does not fetch the repository, run Lean, or require a network connection.

B The exact PWA evaluation recipe

The reference compiler evaluates the global extension without expanding a large list of affine matrices. Locate a point in a rectangular cell with lower corner a , upper corner b , and normalized coordinates $t_i = (x_i - a_i)/(b_i - a_i)$. Choose a deterministic ordering $t_{\sigma(1)} \geq t_{\sigma(2)} \geq t_{\sigma(3)}$. Let $v_0 = a$ and form v_j by changing coordinates $\sigma(1), \dots, \sigma(j)$ to their upper endpoints. Then

$$F(x) = (1 - t_{\sigma(1)})F(v_0) + (t_{\sigma(1)} - t_{\sigma(2)})F(v_1) + (t_{\sigma(2)} - t_{\sigma(3)})F(v_2) + t_{\sigma(3)}F(v_3). \quad (\text{B.1})$$

All four coefficients are nonnegative and sum to one. At ties, the formula agrees on the common simplex face. At rectangular boundaries, it agrees with the adjacent cell formula. This makes endpoint behavior explicit, including points exactly on $1/3$, $2/3$, or a terminal slab boundary.

The Python compiler first checks terminal slab membership at a vertex, then active emergency membership, then active low-coordinate digit membership. Because the relevant prescribed sets are disjoint, this order is a convenient implementation convention rather than a mathematical priority rule. Vertices outside all prescribed sets get the rejecting sink value. The cache stores rational vertex values only.

C A finite-certificate pseudocode contract

A proposed safety certificate contains a mesh index n , an ordering of the vertices consistent with the public grid convention, and a bit for each vertex. An untrusted producer may generate it by any method. A trusted checker can use the following contract.

```

N := D * 2^n
V := all d-dimensional N-grid vertices, plus the exact source if absent
T := grid vertices belonging to a target box
lambda := (L + 1) / N
reject unless every certificate entry is 0 or 1
reject unless the source entry is 1 and every target entry is 0
for each ordered pair (u, v) in V x V:
    if ||v - f(u)||_infinity < lambda:
        reject if bit[u] = 1 and bit[v] = 0
accept, certifying rho >= (L + 1) / (2*N)

```

The inputs f, H, L, D are not arbitrary unchecked text. The checker assumes a validated map description, a valid global Lipschitz bound, a target inside the cube, and a denominator-clearing integer. These form its specification. The companion implementation checks several syntactic conditions but cannot prove an externally supplied Lipschitz bound for an arbitrary Python callable. For maps produced by the TM compiler, [lemma 27.2](#) supplies the bound.

The accepted certificate guarantees safety for every per-step budget strictly below the reported radius bound. This final strictness is not an implementation detail: critical-value attainment can fail, and the radius lower bound alone does not settle the equality case.

D Source and verification notes

The repository was inspected at the pinned commit printed in [Section 1](#). Bibliography entries for repository sources use that commit rather than a moving branch. The references on weakly

computable reals, probabilistic-program hardness, exact Clifford+ T arithmetic, and max-relative entropy provide the established context; the proofs of the specific budget and compiler statements appear in full in this manuscript.

No claim is made that the literature search establishes first publication of every consequence proved here. In particular, general c.e.-to-Diophantine existence, fair-coin representation of lower semicomputable output laws, and the entropy formulas are not advertised as new discoveries. The primary new deliverable of this work is the explicit synthesis of these ideas with the canonical finite normalization certificate, including its verified implementation and its carefully separated infinite semantics.

E Proof audit, provenance, and non-claims

E.1 Dependencies

The core construction and quantitative results use elementary finite graph theory, real infima, rational affine geometry, and the standard existence of a binary universal Turing machine. MRDP is used only to pass from an enumerable finite-certificate predicate to one fixed polynomial. The recurrent ReLU corollary additionally uses the cited max–min representation theorem. Earlier dynamics literature motivates and contextualizes the questions; its qualitative robustness results are not substituted for the proofs of the explicit estimates here.

E.2 Points checked in the mathematical argument

The real infimum need not be attained. The source is never inadvertently rounded. Target rounding works for closed boxes with aligned endpoints, including degenerate ones. The graph is complete, so a bottleneck minimum and finite cut minima exist. Strict versus non-strict edge thresholds are kept separate. Integer polynomial subtraction is signed. The quartic family has varying arity, while the MRDP polynomial has fixed arity.

The Turing update formulas map entire guard boxes into the cube. Terminal slabs have genuine neighborhoods, and the extension is continuous across all faces. The rejected computation has a finite-prefix error estimate followed by an invariant rejecting slab. Emergency timing is indexed so that a first rejection at T can be bypassed at time T , not one step too late. The universal map is fixed before the enumerable pair is selected. The original machine’s halting time is never silently identified with the universal interpreter’s rejection time.

E.3 Limits of the claim

No named classical open conjecture is declared solved. The proposed three-dimensional joint theorem package is presented with proofs, but without an exhaustive novelty certification or independent peer review. No minimal dimension, optimal Lipschitz constant, optimal polynomial degree, optimized network size, or finite-fold MRDP result is asserted. No ordinary polynomial map, smooth flow, chemical network, or physical device is claimed to implement the construction without additional work.

The proofs and code were prepared with ChatGPT for this request. All reported finite tests were actually run on the included code. No Lean verification, repository-wide audit, expanded universal

transition table, extracted ReLU weight list, or expanded fixed MRDP polynomial is included. The archive contains the precise ingredients that were produced, rather than placeholders masquerading as those further artifacts.

F [write] Provenance of this report

This report was built from three manuscripts of batch 60, delivered in the arrival commit `ccfb084ad` and placed in the placement commit `7498484af`. All three are dated 30 September 2026 and pin ProveIt at `f608f1cb3c5be8a736df1328c01b293aabfccf4e`.

No.	Archive	Contribution
07	<code>quantum_diophantine_research</code> (22-page A4 PDF, not shipped)	Rational Gram packing, undecidability for seven rational 4×4 outcomes, bijective bounded-history quartics, the fixed MRDP polynomial for <code>QMort_{4,7}</code> , probability gaps and perturbation bounds, nonnegative decidability; 32,418 exact checks; ten questions.
08 (base)	<code>ProveIt_Diophantine_Quantum_Normalization</code> (29-page Letter PDF, not shipped)	The front matter, Part I, Sections 11, 12 and 33; the explicit single-fold normalization quartic; twelve questions.
09	<code>ProveIt_Diophantine_Barrriers</code> (26-page Letter PDF, not shipped)	Part III: horizon-free radius approximation, cut duality and complete quartic safety certificates, the three-dimensional universal trichotomy and its arithmetic classification, transfer and ReLU corollaries; 5,660 exact checks; ten questions.

Where the merge had to choose:

- Manuscript 08 is the base: its framework (output laws of classical and quantum programs) contains 07’s quantum setting as one kind of program, and 09 is a separate spine. 08’s section order is kept, except that its quantum front end and operator budget open Part II beside 07 rather than closing Part I.
- 07’s and 09’s introductions are split: their repository paragraphs are printed in Section 1, the rest opens their Part. 07’s implementation, formalization, question and conclusion sections, and 09’s test, formalization, question and conclusion sections, are printed with 08’s in Part IV and the appendices; 09’s worked example stays in Part III. The three abstracts and status statements are printed in Section 1.
- Every result, proof, example, remark, question and limitation of the three sources is printed. No result was duplicated across the sources, so no second routes arise.
- Labels were prefixed `pqc:` (08), `pqc:qm:` (07) and `pqc:cb:` (09); labels added here are `pqc:conv:`, `pqc:part:`, `pqc:app:provenance`, `pqc:qm:rem:chainformal` and the `pqc:q:`, `pqc:qm:q:`, `pqc:cb:q:` labels on research questions.
- Symbols and macros: Section 2.4. File names: shipped names throughout; 08’s archive table and 07’s and 09’s reproduction paragraphs were rewritten for them.
- Bibliography: the three lists are merged. 07’s and 08’s entries named `matiyasevich` are different works (the 1993 book, the 1971 paper) and both are kept; 07’s entry for ProveIt’s

MRDP guide is kept apart from 08's entry for `MRDP.lean`. 09 cited the same two repository files at the moving `main` branch; those citations now point to the pinned entries, with 09's remark that the guide was last modified on 24 September 2026.

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